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Dissertation for Degree of Doctor
Supervisor: Prof. Jae-Woo Lee

GenAI Optimization Approach for Large Scale Aircraft Design under Uncertainty

Submitted by
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**Department of Aerospace Information
Engineering
Graduate School of Konkuk University**

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Abstract

GenAI Optimization Approach for Large Scale Aircraft Design under Uncertainty

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Design optimization is essential for large-scale engineering problems, such as aircraft design. Real-world design processes inherently involve uncertainty, and evaluating the probability of performance under uncertain conditions typically requires repeated analyses or simulations. These requirements make uncertainty-based aerodynamic shape optimization computationally expensive. The optimal solution depends on how the design space is represented, and the dimensionality of the design variables strongly influences the efficiency of the optimization process. In aircraft design, the design space is often high-dimensional and complex, making it difficult to obtain high-quality solutions and significantly increasing computational cost. These limitations highlight the need for an uncertainty-based optimization framework that can produce high-quality solutions efficiently and adapt to optimization problems of varying sizes and complexities.

This dissertation aims to improve uncertainty-based optimization efficiency, specifically in terms of quality solutions, computational speed, and adaptability to different optimization problem. To achieve this, a Generative-AI (GenAI) optimization framework is proposed. A Variational Autoencoder (VAE) is employed

to reduce complex, high-dimensional geometric design spaces into a compact, lower-dimensional latent representation that captures physically meaningful features while effectively covering the original design space. A Mixture Density Network (MDN) is then trained to predict the uncertainty in aerodynamic performance arising from variations in flight conditions within the latent space, enabling efficient estimation of probability distributions. These models are integrated into an uncertainty-based optimization process, allowing robust and reliability-based optimization to be conducted in the reduced-dimensional design space with significantly lower computational effort.

The proposed framework is demonstrated through two case studies: airfoil design and complex aircraft design using a blended-wing-body (BWB) configuration. The results demonstrate that the method enables the generation of feasible and diverse designs, accurate prediction of aerodynamic performance uncertainty, and substantial improvements in optimization efficiency and solution quality compared to conventional approaches. Furthermore, the proposed approach can address optimization problems with a large number of design variables, demonstrating its adaptability and practical applicability to aircraft design.

These findings indicate that the proposed approach enables uncertainty-based optimization for both simple and complex large-scale problems while achieving substantial improvements in computational efficiency and solution quality. Overall, the proposed methodology represents a significant step forward in aerospace optimization research.

Keyword : Generative AI, Probabilistic machine learning, Large-scale design optimization, Design under uncertainty, Airfoil design, Blended wing body aircraft design

1. Introduction

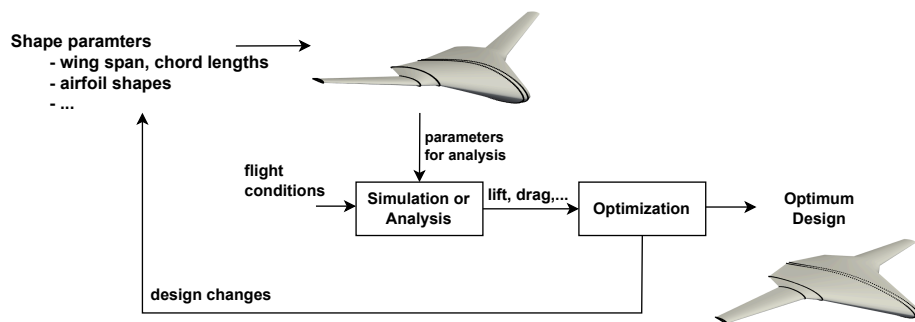
The world around us is composed of complex systems such as aircraft, automobiles, and everyday technologies, all of which involve thousands of design variables. Engineers continually strive to make these systems better, simpler, lighter, faster, stronger, more efficient, and more sustainable. In this context, the design process plays a critical role, influencing nearly every aspect of modern life, from the production of clean water to the technologies we use and the environments in which we live.

The design process involves defining the problem formulation, including what is being designed, which parameters are considered, and what requirements must be satisfied. It is inherently iterative and requires substantial trial and error as the design is refined based on these factors. Given these demands, automating the design process becomes essential, and this is where optimization plays a central role [1]. Optimization is a powerful and effective class of engineering tools that has been validated through decades of successful application [2].

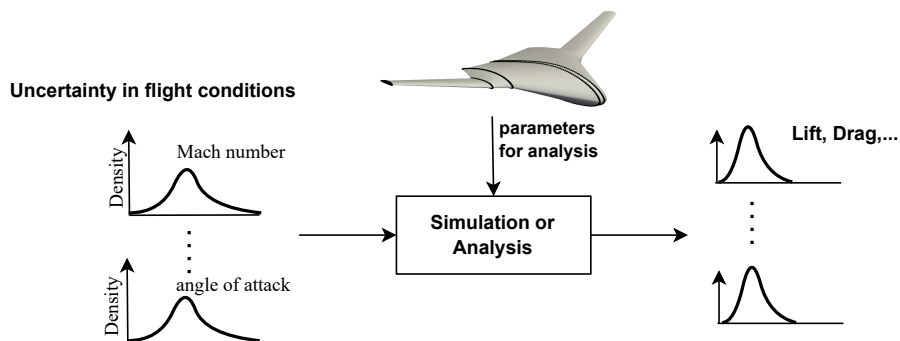
In the field of aircraft design, complex problems that encompass multiple aspects of the aircraft, including geometry, aerodynamics, structural weight, and performance, must be carefully considered. An illustration of the general aircraft design optimization process is presented in <Figure 1-1>. This process consists of iteratively updating design parameters, including the shape of the aircraft, and applying computational analyses and optimization techniques to determine the optimal design configuration.

In the real world, a wide range of uncertainties inherently exists, arising

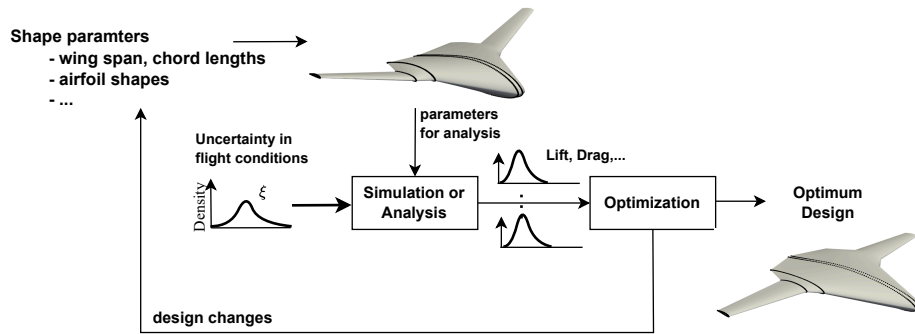
both from the system itself—such as the aircraft—and from the environmental and operational conditions in which it operates [3]. These sources of randomness are important to consider in the design process and can be addressed through uncertainty propagation and probabilistic analysis of the aircraft system, which requires substantial data, as illustrated in <Figure 1-1>. This is where uncertainty-based design optimization [3–5] becomes a crucial aspect of the optimization field, as it accounts for inherent uncertainties during the optimization process, as shown in <Figure 1-3>. Through this process, an optimal design that is robust and reliable in the face of those inherent uncertainties can be achieved.



<Figure 1-1> Illustration of aircraft design optimization



<Figure 1-2> Uncertainty propagation



<Figure 1-3> Aircraft design optimization under uncertainty

Thus, the quality of the best, or optimum, design is strongly influenced by how the problem is represented, including the aircraft shape; by the computational tools employed, such as CFD; and by the search algorithms used throughout the process, particularly when dealing with uncertainties. In this context, an efficient optimization approach plays a crucial role in the overall design process.

Efficient optimization makes solutions significantly easier to find and evaluate, yielding high-quality results while streamlining the overall process [6]. As the optimization process serves as a general engineering problem-solving framework [7], its efficiency can be defined in various aspects, such as **solution quality**, **computational speed**, and **adaptability** [2, 8]. Thus, to achieve an efficient optimization process, a method or framework must be improved along these aspects. These aspects are influenced by the complexity of the optimization problem, the computational cost of the analyses, and the characteristics of the search algorithms. As these attributes improve, the overall efficiency of the optimization process can be greatly enhanced.

Most existing research has primarily focused on enhancing **computational speed**, aiming to obtain solutions more quickly while maintaining accuracy by

replacing expensive high-fidelity analyses with approximation methods, commonly referred to as surrogate models [9, 10]. Improvements in **solution quality** and **adaptability** are typically achieved by advancing optimization algorithms, including gradient-based methods, global optimization techniques such as evolutionary algorithms, and Bayesian optimization approaches [7].

Another crucial attribute is the representation of the design space, which defines, for example, the shape of the aircraft and is characterized by the design variables, a set of parameters that determine all potential solutions [11]. The formulation of these design variables primarily determines the complexity of the optimization problem, which strongly influences the efficiency of the optimization process. This aspect defines the scale of the problem and the representation of design parameters, such as geometric variables, which are commonly expressed using parameterization methods. For instance, two-dimensional geometries are often represented using the Class–Shape Transformation (CST) method [12], while three-dimensional designs can be modeled using Free-Form Deformation (FFD) techniques [13]. Different representations lead to different design spaces. These design spaces are generally high-dimensional and complex, which makes it challenging to obtain high-quality solutions and significantly increases the computational cost of the optimization process.

This dissertation aims to improve three key attributes of optimization efficiency, namely **solution quality**, **speed**, and **adaptability**, by proposing a novel optimization approach that integrates a new representation of complex design spaces into a compact latent space and employs a probabilistic surrogate model to address real-world uncertainties.

1.1 Challenges in conventional design optimization

The aerodynamic design of an aircraft is greatly influenced by its geometric representation, which is defined through a set of design variables in the optimization process. In this process, the design parameters are fed into computational analyses to evaluate aerodynamic performance, and optimization is carried out iteratively, as illustrated in <Figure 1-3>. There are several drawbacks to this conventional approach for aircraft design optimization. The main factors that affect optimization efficiency in this process are: 1) the geometric representation of design parameters (variables), and 2) the computational analysis tools used.

1. Downsides of geometric representation in optimization;

- Limited design space: In conventional approaches, most aircraft design optimizations are performed using only key design parameters such as wingspan and chord lengths. Although this limited geometric freedom simplifies the optimization problem, it prevents the design process from achieving maximum aerodynamic performance.
- High number of dimensions: A large design space with hundreds of design variables is often required to fully exploit the potential of aerodynamic design [14]. However, this significantly increases the complexity of the optimization problem, and this leads to a design space that is much larger than required for optimization [15, 16].
- Generate infeasible design: Traditional parameterization methods, such as CST, FFD, and B-spline [17], often produce abnormal shapes when using a large number of dimensions, introducing unnecessary difficulties

in the aircraft design process.

2. Downsides of using computational analysis tools in optimization;

- High computational cost: The performance response of an aircraft is analyzed using computational tools or programs. During the optimization process, numerous design changes require repeated analyses until the optimal design is found. Furthermore, when dealing with uncertainties, a large amount of data is required to model them accurately [3], which greatly increases the computational cost of the optimization process.

An efficient optimization approach must overcome these challenges by generating high-quality solutions, identifying them quickly, and adapting to problems of any scale or size. These challenges can be addressed by learning from experience, a concept made possible through machine learning techniques. Once properly trained, such models can automatically capture significant features of the design space, allowing the dimensionality to be reduced into a compact lower-dimensional representation without producing abnormal shapes, even within large design spaces. In addition, probabilistic machine learning enables uncertainty modeling, although it typically requires large amounts of data. Thus, the integration of generative AI and probabilistic surrogate models makes it possible to enhance conventional uncertainty-based optimization approaches, as proposed in this dissertation.

1.2 Research objectives and contributions of the dissertation

As I proposed an optimization approach, I consider the following three aspects of optimization efficiency, as discussed in [2, 7]:

- **Quality of solutions:** Does the framework generate high-quality solutions that meet performance objectives and satisfy all constraints?
- **Speed:** Given a design problem, does the framework produce solutions quickly?
- **Adaptability:** Can the framework effectively adapt to various problems that differ in scale, complexity, or domain?

Building upon these requirements and the challenges of the conventional optimization approach, the following research objectives are addressed.

- To develop a generative model that represents the high-dimensional design space in a compact lower-dimensional space without losing its effectiveness and that captures design variability without generating infeasible geometric shapes.
- To develop a surrogate model capable of predicting performance uncertainties, such as aerodynamic performance, from a lower-dimensional representation of the design space.
- To develop an efficient optimization approach by integrating a generative model and a surrogate model for different optimization formulations, including deterministic, robust, and reliability-based optimization.

In response to the aforementioned developments, challenges, and requirements, the contributions of this dissertation focus on problem-solving, as demonstrated in the case study sections.

Contribution 1: This research demonstrates how complex, high-dimensional design spaces in airfoil and aircraft design can be reduced to compact lower-dimensional representations without losing effectiveness by employing deep generative models.

Contribution 2: Abnormal shape generation, which often burdens the optimization process, is eliminated by learning from existing systematic and meaningful data representations of high-dimensional spaces.

Contribution 3: The uncertainty of aerodynamic performance can be accurately predicted from lower-dimensional representations by modeling a probabilistic surrogate model, thereby eliminating the need for repeated computations.

Contribution 4: A constraint-based geometric sampling strategy is proposed to generate diverse aircraft shapes without producing infeasible designs for deep generative model training. This method can be effectively applied not only to aircraft but also to wing-like shapes such as propeller blades and wind turbine blades.

Contribution 5: An efficient optimization approach is developed that can adapt to different scales of optimization problems, from airfoil design to complex aircraft design, through generative modeling.

1.3 Outlines of dissertation

The dissertation is structured as follows

Chapter 2: reviews the existing literature on design optimization, uncertainties, uncertainty-based optimization, dimensionality reduction, surrogate modeling, and generative AI and its applications, all of which are related to the proposed optimization approach.

Chapter 3: describes the step-by-step development of the GenAI-enhanced optimization approach, explaining each module in detail and the required metrics for evaluation.

Chapter 4: presents the case study of the proposed approach for airfoil

design, demonstrating how generative AI and probabilistic surrogate modeling overcome the limitations of conventional optimization methods.

Chapter 5: demonstrates the adaptability of the proposed optimization approach to a more challenging problem, aircraft design, presenting notable results in shape generation and improved optimization efficiency in both deterministic and uncertainty-based optimization.

Chapter 6: provides a conclusion summarizing the main contributions, key findings, limitations of this work, and potential future research directions.

2. Literature Review and Related Works

This section reviews related areas of this research work, including design optimization, uncertainty-based design optimization, generative AI, and probabilistic surrogate modeling.

2.1 Overview of design optimization in engineering

In the engineering design process, design optimization has served as a core mechanism, providing a systematic framework for improving a design's performance and efficiency [7]. In aerospace applications, design optimization is essential, as multiple disciplines are considered throughout the process, such as in MDO [18, 19]. The optimal design parameters are formalized through the optimization formulation, in which the design process involves the minimization or maximization of an objective function $f(\mathbf{x})$ over a design vector $\mathbf{x} \in \mathbb{R}^D$, subject to functional constraints. This can be written as a general optimization formulation;

$$\min_{\mathbf{x} \in \mathcal{D}} f(\mathbf{x}) \quad \text{subject to} \quad g_i(\mathbf{x}) \leq 0, \quad h_j(\mathbf{x}) = 0 \quad (2.1)$$

where, g_i is the inequality constraints, and h_j is the equality constraints.

The design of aerospace systems has long applied this framework to identify optimal configurations that satisfy both design and operational requirements. The primary challenge in applying this framework to complex aerospace systems is computational intractability. The function evaluations $f(\mathbf{x})$ and $g_i(\mathbf{x})$

require the execution of high-fidelity simulations, such as Computational Fluid Dynamics (CFD), which are computationally expensive. The iterative process of optimization requires multiple analyses, sensitivity analyses, and thorough design updates while searching for the optimum design using gradient-based or global optimization approaches, which are costly for large-scale optimization problems. This challenge motivates the development of approximation and machine learning techniques for efficient optimization.

2.1.1 Optimization algorithms and efficiencies

The efficacy of an optimization method is determined by its metrics among **speed** (computational efficiency), **solution quality** (proximity to the global optimum), and **adaptability** (scalability to complex design problems) [7, 20, 21]. The optimal design is searched through optimization algorithms, which can be divided into gradient-based optimization and gradient-free (global) algorithms.

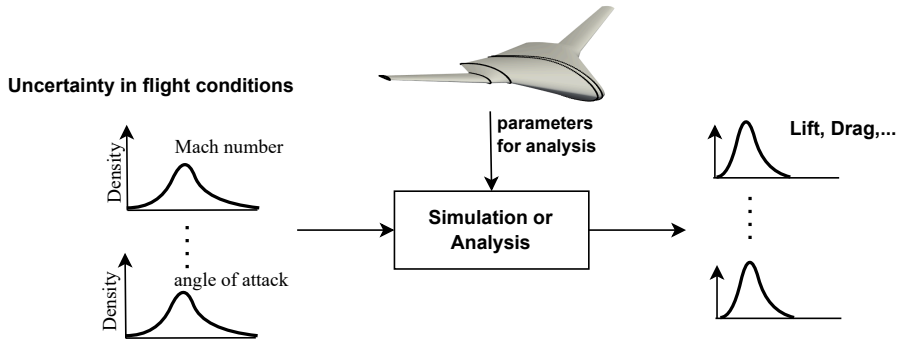
- **Gradient-based search algorithms:** These methods, often employing either the gradient of a function or adjoint-based sensitivity analysis, exhibit rapid local convergence and computational efficiency [22, 23]. However, their effectiveness is limited in non-convex design spaces, where they are highly susceptible to convergence to sub-optimal local optima (minimum or maximum) [7, 20].
- **Gradient-free search algorithms:** Global optimization methods, such as Evolutionary Algorithms (EAs), such as Genetic Algorithms (GA), ant-colony optimization (ACO), particle-swarm optimization (PSO), and Bayesian Optimization (BO), provide robust exploration capabilities and increase the likelihood of locating global optima [20, 21]. Yet, their stochastic nature demands a large number of evaluations, which, when coupled

with simulations or analysis methods, leads to significant computational cost. [9, 10]. Consequently, recent efforts focus on developing hybrid or surrogate-based approaches to retain global search efficiency while reducing computational expense [16, 24].

2.1.2 Aerodynamics shape optimization (ASO)

The design of aircraft is governed by Multidisciplinary Design Optimization (MDO) [25, 26], a framework that formally integrates coupled disciplines such as aerodynamics, structures, propulsion, mission, and performance [19, 27]. Aerodynamic Shape Optimization (ASO) plays a pivotal role within MDO by defining the geometry (shape) of the aircraft [28, 29].

In ASO optimization, the representation of shape parameters, analysis methods, and optimization algorithms is derived. While many studies address simplified 2D problems (e.g., 2D airfoils [30, 31]), scaling to complete 3D configurations (e.g., aircraft wings, tails) presents unique challenges. These include a high number of design variables, the generation of abnormal shapes, nonlinear coupling between aerodynamic and structural effects, and exponential growth in computational cost due to high-fidelity CFD simulations[32, 33]. Therefore, optimization frameworks that can robustly overcome these challenges and scale to this level of complexity are essential for achieving maximum efficiency in aircraft design.



<Figure 2-1> Illustration of uncertainty propagation with aleatory uncertainty (flight conditions) for aircraft design

2.2 Uncertainty-based optimization (UBO)

2.2.1 Random variables and uncertainty propagation

Engineering systems are inherently subject to random factors (ξ), including aleatoric (e.g., flight conditions, material properties) and epistemic uncertainties (e.g., model inadequacy, lack of data) [3, 34]. Quantifying their stochastic impact is critical for ensuring reliability and robustness in system design. This uncertainty analysis focuses on quantifying the uncertainty (UQ) characteristics of the system output $P(\mathbf{x}, \xi)$ resulting from model input and model uncertainties propagated through computational simulation [3], as illustrated in <Figure. 2-1>.

Uncertainty can be modeled by non-probabilistic methods, which address uncertainty when data are insufficient by providing bounds (interval method) [35, 36] or membership degrees (fuzzy method) [37–39], rather than likelihoods. In contrast, probabilistic methods address uncertainty when variables are described by Probability Density Functions (PDFs), which require substantial data, enabling formal reliability and risk quantification [3, 34, 40, 41].

While dealing with PDF modeling for uncertainty propagation, the Monte Carlo Simulation (MCS) remains a conceptually simple computational algorithm that relies on repeated random sampling to obtain numerical results. However, it becomes computationally infeasible for large-scale problems due to the large number of samples required for PDF estimation [42]. Hence, sample-efficient methods such as Polynomial Chaos Expansion (PCE) and Stochastic Collocation (SC) are often employed [43, 44].

2.2.2 Design optimization under uncertainty

Uncertainty-based optimization (UBO) incorporates probabilistic performance directly into the optimization process. This formulation focuses on robust designs, which are less sensitive to random variables, and reliable designs, which have lower probabilities of failure with respect to design constraints.

- **Robust design optimization (RDO):** Focuses on minimizing the sensitivity of performance P to uncertainty ξ :

$$\min_{\mathbf{x}} \left(E[P(\mathbf{x}, \xi)] + w \cdot \sqrt{Var[P(\mathbf{x}, \xi)]} \right) \quad (2.2)$$

where w balances nominal performance and robustness, $E[P(\mathbf{x}, \xi)]$ is the expected value (mean value) and $Var[P(\mathbf{x}, \xi)]$ is the variance of the performance function. [4, 45].

- **Reliability-based design optimization (RBDO):** RBDO aims to find the optimum design that minimizes a cost and satisfies an allocated reliability target with respect to performance function(s), while accounting for various sources of uncertainty (e.g., material properties, geometric toler-

ances, and loading conditions).

$$P_f[g_j(\mathbf{x}, \boldsymbol{\xi}) \geq 0] \leq \Phi(-P_{freq}) \quad (2.3)$$

where $P_{freq} = \Phi(-\beta_{target})$ represents required or failure reliability, β_{target} represents target reliability index, Φ represents the cumulative distribution function of standard normal distribution [40, 46, 47].

2.2.3 Computational challenges of UBO

UBO requires repeated evaluations of analysis or simulation for statistical moments and failure probabilities by modeling PDF functions within the optimization loop, often leading to a nested, bi-level formulation [48, 49]. This compounding of computational expense highlights the need for efficient surrogate models to make RDO and RBDO feasible for large-scale problems [50–52].

2.3 Geometric shape representation

2.3.1 Downsides of conventional parameterization

Conventional geometric parameterization methods, such as Class-Shape Transformation (CST) [12], Free-Form Deformation (FFD) [13], and B-Splines/NURBS [17], define the design vector $\mathbf{x}$ directly in physical space. Achieving high geometric variability typically requires hundreds or thousands of parameters, resulting in high-dimensional and redundant design spaces [15]. These spaces often contain infeasible designs, necessitating geometric filtering or penalty functions [53], which degrade optimization efficiency.

2.3.2 Learning lower-dimensional representation

To overcome the high dimensionality and infeasibility of designs, dimensionality reduction (DR) is used to map $\mathbf{x} \in \mathbb{R}^D$ to a compact latent space $\mathbf{z} \in \mathbb{R}^k$. Recent developments in deep generative models, such as autoencoders (AEs), variational autoencoders (VAEs), and generative adversarial networks (GANs), have enabled the learning of nonlinear and structured mappings [16, 21, 24].

- **Variational Autoencoders (VAEs):** Learn a smooth, continuous latent distribution through Kullback–Leibler regularization in the Evidence Lower Bound (ELBO) loss [54]. VAEs have been applied to airfoil generation [55], wind turbine optimization [56], and aeroacoustic design [57].
- **Generative Adversarial Networks (GANs):** Enforce geometric variability through adversarial learning of the shape distribution. Examples include BézierGAN for aerodynamic shapes [31, 58].

The generative model learns a lower-dimensional representation of the design data in a latent space, reducing the search space to a compact form and enabling efficient optimization.

2.4 Machine learning and design optimization

2.4.1 Deterministic surrogate modeling and latent space optimization (LSO)

The combination of DR via generative models and surrogate modeling via neural networks or Kriging defines latent space optimization (LSO) [59–61]. LSO shifts the computationally expensive optimization from the physical space to

the latent space:

$$\min_{\mathbf{z}} \hat{f}(\mathbf{z}) \quad (2.4)$$

This approach significantly accelerates optimization by replacing analyses with surrogate models in reduced dimensions. However, existing LSO frameworks [60, 62] are deterministic and provide only point estimates, making them unsuitable for uncertainty-based optimization.

2.4.2 Probabilistic machine learning models for UDO

To enable Uncertainty-Based Optimization (UBO) within LSO, the surrogate model can be made probabilistic to estimate PDF functions. Probabilistic machine learning models predict full distributions $p(y|\mathbf{x})$, enabling the propagation of performance uncertainty. **Bayesian Neural Networks (BNNs):** Quantify model uncertainty using distributions over network weights, approximated via variational inference or dropout [63, 64]. **Mixture Density Networks (MDNs):** Predict Gaussian mixture parameters and represent conditional PDFs.

$$P(Y|\mathbf{X}) = \sum_{i=1}^m \alpha_i(\mathbf{X}), \mathcal{N}(Y|\mu_i(\mathbf{X}), \sigma_i^2(\mathbf{X})) \quad (2.5)$$

where, $p(Y|\mathbf{X})$ is the conditional probability density function, $\mathcal{N}(Y|\mu_i(\mathbf{X}), \sigma_i^2(\mathbf{X}))$ is the Gaussian mixture model, α is the mixing coefficients. This process enables direct computation of statistical moments for RDO/RBDO [65–67].

2.5 Related works

This section reviews related works in the literature, summarized in <Table 2-1>. Numerous papers have been considered, ranging from aerospace applications to wind and electrical forecasting, all relevant to the methodology of this disser-

<Table 2-1> Related works in literature: shape representation, analysis method, optimization, application, and reference.

Shape representation	Analysis method	Optimization	Application	Reference
SVD	CFD (analysis)	None	Airfoil/wing	[16]
B-spline	CFD (analysis)	Deterministic	Airfoil	[68]
GAN	GPR	Deterministic	Airfoil/wing	[15, 69]
GAN (BezierGAN)	CFD + GPR	Deterministic (BO)	Airfoil	[31, 58]
GAN (B-spline GAN)	CFD + GPR	Deterministic (LSO)	Airfoil	[70]
VAE	CFD (analysis)	Deterministic (LSO)	Airfoil	[57]
VAE	GPR	RDO (LSO)	Airfoil	[71]
VAE	DNN	None	Airfoil	[55]
VAE-GAN	DNN	None / Deterministic	Airfoil / blade	[56, 72]
VAE	DNN	Deterministic (LSO)	Materials design	[59, 60]
GAN	DNN	Deterministic	Airfoil	[62]
CST	Kriging + MF-EI	RDO	Airfoil	[48]
GAN	None	None	3D Propeller	[73]
GAN	GPR	Deterministic (LSO)	3D Wing	[74]
AE	GPR	Deterministic (LSO)	3D Wing	[75]
VAE	DNN	Deterministic (LSO)	3D Car	[76]
AE	Analysis	RBDO	2D Truss	[77]
-	DNN	RBDO / UQ	2D Truss	[78]
-	BNN	None / UQ (Reliability analysis)	Forecast (electrical)	[63, 64]
-	MDN	None / UQ (Reliability analysis)	Bridge design	[79]
-	MDN	RDO / UQ	Forecast (Wind/energy)	[67, 80, 81]
PCA (DR)	DBN	RDO	Airfoil	[24]
VAE	MDN	RDO	Materials design	[66]

GPR = Gaussian Process Regression; DNN = Deep Neural Network; DBN = Deep Brief Neural Network, BNN = Bayesian Neural Network; MDN = Mixture Density Network; LSO = Latent Space Optimization; RDO = Robust Design Optimization; RBDO = Reliability-Based Design Optimization.

tation. The review is organized into shape representation, analysis methods, optimization, and application areas.

Shape representation has traditionally relied on conventional methods [48];

however, these approaches often produce abnormal shapes. To address this, advanced techniques such as principal component analysis (PCA) [24] and singular value decomposition (SVD) [16] have been effectively applied to the design of airfoils and wings, capturing shape variability with fewer parameters and enabling optimization in reduced-dimensional spaces. More recently, deep learning-based methods such as autoencoders (AEs) [82], variational autoencoders (VAEs) [54], and generative adversarial networks (GANs) [83] have shown strong potential for learning compact representations of design variables. GAN-based methods [15, 31, 58, 70] can generate smooth aerodynamic shapes from a low-dimensional latent space and have been applied to airfoil optimization. Another popular method is the VAE, which learns a structured and continuous lower-dimensional latent space for intuitive shape generation [55, 57, 71]. These methods have been used primarily for two-dimensional airfoil optimization, leveraging surrogate modeling techniques—particularly Kriging, Gaussian Process Regression (GPR), variable fidelity modeling [84], and Deep Neural Networks (DNNs) [85, 86]—to reduce the cost of simulation-based optimization.

Additionally, uncertainties arising from operations, manufacturing, and model fidelity can degrade performance, making RDO [87] and RBDO [46] essential in aerospace design. These methods incorporate performance PDFs into objectives and constraints and often employ bi-level frameworks [48] that require costly Monte Carlo simulations. For example, [48] used Multi-Fidelity Hierarchical Kriging with Multi-Fidelity Expected Improvement for robust airfoil design, [88] employed a self-organizing map neural network for turbine optimization, and [24] combined PCA with a Deep Belief Network (DBN) for surrogate modeling. Other surrogate models, such as Polynomial Chaos Expansion (PCE), are also sample-efficient for modeling under uncertainty; however, a major draw-

back is that the number of samples grows exponentially with the number of inputs [52], making computation impractical and leading to the curse of dimensionality [89].

Recent advances in probabilistic machine learning have increased the use of Bayesian Neural Networks (BNNs) and Mixture Density Networks (MDNs) for quantifying uncertainty and predicting statistical properties and PDFs with limited data [66], even in high-dimensional settings. These models have been applied in reliability analysis—such as BNNs for uncertainty propagation [63] and failure probability prediction [64]—and in MDNs for wind prediction [80], surrogate modeling of wind turbine blades [67], reliability analysis [64, 81], and robust design optimization for materials [66].

Furthermore, most shape optimization studies in the literature focus on 2D airfoil designs; only a limited number address 3D wings (e.g., [73–75]). These methods typically focus on deterministic optimization and are applied to propeller blades and wing design, often using point clouds to learn the lower-dimensional space while overlooking uncertainty quantification. Few efforts integrate generative models for dimensionality reduction with probabilistic surrogates to address uncertainty-based design optimization in aerodynamic shape optimization (ASO). For example, [24] employed PCA with belief networks, and [66] applied VAE–MDN to materials design, implicitly modeling uncertainties. However, uncertainty prediction and the estimation of reliability or failure probability from the latent space, with explicit consideration of uncertainty sources, remain largely unaddressed—particularly for aerodynamic shape optimization and especially for three-dimensional aircraft design.

2.6 Research Gap

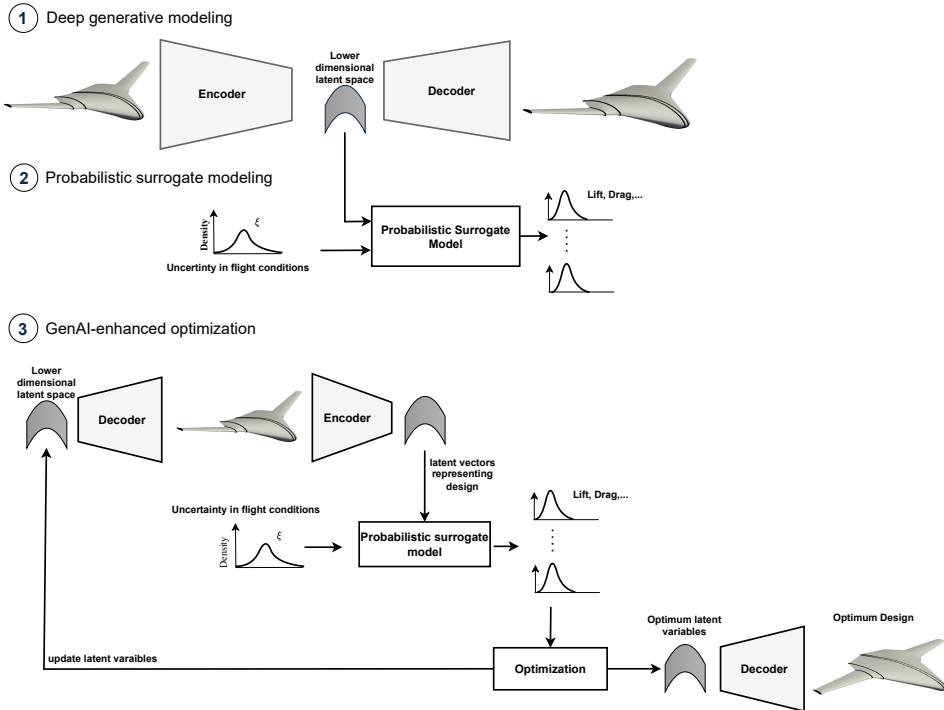
The literature clearly establishes the importance of: (1) generative models for low-dimensional geometric representation (**Contribution 1,2**), and (2) probabilistic surrogates for uncertainty quantification (**Contribution 3**).

However, a significant research gap remains in integrating these principles for complex aircraft systems. Most existing studies are limited to component-level, 2D applications [15, 24, 55, 62, 71], with only a few addressing 3D parameterization [73–75]—and these use deterministic surrogate models. No published work demonstrates the combined use of VAE and MDN for design optimization under uncertainty directly through a compact, lower-dimensional latent space for 3D aircraft configurations.

This dissertation addresses this gap by developing a generative AI probabilistic optimization framework that unites VAEs for design generation with MDNs for probabilistic performance prediction. This framework enables robust and reliable design optimization that is both computationally feasible and scalable to full aircraft design problems (**Contribution 5**), representing a significant advancement in aerospace optimization research.

3. GenAI Optimization Approach

This section presents our generative AI optimization approach, designed to overcome the limitations of conventional optimization methods. An overview of the design optimization process is shown in <Figure 3-1>. It involves three main steps: (1) Deep generative modeling to learn a lower-dimensional representation of the complex design space, (2) Probabilistic surrogate modeling to predict performance uncertainties from input variations, using limited data from the compact latent space, and (3) Performing design optimization within the latent space for deterministic, robust, and reliable designs.



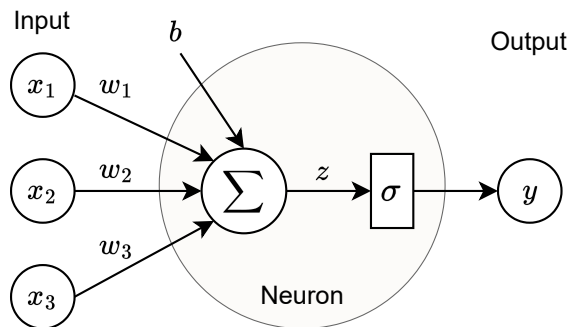
<Figure 3-1> Overall Process

The following sections explain each step in detail. First, we provide an overview of deep neural network architectures and their operation in Section 3.1, followed by deep generative modeling in Section 3.2, and the training and evaluation of the probabilistic surrogate model in Section 3.3. Subsequently, we present the methodology for conducting robust and reliable design optimization based on the trained models in Section 3.4.

3.1 Overview of deep neural network

The core computational mechanism behind both deep generative modeling and probabilistic surrogate modeling is the deep neural network (DNN). DNNs are powerful and able to perform highly complex learning tasks [90, 91], such as uncovering hidden patterns, nonlinear regression, classification, and generative modeling.

A DNN is composed of input, hidden, and output layers of interconnected nodes, or neurons. Neural networks were developed as generalized mathematical models inspired by biological nervous systems [92]. In a single neuron, as shown in <Figure 3-2>, there is a mathematical operation to compute the



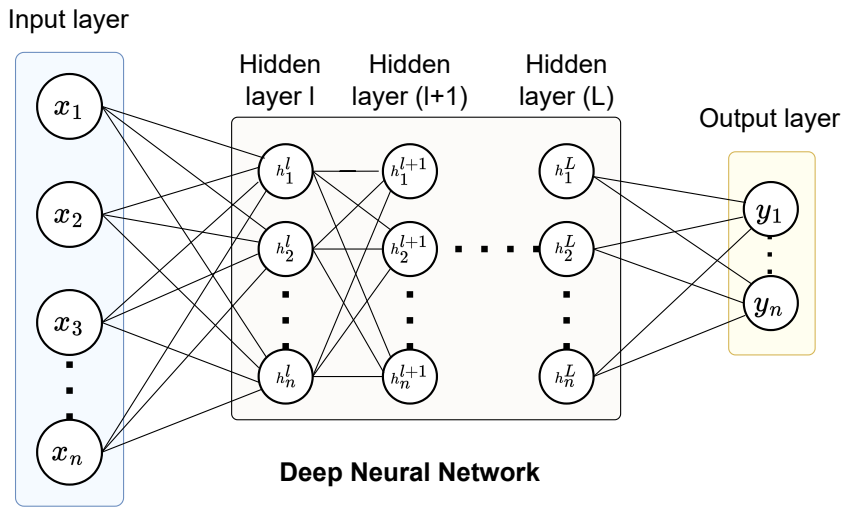
<Figure 3-2> Schematic diagram of single neuron in a neural network

neuron's output. This operation can be expressed as follows:

$$z = \sum \mathbf{W} \mathbf{x}^T + b = w_1x_1 + w_2x_2 + w_3x_3 + b \quad (3.1)$$

$$y = \sigma(z) \quad (3.2)$$

where $\mathbf{W}$ is the vector of weight values, $\mathbf{x}$ is the vector of input values, w_1, w_2, w_3 are the individual weight values, x_1, x_2, x_3 are the input values, and b is the bias term. The weighted sum is then passed through a nonlinear activation function, $\sigma(\cdot)$, which determines whether the neuron fires and produces an output (y). A commonly used activation function is the Rectified Linear Unit (ReLU).



<Figure 3-3> Schematic diagram of deep neural network (DNN)

An example of a DNN architecture is illustrated in <Figure 3-3>. A deep neural network extends the concept of a single neuron by stacking multiple layers of interconnected neurons between the input and output layers. Each layer transforms the data through a series of weighted summations and nonlinear activations, enabling the network to learn complex, nonlinear relationships

between inputs and outputs. The mathematical operation of a DNN can be represented as a sequence of linear and non-linear transformations applied layer by layer, known as forward propagation. For an input vector x , the operation at the l^{th} layer can be expressed as

$$\mathbf{z}^{(l)} = \mathbf{W}^{(l)} \mathbf{a}^{(l-1)} + \mathbf{b}^{(l)}, \quad (3.3)$$

where $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ are the weight matrix and bias vector of the l^{th} layer, respectively, and $\mathbf{a}^{(l-1)}$ is the output from the previous layer ($\mathbf{a}^{(0)} = x$). The activation of each neuron in the l^{th} layer is then obtained by applying a non-linear activation function $\sigma(\cdot)$ as

$$\mathbf{a}^{(l)} = \sigma(\mathbf{z}^{(l)}). \quad (3.4)$$

Repeating this process for all layers, the output of the network can be written as

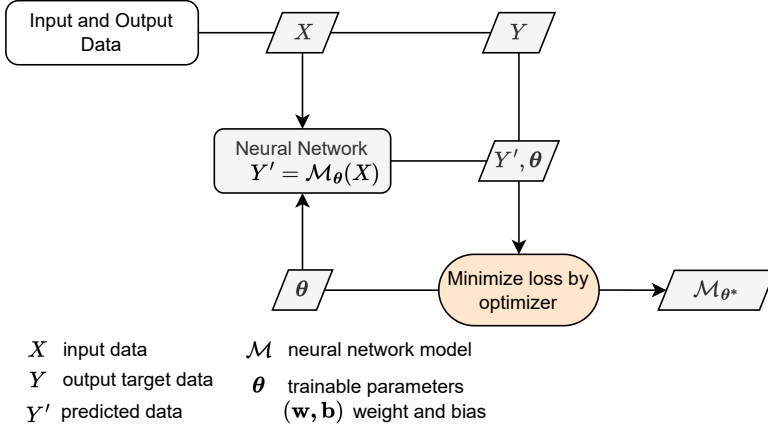
$$\mathbf{Y}' = \mathbf{a}^{(L)} = \sigma^{(L)} \left(\mathbf{W}^{(L)} \sigma^{(L-1)} \left(\dots \sigma^{(1)} (\mathbf{W}^{(1)} x + \mathbf{b}^{(1)}) \dots \right) + \mathbf{b}^{(L)} \right), \quad (3.5)$$

where L denotes the total number of layers in the network. This formulation allows the DNN to learn complex mappings between input (X) and output data (Y') through successive layers of linear transformations and non-linear activations.

Since this process initially only predicts the output data Y' , the weights and biases ($\mathbf{W}, \mathbf{b}$) are iteratively updated to better predict the target output data Y . This is achieved by minimizing a loss function, which can be calculated as shown in Equation 3.6.

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (Y - Y')^2 \quad (3.6)$$

where N is the total number of samples.



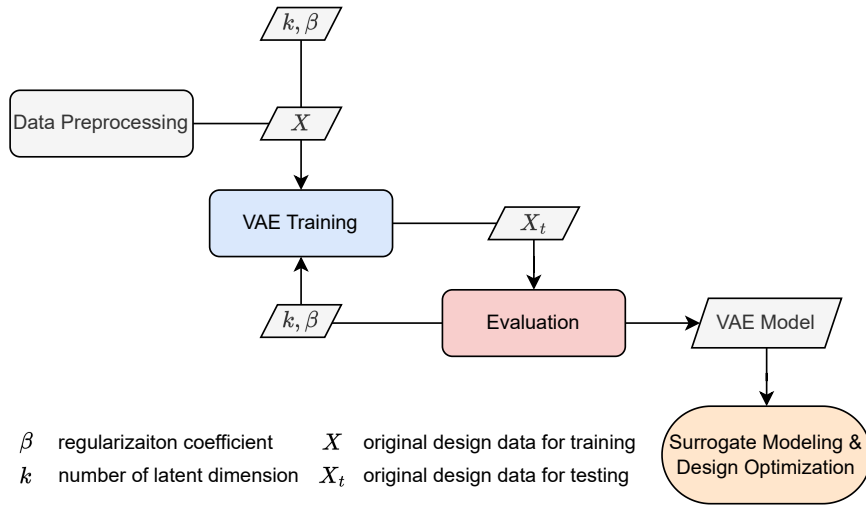
<Figure 3-4> Training process of deep neural network

This procedure is known as the *training process*, as illustrated in <Figure 3-4>, where the objective is to obtain a neural network model ($\mathcal{M}$) with optimal parameters (θ^*), representing the weights and biases ($\mathbf{W}, \mathbf{b}$) of the network. The training is performed by minimizing the loss function defined in Equation (3.6) through the backpropagation algorithm.

Through the use of deep neural networks, both generative modeling and probabilistic surrogate modeling can be effectively implemented for GenAI optimization approach.

3.2 Deep generative modeling

This section explains the training process used to learn a lower-dimensional latent representation from a complex design space, as utilized in surrogate



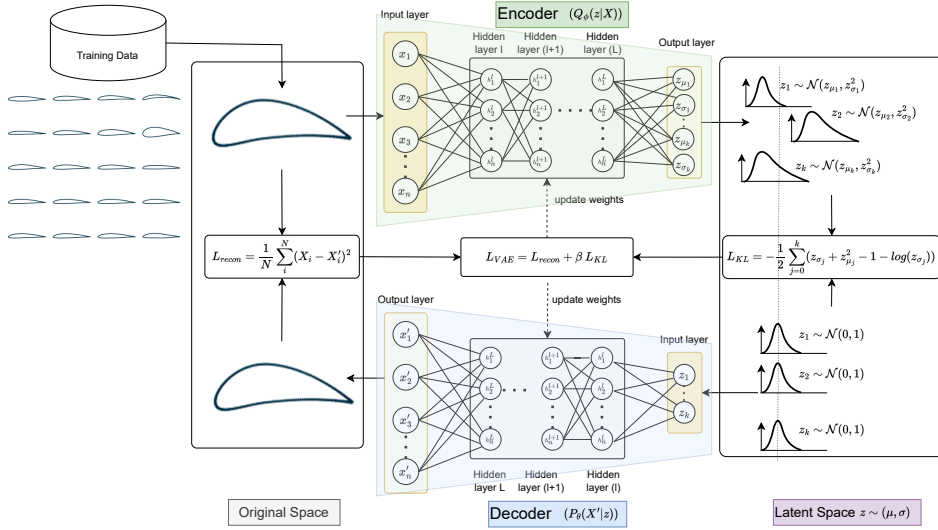
<Figure 3-5> Process and data dependencies for deep generative modeling

modeling and design optimization shown in <Figure 3-5>.

The process includes three main steps: (1) Data pre-processing, where the original design data—which may be complicated and unordered—are unified into a consistent format and normalized for use in training the generative model; (2) Training the generative model, specifically a variational autoencoder (VAE), to learn the latent representation of complex design data (details of the VAE are provided in Section 3.2.1); and (3) Model evaluation, in which we assess the model’s ability to obtain a disentangled latent representation using test data (X_t).

The evaluation focuses on key variables, including the number of latent dimensions (k) and the regularization coefficient (β), and is based on generative model evaluation metrics: (1) generative quality, (2) reconstruction quality, and (3) compactness, as described in Section 3.2.2. Once the model meets these criteria, it is used for surrogate modeling and the design optimization process.

3.2.1 VAE training

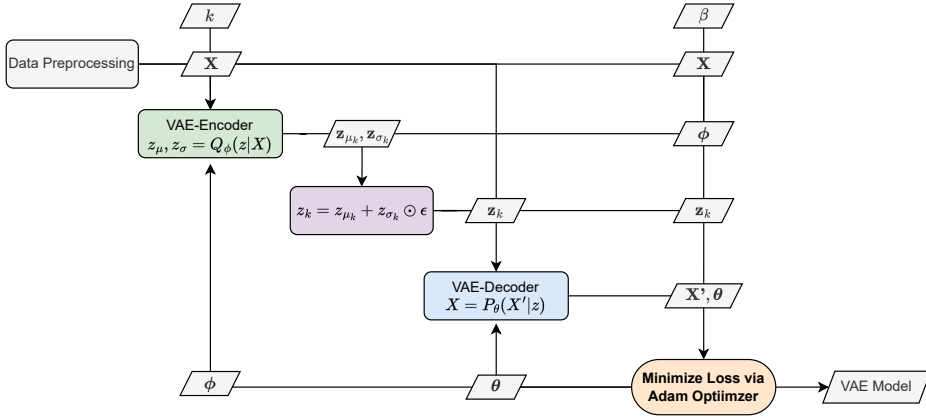


<Figure 3-6> Variational Autoencoders (VAE)

A VAE, first introduced by [54], is a probabilistic extension of autoencoders that learns a reduced-dimensional latent representation of design parameters using two neural networks: an encoder and a decoder, with a latent space in between, as illustrated in <Figure 3-6>. It can be trained using the procedures shown in <Figure 3-7>.

During training, the encoder (Q_ϕ) performs nonlinear dimensionality reduction by compressing the high-dimensional input data (X , $X \in \mathbb{R}^N$) into two k -dimensional vectors: $z_\mu \in \mathbb{R}^k$ and $z_\sigma \in \mathbb{R}^k$, which represent the mean and standard deviation of a multivariate Gaussian distribution, respectively. Here, k is the dimension of the latent space and is a hyperparameter that must be chosen to adequately capture the features of the input dataset (X). These vectors are parameterized by a set of trainable parameters, ϕ , which are optimized by minimizing the loss function.

A lower-dimensional latent representation $z \in \mathbb{R}^k$ is obtained by sampling



<Figure 3-7> Training Procedure of VAE

from the predicted mean and standard deviation. The latent space can be expressed as Eq. (3.7), also called a reparameterization trick [54], which enables backpropagation during training.

$$z = z_{\mu} + z_{\sigma} \odot \epsilon \text{ with } \epsilon \sim \mathcal{N}(0, I) \quad (3.7)$$

where z is the latent vector encoding of the input design space; z_{μ} and z_{σ} are the predicted mean and standard deviation from the encoder; $\odot$ denotes element-wise multiplication; and ϵ is a random variable sampled from a fixed standard normal distribution. The decoder (P_{θ}) maps the latent space z back to a reconstruction of the input data (X') and is parameterized by another set of trainable parameters, θ .

$$\mathcal{L}_{VAE} = \mathcal{L}_{\text{recon}}(x, x') + \beta \mathcal{L}_{KL}(z, \mathcal{N}(0, I)) \quad (3.8)$$

The loss function of a VAE consists of two terms. The first term, the reconstruction loss $\mathcal{L}_{\text{recon}}$, ensures that the input data can be accurately reconstructed from the latent representation and is typically expressed as the mean

squared error: $\mathcal{L}_{\text{recon}} = \frac{1}{N} \sum_{i=1}^N (x_i - x'_i)^2$. The second term is the Kullback–Leibler (KL) divergence, which measures the difference between the distribution of the latent vector and the standard Gaussian prior $\mathcal{N}(0, I)$; this is given by $\mathcal{L}_{\text{KL}} = D_{\text{KL}}(Q_{\phi}(z|x) || \mathcal{N}(0, I))$, as shown in Eq. (3.9). During training, the VAE learns the parameters of the encoder (ϕ) and decoder (θ) by minimizing the total loss function in Eq. (3.8), where β is the regularization coefficient for the latent vectors.

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}(Q_{\phi}(z|x) || \mathcal{N}(0, I)) = -\frac{1}{2} \sum_{j=0}^k (z_{\sigma_j} + z_{\mu_j}^2 - 1 - \log z_{\sigma_j}) \quad (3.9)$$

In VAE training, the parameters ϕ and θ represent the $(\mathbf{W}, \mathbf{b})$ of the encoder and decoder networks, respectively. However, the most critical hyperparameters, the latent dimension k and the regularization coefficient β , must be carefully tuned using appropriate evaluation metrics in Section. 3.2.2 to obtain a disentangled latent representation.

3.2.2 Evaluation metrics for generative model

The evaluation of the VAE model aims to obtain a disentangled latent-space representation. The latent space is a compact representation of geometric variability, where each dimension captures distinct geometric features. Each latent dimension is modeled as a standard normal distribution.[93] states that a disentangled representation is one in which individual latent variables are sensitive to changes in individual generating factors, while being relatively insensitive to other changes. Disentanglement is impossible without first learning a sufficiently expressive latent representation capable of good reconstruction. To evaluate the disentangled latent representation, the following questions need

to be addressed [11]:

- **(1) Generative quality** Are the shapes represented in the latent space z similar to those in the original design space X ?
- **(2) Reconstruction quality** Given a latent vector z , can the model accurately reconstruct the original design parameters X that generated it?
- **(3) Disentanglement/Compactness** Does the latent vector capture the relevant variations while remaining insensitive to changes in unrelated generative factors?

(1) Generative quality is related to the typically evaluated using the Maximum Mean Discrepancy (MMD) metric [73, 74, 94]. MMD is a statistical measure that quantifies the difference between two probability distributions P and Q based on samples drawn from them. In the context of generative models, MMD compares the distribution of generated designs (X') with the distribution of the original dataset (X). A small MMD value indicates that the generated data distribution is close to the real data distribution, demonstrating that the model has successfully captured the characteristics of the original design space. MMD metrics is evaluated by Eq. (3.10).

$$\widehat{\text{MMD}}^2(X, X') = \frac{1}{m^2} \sum_{i=1}^m \sum_{j=1}^m k(x_i, x_j) + \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n k(x'_i, x'_j) - \frac{2}{mn} \sum_{i=1}^m \sum_{j=1}^n k(x_i, x'_j) \quad (3.10)$$

where $k(\cdot, \cdot)$ is a positive-definite kernel function, commonly chosen as the Gaussian radial basis function (RBF) kernel:

$$k(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2\sigma^2}\right) \quad (3.11)$$

(2) Reconstruction quality is evaluated using reconstruction error, which measures how the actual design (X) differ from the design (X') that we reconstructed from the lower-dimensional space. In this research, Mean square error (MSE) is used for the reconstitution error shown in Eq. 3.12.

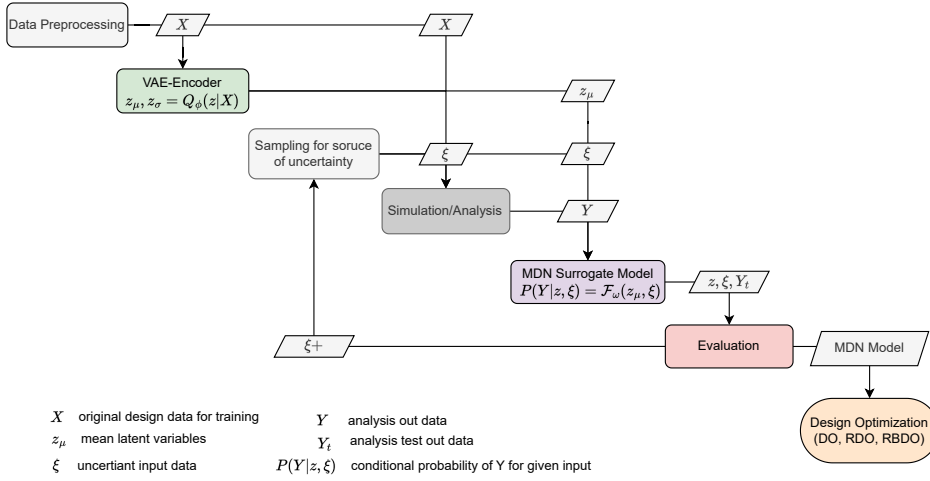
$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (X - X')^2 \quad (3.12)$$

(3) Disentanglement is evaluated by the independence of the latent variables, correlation coefficients can be used. In this research, we use the determinant of the correlation matrix [95], [71], [56], denoted by $\det \mathbf{R}$, where $\mathbf{R} = (R_{ij})_{d \times d}$:

$$R_{ij} = \frac{C_{ij}}{\sqrt{C_{ii}C_{jj}}}, \quad R_{ii} = 1 \quad (3.13)$$

for all $1 \leq i \neq j \leq d$, where C_{ij} denotes the (i, j) component of the covariance matrix C . Note that $\det \mathbf{R} = 100$ when all latent variables are completely uncorrelated. We report the value of $\det \mathbf{R}$ as a metric to quantify the independence of the latent variables.

The three metrics above are highly sensitive to the key hyperparameters: the number of latent dimensions (k) and the regularization coefficient (β) in the VAE loss function. We evaluated these metrics by conducting a trade study of these hyperparameters across different run cases and selected the model with the best performance metrics for use in surrogate modeling and design optimization, as shown in <Figure 3-5>.



<Figure 3-8> Process and data dependencies for surrogate modeling

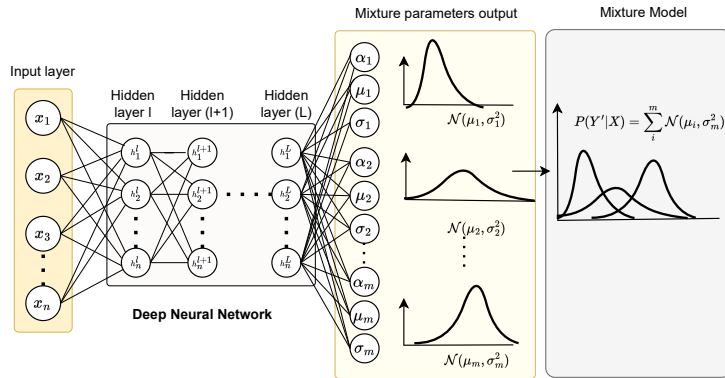
3.3 Probabilistic surrogate modeling from latent space

This section presents surrogate modeling using the latent representation from the generative model (VAE). In this process, we explicitly model aleatory uncertainty to account for variability in operating conditions, as shown in <Figure 3-8>.

The process includes three main steps. (1) We use the VAE encoder ($z_\mu, z_\sigma = Q_\phi(X)$) to obtain the lower-dimensional latent vector mean (z_μ). Directly using sampled latent vectors involves a stochastic process [60], which can destabilize downstream processes. To mitigate this, we use the mean (z_μ) of the latent distribution as input to the probabilistic surrogate model. The input design (X) is taken from the training dataset used for latent representation learning and is then fed into simulations to obtain the corresponding analysis responses. (2) Uncertainty sampling and simulation are performed to obtain the response of the generated design (X) using explicitly defined sources of uncertainty (ξ). The sources of uncertainty may vary depending on the problem; in this re-

search, we consider flight conditions such as variations in Mach number M and angle of attack α . These uncertain parameters are used to perform aerodynamic analysis on the given input design X . The resulting outputs Y are subsequently used for surrogate modeling. In this work, sources of uncertainty are sampled from a normal distribution. (3) Once the analysis data are prepared, surrogate modeling is performed using a Mixture Density Network (MDN). Instead of using the original design space (X), we use latent variables (z_μ) and uncertain parameters (ξ) as inputs, with outputs (Y) obtained from the analysis model. The MDN training procedure is explained in Section 3.3.1, and its evaluation is described in Section 3.3.2, where model accuracy is assessed by increasing the number of uncertainty samples (ξ). After evaluating the model with various performance metrics for different data sizes, the surrogate model is employed for design optimization.

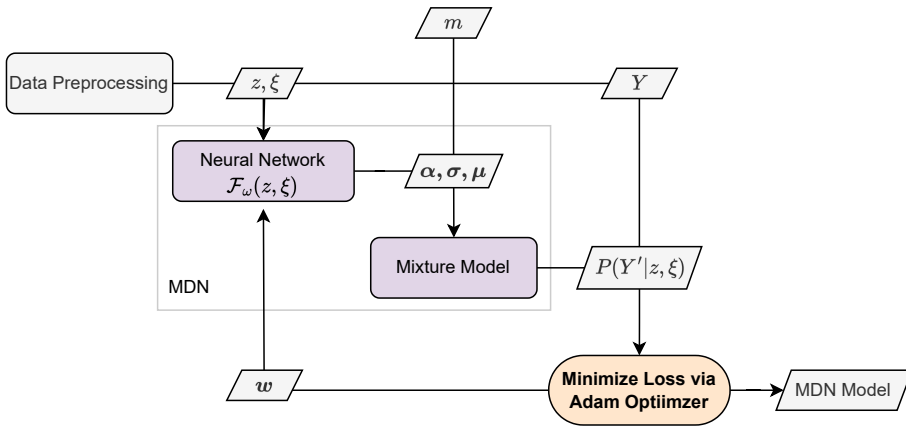
3.3.1 Training mixture density network (MDN) surrogate model



<Figure 3-9> Mixture density network (MDN)

A Mixture Density Network (MDN), first introduced by [65], is a class of neural network models that provides a general framework for modeling conditional

density functions. The MDN models the conditional probability distribution for a given input (x), enabling the prediction of uncertainty by estimating the probability density of target values using $P(Y|x)$. Because MDNs predict the conditional density function, they can address and manage uncertainty within their architecture. An MDN is a supervised learning model that combines a neural network (NN) with mixture components to capture both aleatory and epistemic uncertainty. An example of an MDN $\mathcal{F}_w$ is shown in <Figure 3-9>, which includes one input layer, three hidden layers, and output layers that provide the mixture parameters: mixing coefficients α , mean values μ , and standard deviations σ , all with trainable parameters $\mathbf{w}$. The MDN predicts a conditional probability density $P(Y|X)$, which is a combination of mixture distributions [65].



<Figure 3-10> Mixture density network training procedures

The training procedure of the Mixture Density Network (MDN) is illustrated in <Figure 3-10>. In this process, the input parameters consist of the random variables X , which are fed into a neural network $\mathcal{F}_w(z, \xi)$. Rather than predicting a single deterministic output, the neural network predicts the parameters of a mixture model—specifically, the mixture weights α , means μ , and standard deviations σ .

The mixture model then represents the conditional probability distribution of the target response $P(Y|X)$ using a weighted sum of Gaussian kernels [96], as given in Eq. (3.14):

$$P(y|x) = \sum_{i=1}^m \alpha_i(x), \mathcal{N}(y | \mu_i(x), \sigma_i^2(x)) \quad (3.14)$$

Here, m is the number of mixture components, $\alpha_i(x)$ are the mixing coefficients (which satisfy $\sum_{i=1}^m \alpha_i(x) = 1$), and $\mathcal{N}(y|\mu_i(x), \sigma_i^2(x))$ represents the conditional Gaussian density of the i^{th} mixture component with mean μ_i and standard deviation σ_i .

To ensure valid probabilities, the mixing coefficients are passed through a softmax transformation with the network output a :

$$\alpha_i = \frac{\exp(z_i^a)}{\sum_{j=1}^m \exp(z_j^a)} \quad (3.15)$$

The standard deviations are constrained to be positive by applying an exponential activation to the corresponding network outputs:

$$\sigma_i = \exp(z_i^a) \quad (3.16)$$

Finally, the means μ_i are given directly by the network outputs with a linear activation:

$$\mu_i = z_i^a \quad (3.17)$$

Since the output of an MDN is a conditional probability density, the conventional least-squares loss is not used. Instead, the loss function in Eq. (3.18) is the negative log-likelihood (NLL) of the probability density of the target data (Y) given the inputs (X) [79].

$$\mathcal{L}_{MDN} = -\frac{1}{N} \sum_{n=1}^N \log \left(\sum_i^m \alpha_i(x_n) \mathcal{N}(y | \mu_i(x), \sigma_i^2(x)) \right) \quad (3.18)$$

During the training process, the weights and biases $\mathbf{w}$ are updated by minimizing the loss function until convergence. Once trained, the MDN provides a conditional probability density estimation model capable of representing the entire conditional output distribution, rather than a single deterministic prediction.

3.3.2 Evaluation metrics for surrogate model

For evaluating the probabilistic surrogate model, we use two standard metrics: Mean Squared Error (MSE), see Eq. (3.19), and the coefficient of determination (R^2), see Eq. (3.20).

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (Y_i - Y_i')^2 \quad (3.19)$$

Here, N is the total number of samples, and Y_i and Y_i' are the actual and predicted values from the MDN, respectively. Lower values of the MSE indicate a better fit of the model to the data.

$$R^2 = 1 - \frac{\sum_{i=1}^N (Y_i - Y_i')^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} \quad (3.20)$$

Here, $\bar{Y}$ is the mean of the actual values. The R^2 score ranges from 0 to 1, with higher values indicating a better model fit.

The surrogate model is evaluated by increasing the number of uncertainty samples (ξ) while keeping the latent design variables (z) fixed during the evaluation stage. After assessing model accuracy, the trained surrogate model is used for the optimization process.

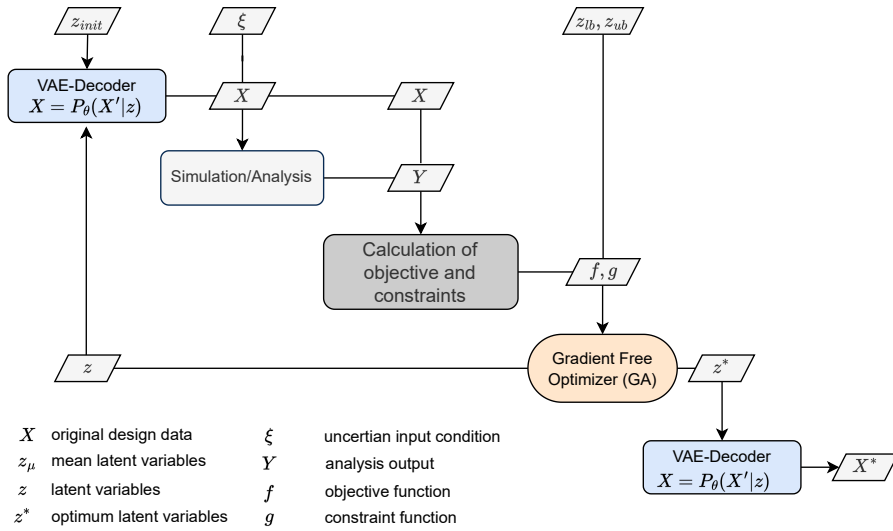
3.4 GenAI optimization approach

This section presents a Generative AI optimization approach designed to overcome the limitations of conventional optimization methods. Two types of optimization strategies are proposed: simulation-based optimization and surrogate-based optimization. In simulation-based optimization, Generative AI is incorporated to improve solution quality by enhancing the shape representation beyond what is possible with conventional approaches. In contrast, surrogate-based optimization aims to improve the overall efficiency of the optimization process—not only by achieving high-quality solutions, but also by significantly reducing computational cost while maintaining accuracy compared to simulation-based approaches.

3.4.1 Simulation-based optimization

In this approach, we employ a generative model to enhance the quality of solutions and overcome the limitations of conventional optimization methods. The process of the GenAI simulation-based optimization is illustrated in <Figure 3-11>.

Here, the latent variables (z) serve as the design variables for the optimization problem. First, the VAE decoder ($P_\theta(X|z)$) transforms the initial latent variables (z_{init}) into the original design parameters (X), such as the airfoil or aircraft shape. Second, an analysis is performed to compute outputs (Y)—for example, lift and drag—based on the design inputs (X) and analysis conditions (ξ), which include Mach number, altitude, and angle of attack. Third, the objective and constraint functions are evaluated using the analysis outputs (Y) and design parameters (X). Finally, the optimizer (GA) searches for the optimal solution within the prescribed lower and upper bounds of the latent variables



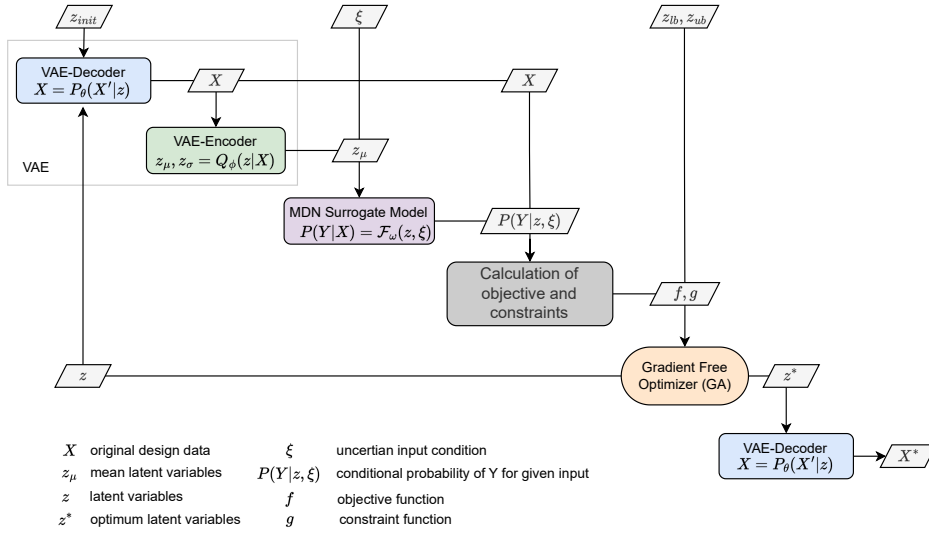
<Figure 3-11> GenAI optimization (simulation-based)

(z_{lb}, z_{ub}) , updating them iteratively. Once the optimal latent variables (z^*) are obtained, the VAE decoder ($P_{\theta}(X|z^*)$) is used to reconstruct the corresponding original design representation (X^*).

3.4.2 Surrogate-based optimization

In this approach, we employ a generative model and a surrogate model to improve the optimization efficiency in terms of both solution quality and reduced computational effort. The process of the GenAI surrogate-based optimization is illustrated in <Figure 3-12>.

First, the latent design variables (z) are provided to the decoder ($P_{\theta}(X|z)$), which generates the corresponding design (X) from the latent variables. The reconstructed design (X) is then analyzed to extract key geometric parameters such as maximum thickness (t/c_{max}), maximum camber (c_{max}), wingspan, and wing area. These geometric features are subsequently used in the calculation



<Figure 3-12> GenAI optimization approach (surrogate-based)

of constraints and the objective function within the optimization formulation to ensure the geometric and aerodynamic integrity of the design.

Second, unlike the simulation-based approach, we use the encoder ($Q_\phi(z|X')$) to predict the latent vectors and determine their mean (z_μ) and standard deviation (z_σ) in the latent space. This encoding–decoding cycle ensures that the candidate designs proposed by the optimizer are valid and consistent with the latent space distribution [66]. As a result, they can be effectively evaluated using the probabilistic surrogate model $\mathcal{F}_w(z_\mu, \xi)$, which is trained on the mean latent vectors (z_μ) and uncertain input parameters (ξ).

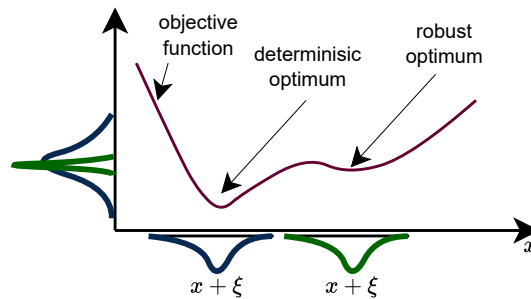
Third, the responses from the surrogate model $\mathcal{F}_w(z, \xi)$ are used to evaluate the objectives and constraints. During optimization, the objective (f) and constraints (g) are iteratively updated and passed back to the optimizer. This process continues until convergence is achieved, at which point the optimal solution (z^*) is obtained, and the corresponding design (X^*) is decoded from the VAE decoder.

The formulation of the objectives (f) and constraints (g) may differ depending on the type of design optimization.

3.4.3 Optimization formulations

In this section, two types of formulations are presented. Since the primary focus of this study is uncertainty-based optimization, robust design optimization (Section 3.4.3.1) and reliability-based design optimization (Section 3.4.3.2) are formulated accordingly. Optimization is performed in the learned latent space z , which is non-convex; consequently, global optimization algorithms such as evolutionary algorithms are often used [11]. We employed a genetic algorithm (GA) via pymoo [97], a multi-objective evolutionary framework known for its effectiveness in avoiding local optima and approaching the global optimum.

3.4.3.1 Robust design optimization formulation



<Figure 3-13> Illustration of robust design optimization

Robust design optimization (RDO) focuses on minimizing the expected variance of the objective function under uncertainty. As illustrated in <Figure. 3-13>, RDO aims to minimize the variation of the objective function caused by

uncertain variables. Substantial data are required to model the statistical mean μ and standard deviation σ for formulating the objective function, as shown in Eq. (3.21).

$$\begin{aligned}
& \min_{\mathbf{z}, \xi} \quad J(\mathbf{z}, \xi) = f(\mu(\mathbf{z}, \xi), \sigma(\mathbf{x}, \xi)) \\
& \text{w.r.t.} \quad \mathbf{z}^L \leq \mathbf{z} \leq \mathbf{z}^U, \\
& \text{s.t.} \quad g_i(\mathbf{z}, \bar{\xi}) \leq 0, \quad i = 1, \dots, N_C
\end{aligned} \tag{3.21}$$

where z is the latent design variable, ξ represents the uncertain parameters, and f and g denote the objective and constraints required for the optimization. In our optimization process, the predicted mean and variance for robust design optimization in Eq. (3.21) can be computed using Eq. (3.22) and Eq. (3.23), respectively, via a Monte Carlo simulation approach, with the MDN surrogate model predicting the probability density of the output parameter (Y).

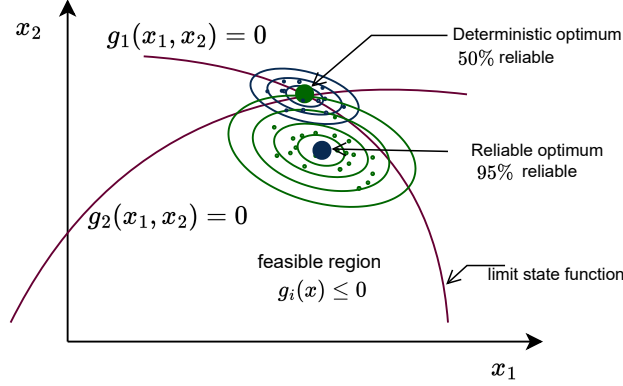
$$\mu_Y = \frac{1}{N_{\text{mcs}}} \sum_{i=1}^{N_{\text{mcs}}} \mathcal{F}_w(z, \xi_i) \tag{3.22}$$

$$\sigma_Y^2 = \frac{1}{N_{\text{mcs}}} \sum_{i=1}^{N_{\text{mcs}}} (\mathcal{F}_w(z, \xi_i) - \mu_Y)^2 \tag{3.23}$$

Here, μ_Y and σ_Y are the mean and standard deviation of the quantity of interest for the objective function (f), such as lift or drag coefficient, and are the values output by the MDN surrogate model. N_{mcs} is the number of Monte Carlo samples, and ξ_i denotes the i^{th} sample's value of the uncertain parameter. In this formulation, the constraints (g) are considered deterministic and are calculated using the mean values of the uncertain parameters ($g(z, \bar{\xi})$).

Using the formulation in Eq. 3.21, we are able to perform robust design optimization with our approach as explained in Section 3.4.2.

3.4.3.2 Reliability-based design optimization formulation



<Figure 3-14> Illustration of reliability-based design optimization

Reliability-based design optimization (RBDO) aims to minimize the likelihood of constraint violations caused by uncertain variables and parameters. An illustration of RBDO is shown in <Figure. 3-14>, where two constraints are known as limit state functions. A deterministic optimum is only 50% reliable when uncertain conditions occur, whereas the reliability optimum shifts the design toward feasibility and greater reliability by ensuring that the probability of constraint failure does not exceed a specified threshold. The reliability-based design optimization can be formulated as shown in Eq. (3.24).

Reliability-based design optimization in Eq. (3.24) is performed by estimating the failure probability, which is calculated for a given constraint using the Monte Carlo simulation (MCS) approach, as shown in Eq. (3.25).

$$\begin{aligned}
 & \min_{\mathbf{z}, \xi} \quad J(\mathbf{z}, \xi) = f(\mu(\mathbf{z}, \xi)) \\
 & \text{w.r.t.} \quad \mathbf{z}^L \leq \mathbf{z} \leq \mathbf{z}^U, \\
 & \text{s.t.} \quad \mathbf{P}_{f_i}[\mathbf{G}_i(\mathbf{z}, \xi) \geq \mathbf{0}] \leq \mathbf{P}_{f_{\text{req}}}, \quad i = 1, \dots, N_C
 \end{aligned} \tag{3.24}$$

Here, P_{f_i} denotes the failure probability associated with the i^{th} constraint. $P_{f_{\text{req}}}$ is the required failure probability, which can be calculated from the cumulative distribution function (Φ) of the standard normal distribution using the reliability index β as $P_{f_{\text{req}}} = \Phi(-\beta)$.

$$P_f = \frac{1}{N_{\text{mcs}}} \sum_{i=1}^{N_{\text{mcs}}} I_f[G(x_i)] \quad (3.25)$$

Here, I_f is the indicator function such that $I_f = 1$ when $G(x) \leq 0$ and $I_f = 0$ when $G(x) > 0$, $G(x)$ can be calculated depending on the constraints requirements and prediction of the MDN surrogate model, which can be formulated as in Eq. (3.26):

$$G(x_i) = \mathcal{F}_w(z, \xi) - QoI_{\text{req}} \quad (3.26)$$

where $G(x_i)$ refers to the constraint for the i^{th} MCS sample, and QoI_{req} refers to the target or required constraint value of a quantity of interest, such as the required lift coefficient, depending on the problem formulation. Using the formulation in Eq. 3.24, we are able to perform reliability-based design optimization with our approach.

4. Airfoil Design via GenAI Optimization Approach

This chapter presents a case study using a generative AI optimization approach for subsonic airfoil design with 198 design variables. We begin by describing the problem definition in Section 4.1, then discuss deep generative modeling to learn a compact representation of airfoil design in Section 4.2, followed by surrogate modeling for uncertainty quantification from the latent representation in Section 4.3. In Section 4.4.5, we demonstrate how our approach overcomes the limitations of conventional optimization methods, along with the corresponding results.

4.1 Problem definition

In the process of aircraft design, selecting or designing airfoil shape is a critical stage that significantly affects the overall aircraft performance. Optimizing the aerodynamic shape of an airfoil to meet target performance specifications is therefore a key challenge for designers. Airfoil design is optimized by adjusting a set of variables—specifically, the surface coordinates—to maximize performance. The number of design variables is important for both design space representation and the computational cost of optimization[58]. Directly controlling the surface coordinates can produce unreliable and abnormal designs, which hinder optimization. To address this, researchers have developed parametrization methods such as Class Shape Transformation (CST)[12] and

other approaches like Bézier curves[98]. These methods generate airfoil surface coordinates by manipulating control points, allowing the airfoil shape to be represented with fewer variables. However, these approaches can still produce unrealistic shapes. Additionally, uncertainty-based design optimization often requires repeated simulations, greatly increasing computational cost. These drawbacks significantly impact the overall efficiency of the optimization process.

To address the challenges of generating unrealistic shapes, the curse of dimensionality, and high computational cost, this study employs a GenAI-enhanced optimization approach. In this case study on airfoil design, a VAE is trained to learn a lower-dimensional latent space, and uncertainty prediction is performed using a probabilistic surrogate model. Optimization is then performed using the trained model, demonstrating improvements in finding high-quality solutions and sample efficiency for uncertainty-based optimization.

We follow the steps described in Chapter 3.

4.2 Deep generative modeling

As we explained in Chapter. 3.Sec. 3.2, we follow these steps for generative modeling of airfoil design.

4.2.1 Data pre-processing for airfoil design

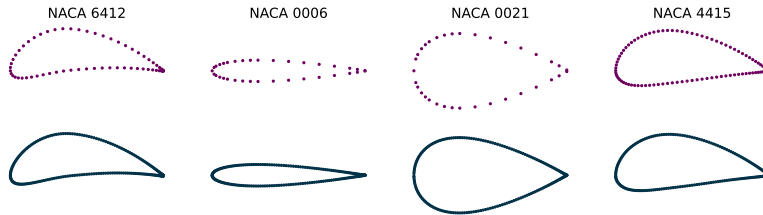
Training the generative model requires shape data for airfoil designs. In this study, we use the UIUC airfoil dataset[99], which contains more than 1,450 two-dimensional airfoils designed for a wide range of applications—including unmanned aerial vehicles, transport aircraft, and wind turbine blades. This dataset is widely used in research due to its large number of airfoils and variety

of types, making it suitable for data-driven design approaches [15, 16, 62].

However, the dataset contains inconsistencies in the coordinate formats of different airfoils. To ensure consistency in geometric representation, all coordinate files are interpolated to generate surface points at identical, cosine-spaced chord-wise locations, following the method described in [57] and using Eq. (4.1). Each airfoil is discretized into 198 points—99 on the upper surface and 99 on the lower surface—excluding the duplicated leading edge point. The trailing edge points are retained on both surfaces to account for finite trailing edge gaps or thicknesses. This uniform discretization allows only the y -coordinates of the surface points to be used as input features for the VAE, simplifying the input structure. The processed surface coordinates are then used to train the VAE, which learns a compact and meaningful latent space representation of the airfoil shapes.

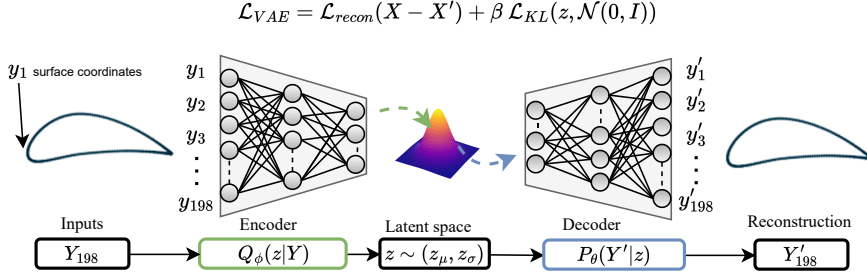
$$x_i = \frac{1}{2} \left(\cos \left(\frac{2\pi(i-1)}{198} \right) + 1 \right), \quad i = 1, 2, \dots, 198. \quad (4.1)$$

A comparison of NACA airfoils from the UIUC dataset before and after preparation is shown in <Figure. 4-1>. After preparation, the airfoil coordinates are uniformly distributed for all airfoil shapes while maintaining the original geometries.



<Figure 4-1> Comparison of NACA airfoils in the UIUC dataset before and after the preparation. Upper row shows the original airfoil coordinates and the lower row shows the prepared airfoil coordinates

4.2.2 Training VAE model



<Figure 4-2> Architecture of VAE for airfoil design

<Table 4-1> Airfoil VAE input, latent representation, and output structure.

Encoder (Q_ϕ)		Latent space (z)	Decoder (P_θ)	
Input	Output	Latent dim ($k = 10$)	Input	Output
$(X \in \mathbb{R}^N)$	$(z_\mu \in \mathbb{R}^k, z_\sigma \in \mathbb{R}^k)$	$(z \in \mathbb{R}^k)$	$(z \in \mathbb{R}^k)$	$(X \in \mathbb{R}^N)$
y_1				y'_1
y_2				y'_2
y_3	z_{μ_1}, z_{σ_1}	$z_1 \sim \mathcal{N}(z_{\mu_1}, z_{\sigma_1}^2)$	z_1	y'_3
y_4	z_{μ_2}, z_{σ_2}	$z_2 \sim \mathcal{N}(z_{\mu_2}, z_{\sigma_2}^2)$	z_2	y'_4
y_5	μ_{z_3}, σ_{z_3}	$z_3 \sim \mathcal{N}(z_{\mu_3}, z_{\sigma_3}^2)$	z_3	y'_5
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
y_{195}	z_{μ_9}, z_{σ_9}	$z_9 \sim \mathcal{N}(z_{\mu_9}, z_{\sigma_9}^2)$	z_9	y'_{195}
y_{196}	$z_{\mu_{10}}, z_{\sigma_{10}}$	$z_{10} \sim \mathcal{N}(z_{\mu_{10}}, z_{\sigma_{10}}^2)$	z_{10}	y'_{196}
y_{197}				y'_{197}
y_{198}				y'_{198}

This section presents training of the VAE to learn a compact latent representation of complex geometry. We used a mirrored VAE architecture with a neural network comprising three hidden layers, as shown in <Figure 4-2>. The VAE is trained on airfoil surface coordinates—198 per airfoil from 1,450 airfoils—to learn a compact latent representation, as described in Section 4.2. We use 90% of the data for training and validation, and the remaining 10% for test-

<Table 4-2> Optimal hyperparameter

Hyperparameter	Encoder (Q_ϕ)/Decoder (P_θ)
Hidden layer	3
Hidden units	501, 269, 425
Activation	relu, relu, relu
Num. latent dims (k)	10
β	1×10^{-5}
Learning rate	5×10^{-5}
Batch size	512
Epochs	8000

ing. The model is implemented in TensorFlow [100] using the Adam optimizer. The VAE architecture for learning airfoil coordinates is shown in <Table 4-2>.

In the VAE training process, the encoder (Q_ϕ) performs non-linear dimensionality reduction by compressing the high-dimensional input data ($X \in \mathbb{R}^N$). Here, we use airfoil surface coordinates having $N = 198$ dimensions. The encoder maps the input into two k -dimensional vectors, $z_\mu \in \mathbb{R}^k$ and $z_\sigma \in \mathbb{R}^k$, which represent the mean and standard deviation of a Gaussian distribution, respectively. These vectors are parameterized by a set of trainable parameters ϕ , which are optimized by minimizing the loss function.

The lower-dimensional latent space $z \in \mathbb{R}^k$ is obtained by sampling from the predicted mean z_μ and z_σ of a normal distribution. The decoder (P_θ) maps the latent space z back to a reconstruction of the input data (X') and is parameterized by the trainable parameters (θ) of the decoder network. During the training process, VAE learns the parameters of encoder (ϕ) and decoder (θ) by minimizing the total loss function in Eq. (3.8).

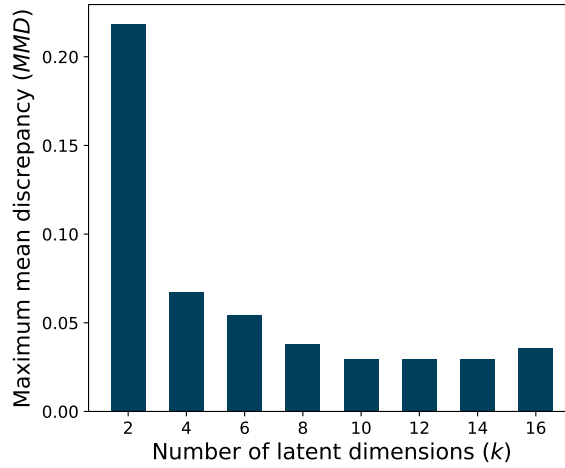
The detailed input, latent space, and output for the airfoil VAE are shown in <Table 4-1>. The encoder (Q_ϕ) receives airfoil surface coordinates Y_N as input, represented by $\mathbb{R}^{198}$, and produces two k -dimensional latent vectors, z_{μ_k} and z_{σ_k} , where $k = 10$ and $\mathbb{R}^{10}$, as determined by evaluation metrics ex-

plained in Section 4.2.3.2. The latent space forms a compact representation of the airfoil surface coordinate data as a multivariate Gaussian distribution, $z_k \sim \mathcal{N}(z_{\mu_k}, z_{\sigma_k}^2)$. The decoder (P_θ) then reconstructs the original airfoil surface coordinates Y'_N , also in $\mathbb{R}^{198}$, from the learned latent space (z_k), which is the output of the decoder network.

4.2.3 Evaluation of VAE model

This section evaluates the latent representation learning model (VAE) according to the evaluation metrics described in Chapter 3, Section 3.2.2, to ensure a disentangled latent representation. The evaluation uses the metrics introduced in the methodology section.

4.2.3.1 Effect of number of latent dimensions (k) on generative quality



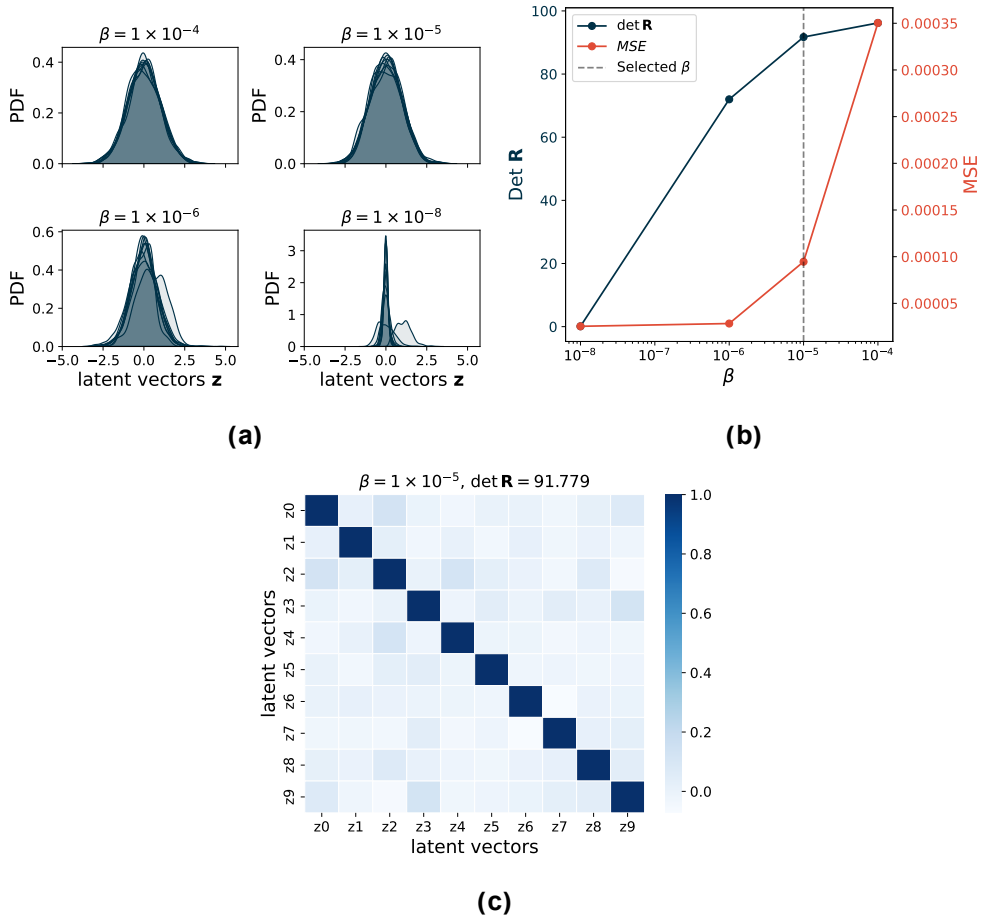
<Figure 4-3> Maximum mean discrepancy (MMD) of VAE models with different latent dimensions (k) at regularization coefficient $\beta = 1 \times 10^{-5}$

First, we evaluated the effect of the number of latent dimensions (k) on the maximum mean discrepancy (MMD), which measures the similarity between

the generative and real data distributions (generative quality). We trained models with different latent dimensions (2, 4, 6, 8, 10, 14, 16), and the results are shown in <Figure. 4-3>. The results indicate that MMD decreases significantly up to dimension 10, after which it stabilizes around 0.0295. This suggests that increasing the number of dimensions beyond 10 does not improve the generative quality of the VAE, while using fewer dimensions may not capture essential underlying features. Thus, the major features of the geometry can be adequately encoded with 10 latent variables. We further evaluated the MSE and correlation coefficient using the optimal VAE network architecture, as explained in <Figure. 4-4>.

4.2.3.2 Effect of regularization coefficient (β) on reconstruction quality and disentanglement

Here, we discuss the effects of β on VAE performance based on evaluation metrics. The optimized VAE model is presented in <Table 4-2>. Specifically, we consider $\beta = 1 \times 10^{-4}$, 1×10^{-5} , 1×10^{-6} , 1×10^{-8} . As shown in <Figure. 4-4a>, increasing β results in a more regularized latent space that aligns more closely with a standard normal distribution. When a very small value, such as $\beta = 1 \times 10^{-8}$, is used, the latent distribution becomes sharply peaked, indicating poor regularization. In contrast, a larger value like $\beta = 1 \times 10^{-4}$ produces a distribution that more closely resembles the desired Gaussian prior, promoting better generalization. However, larger β values lead to increased reconstruction error (MSE) between the input and reconstructed designs, indicating poorer reconstruction performance, as shown in <Figure. 4-4b>. Additionally, a higher value of $\det \mathbf{R}$ implies a more expressive and disentangled latent space, which tends to occur with lower β values. Considering the trade-off between MSE and $\det \mathbf{R}$, we selected $\beta = 1 \times 10^{-5}$, which achieves an MSE of 1×10^{-4}

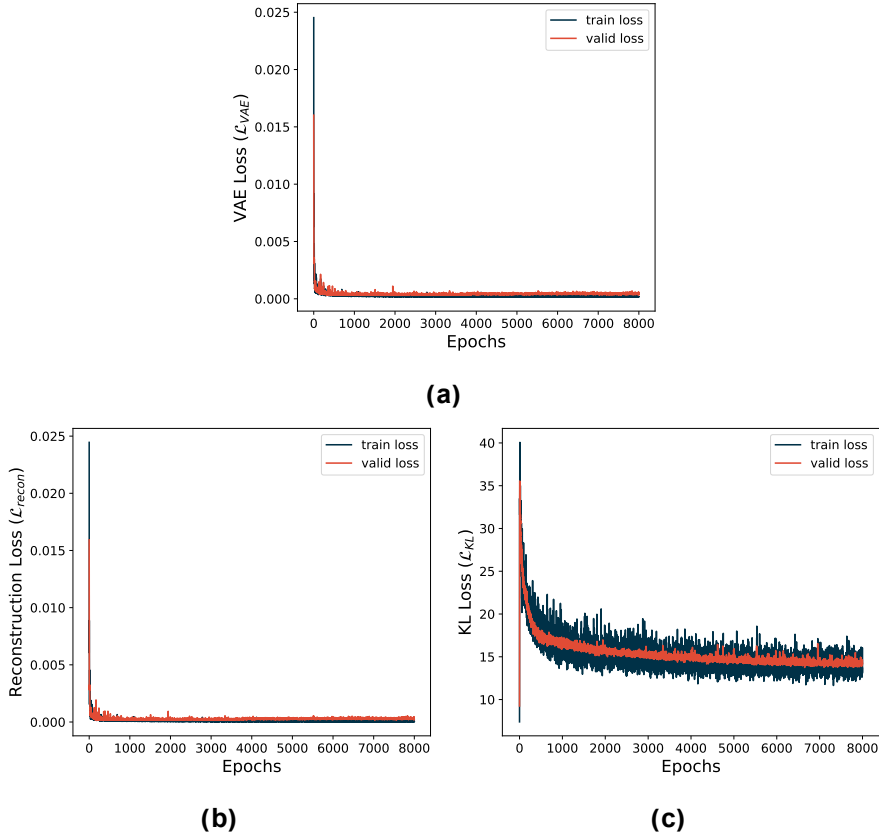


<Figure 4-4> Selection of hyperparameters: (a) PDF of latent variables for different β values; (b) Effects of β on $\det \mathbf{R}$ and FD; (c) Correlation matrix for the latent variables at $\beta = 1 \times 10^{-5}$

and a determinant value of $\det \mathbf{R} = 91.77$, providing a good balance between reconstruction quality and latent variable correlation. <Figure. 4-4c> presents the correlation matrix of the latent variables, where strong diagonal dominance confirms that the latent dimensions are largely uncorrelated.

The convergence of the VAE, as presented in <Table 4-2>, is illustrated in <Figure. 5-12> for both the training and validation datasets. At the start of training, the total loss is approximately 0.025. After 250 epochs, it rapidly decreases to the order of 10^{-5} and then stabilizes for both datasets, as shown in

<Figure. 5-12a> (total loss), <Figure. 5-12b> (reconstruction loss), and <Figure. 5-12c> (KL divergence loss).



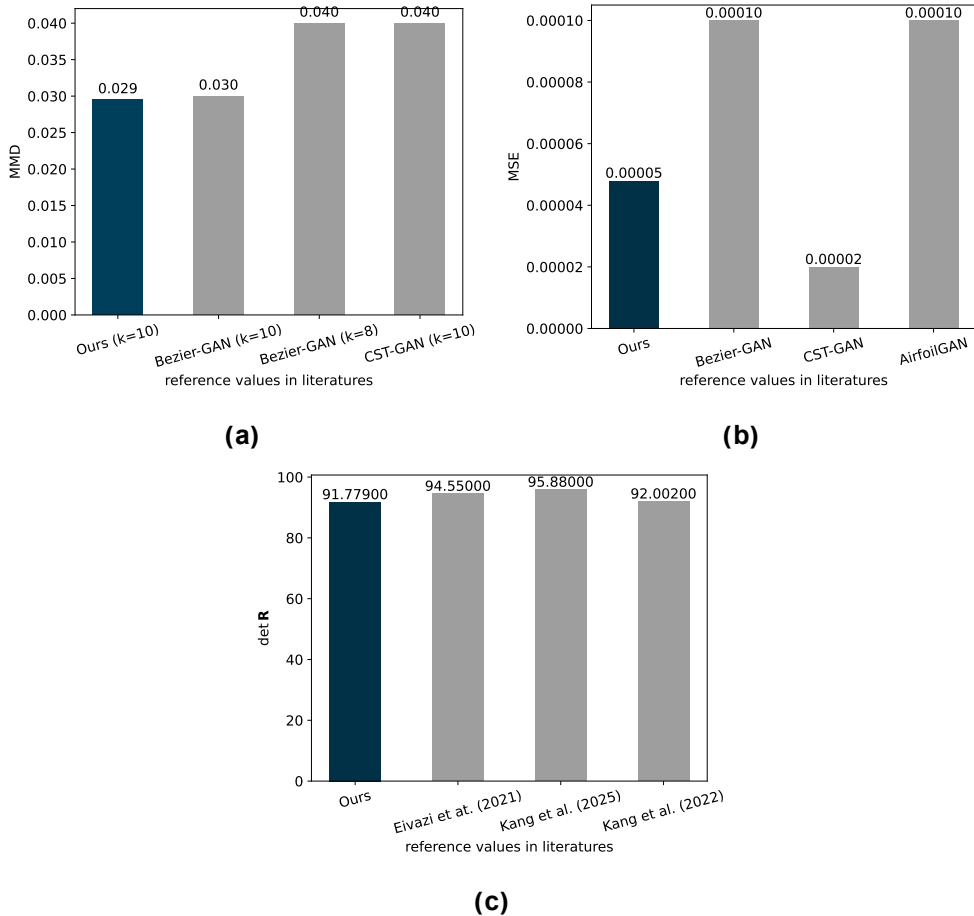
<Figure 4-5> Convergence of VAE training for the optimal model: (a) Total VAE loss $\mathcal{L}_{VAE}$; (b) Reconstruction loss $\mathcal{L}_{recon}$; (c) KL divergence loss $\mathcal{L}_{KL}$

4.2.3.3 Verification of evaluation metrics

Here, we verify our results by comparing with existing research on airfoil parameterization using generative models. Bezier-GAN[31], CST-GAN[101], and Airfoil GAN[72] are included for MMD and MSE comparison, as shown in <Figure. 4-6a> and <Figure. 4-6b>. For latent correlation coefficient comparison, we reference Eivazi et al.[95], Kang et al.[71], and Kang et al.[102], as shown in <Figure. 4-6c>.

The comparison results show that the MMD value is quite similar to that of Bezier-GAN, a popular airfoil parameterization method. Additionally, our MSE demonstrates better reconstruction quality compared to Bezier-GAN, while CST-GAN outperforms all other methods. The determinant of the correlation matrix ($\det \mathbf{R}$) is also comparable to values reported in the literature, with our model achieving 91.779, close to the airfoil VAE model in [102].

These results confirm the reliability of the proposed generative model for subsequent processes and future investigations.

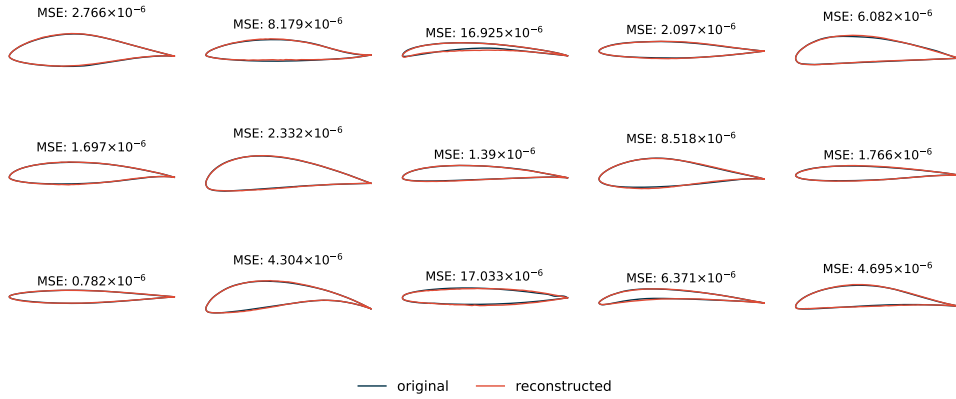


<Figure 4-6> Comparison of evaluation metrics with existing literature: (a) Maximum mean discrepancy (MMD); (b) Reconstruction error (MSE) (c) Determinant of correlation ($\det \mathbf{R}$)

4.2.4 Performance of VAE model

4.2.4.1 Reconstruction quality

We assess the reconstruction quality of the generative model. <Figure 4-7> shows the reconstruction performance for five test airfoils, achieving a mean squared error (MSE) of approximately 1×10^{-6} . Each airfoil was reconstructed using the encoder–decoder process, with MSE calculated for each airfoil in the test dataset.



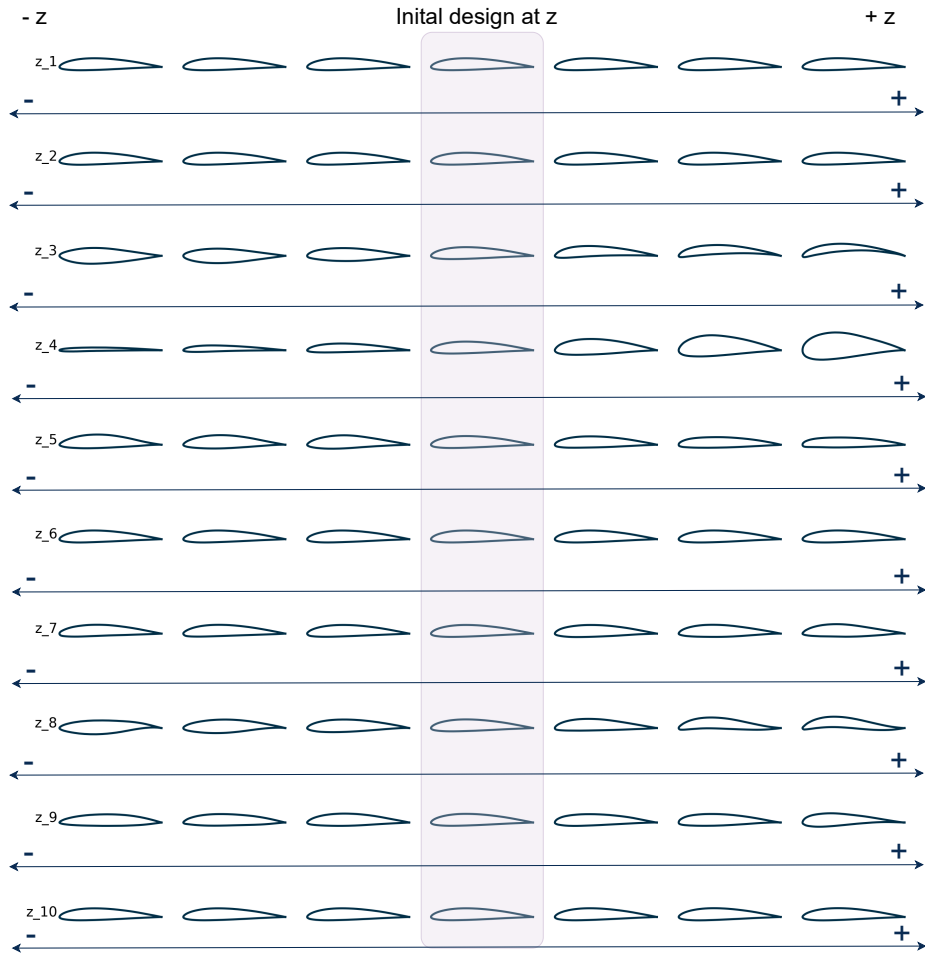
<Figure 4-7> Comparison of original and reconstructed airfoil in the test datasets

<Table 4-3> Reconstruction quality of VAE

Reconstruction Error	Training dataset	Testing dataset
MSE	9.460×10^{-5}	2.250×10^{-4}

4.2.4.2 Disentanglement

Here, we evaluate the disentanglement of the latent space, demonstrating that each latent vector is independent and interpretable with respect to the original



<Figure 4-8> Designs synthesized by varying the latent dimensions, Each row is a traversal of a single latent dimension

data. The low-dimensional latent space is advantageous for new design, as traversing the latent space z along different axes gives rise to significantly different geometric changes. <Figure 4-8> illustrates this by showing the effect of moving along 10 different latent axes, with each axis representing a distinct change in airfoil design.

The middle design is the initial design (X) input to the encoder ($Q_\phi(z|X)$) to obtain the latent variables (z) for that airfoil. Each latent axis is then changed

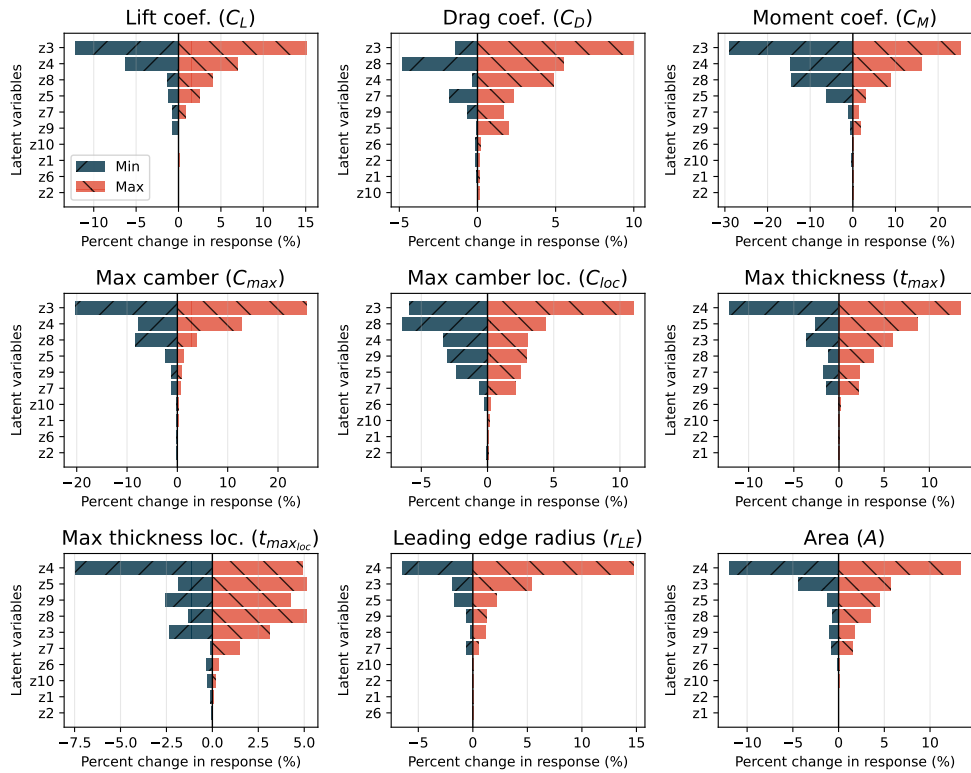
incrementally, while the other dimensions are held constant, and the decoder ($P_\theta(X'|z)$) is used to generate the corresponding design space (X).

As shown in <Figure. 4-8>, the first latent dimension represents camber changes, the second dimension affects airfoil thickness, the third represents leading edge radius, and the fifth governs other shape changes. This demonstrates that our VAE has learned a disentangled, lower-dimensional representation of airfoil shapes suitable for downstream tasks such as design optimization.

4.2.5 Sensitivity analysis

Sensitivity analysis is performed to identify the effect of each latent variable on the airfoil shape and its aerodynamic performance. The airfoil geometric parameters, including maximum thickness (t_{max}), maximum thickness location (t_{maxloc}), maximum camber (C_{max}), maximum camber location (C_{maxloc}), leading-edge radius (r_{LE}), and airfoil area (A), as well as the aerodynamic parameters—lift coefficient (C_L), drag coefficient (C_D), and moment coefficient (C_M)—are evaluated by varying each latent variable within $\mu \pm \sigma$. The percentage change in each response relative to the baseline airfoil was computed, and tornado diagrams were generated to visualize the sensitivity of each response variable to individual latent dimensions shown in <Figure 4-9>. The aerodynamic analysis is conducted at a Reynolds number of $Re = 4.45 \times 10^6$, an angle of attack of $\alpha = 2.79^\circ$, and a Mach number of $M = 0.1962$ using NeuralFoil. The details of the analysis tool are explained in Section 4.3.1.

The sensitivity analysis presented in <Figure 4-9> shows that different latent variables have distinct, physically meaningful effects on both geometry and aerodynamics. Several latent variables, such as z_3 , z_4 , and z_8 , induce the largest variations in the lift coefficient (C_L), reaching approximately $\pm 15\%$, pri-



<Figure 4-9> Sensitivity of latent variables on geometric and aerodynamic parameters of the airfoil. Tornado diagrams illustrate the percentage change in each response variable when individual latent variables are perturbed within $\mu \pm \sigma$

marily by modifying the airfoil camber, which directly impacts lift. The drag coefficient (C_D) displays moderate sensitivity, varying within approximately $\pm 5 - 10\%$, due to the combined influence of camber and thickness. The pitching-moment coefficient (C_M) is particularly sensitive to z_3 , z_4 , z_8 , and z_5 , which generate the largest percentage changes, up to $\pm 25 - 30\%$. These variables mainly affect the leading-edge radius and trailing-edge angle, resulting in significant changes in pitching moment (C_M).

The geometric responses reveal clear and interpretable patterns. Camber-related metrics, such as maximum camber ($C_{\max}$) and camber location (C_{loc}),

are most strongly influenced by z_3 and z_5 , consistent with their effect on C_L . Maximum thickness ($t_{\max}$) and its location are predominantly governed by z_6 and z_7 , which induce monotonic changes in thickness. The leading-edge radius (r_{LE}) is primarily determined by z_4 , indicating that this latent dimension encodes localized modifications near the airfoil nose. As expected, airfoil area (A) follows a trend similar to maximum thickness because of the coupling between thickness distribution and area.

Overall, the tornado diagrams show that the VAE latent space provides a disentangled representation of airfoil shape variations. Certain latent variables, in particular z_3 , z_4 , and z_5 , exert a dominant influence over both geometric and aerodynamic responses, while others such as z_1 and z_2 have a relatively minor impact. These findings confirm that the latent variables serve as efficient and physically interpretable design parameters, making controllable and predictable shape variation possible during the optimization process.

4.3 Probabilistic surrogate modeling from latent space

This section explains probabilistic surrogate modeling from the latent space, taking into account uncertainties in operating flight conditions. The selection of the analysis tool is described in Section 4.3, the training and evaluation of the surrogate model in Section 4.3.2, and uncertainty quantification using the trained model in Section 4.3.3.

We follow the steps described in Chapter 3, Section 3.3.

4.3.1 Analysis tool and data generation

4.3.1.1 Aerodynamics analysis tool

The purpose of this research is to develop a probabilistic surrogate model that used lower-dimensionality representation model as parameterization and the generation of new designs. Therefore, selecting an appropriate analysis tool is a crucial step in surrogate modeling, as it directly affects the accuracy of aerodynamic performance data and the associated computational cost. XFOIL[103], an interactive program designed for the design and analysis of subsonic isolated airfoils (including viscous and inviscid analysis), has been widely in machine learning applications [15, 55, 62, 104], due to its computational efficiency and reasonable accuracy at low speeds. However, XFOIL is not suitable for flow conditions with Mach numbers greater than 0.6, which fall into the transonic regime [105]. Therefore, for transonic cases, CFD analysis tools such as ADflow[106], SU2[107], or Ansys Fluent[108] are typically required. These tools, however, incur significantly higher computational costs. NeuralFoil[109] [110] is an open-source, Python-based tool that addresses the limitations of traditional aerodynamic analysis by employing a physics-informed neural networks. It is trained on tens of millions of XFOIL simulations to enable rapid aerodynamic predictions. NeuralFoil can analyze a full 360-degree range of angles of attack, Reynolds numbers spanning from subsonic to transonic flow regimes, while accounting for varying turbulence parameters. Furthermore, NeuralFoil has been benchmarked against XFOIL and achieves accuracy within a few percent of XFOIL's predictions, while offering substantial computational speedups [111]. Its capabilities have been demonstrated in various applications [112], [111], and [113].

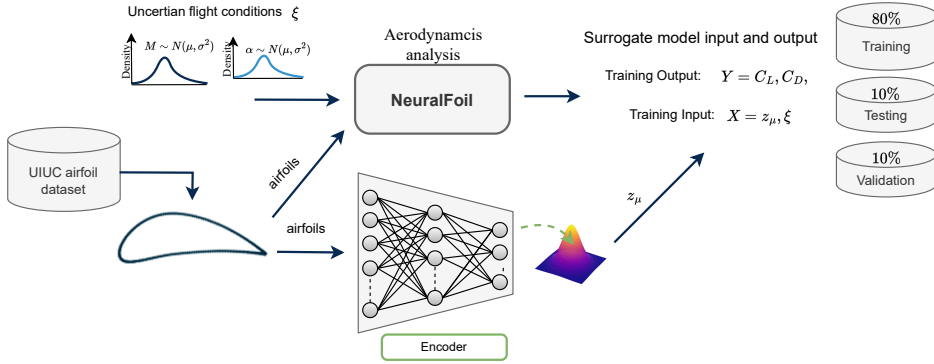
All aerodynamic analyses in this research were conducted using the NeuralFoil

(xxxlarge) model within a consistent subsonic flow regime. Aerodynamic analyses were made at a fixed Reynolds number of $Re = 4.40 \times 10^6$, which represents typical operating conditions for moderate-Reynolds-number airfoil applications. NeuralFoil inherits the transition modeling framework from XFOIL and therefore requires specification of the free-transition amplification factor. To ensure realistic transition behavior under standard low-disturbance atmospheric conditions, the widely used e^9 transition criterion was adopted by setting $N_{\text{crit}} = 9$. This choice prevents unrealistically extended laminar regions and remains consistent with the XFOIL-based physics on which NeuralFoil is trained.

4.3.1.2 Aerodynamics datasets for surrogate modeling

The data generation process is shown in <Figure. 4-10>. We generate aerodynamic data using NeuralFoil to train the MDN surrogate model, following the pre-processing of 1,450 airfoils described in Section 4.2.1. To model aleatoric uncertainty, which captures variability in flight conditions, we adopt a probabilistic sampling strategy for the angle of attack (α) and Mach number (M) in the aerodynamic analysis. For each airfoil, samples are drawn randomly from a normal distribution to account for the variability in flight conditions mentioned in [48]. As a reference case, we utilize the E214 airfoil, which is employed in a UAM prototype—a box-wing aircraft [114] at a cruise design speed of 66.6 m/s at an altitude of 1,000 ft. In addition, we use VAE-encoder to get the latent variables z_μ for each airfoils, so that, the surrogate modeling can be performed directly from the lower dimensional latent space. The input parameters for surrogate becomes latent variables (z_μ) and uncertain flight conditions ξ and the output parameters are lift (C_L) and drag (C_D) coefficients. The uncertain flight conditions sample sizes are used to evaluate the prediction accuracy of the

MDN surrogate model are summarized in <Table 4-4>. We use all airfoils from the data and run this data generation process for different sample size and used 80% for training, 10% for testing and 10% for validation.



<Figure 4-10> Aerodynamics data generation using encoder and analysis tool

<Table 4-4> Aerodynamic analysis conditions and number of samples

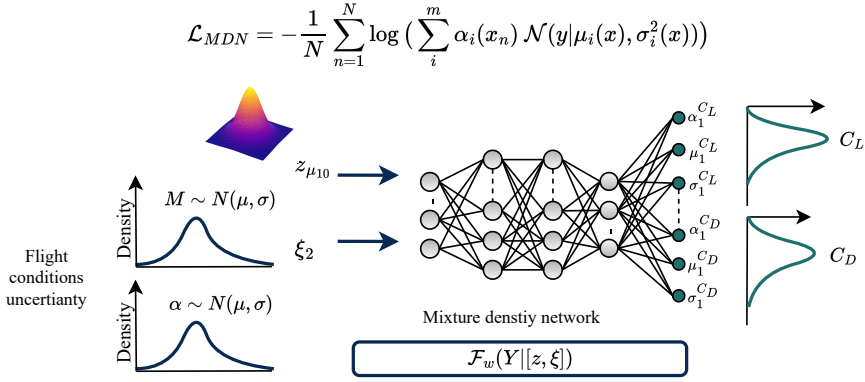
Re	α°	M	Total airfoils	Samples per airfoil
4.45×10^6	$\mathcal{N}(2.79, 0.1^2)$	$\mathcal{N}(0.1962, 0.005^2)$	1450	50, 100, 200

4.3.2 Training and evaluation of surrogate model

<Table 4-5> MDN model optimal hyperparameter

Hyperparameter	MDN($\mathcal{F}_w$)
Hidden layer	4
Hidden units	487, 472, 489, 348
Activation	relu, tanh, relu, relu
Num. of mixtures (m)	1
Dropout	20%
Batch size	258

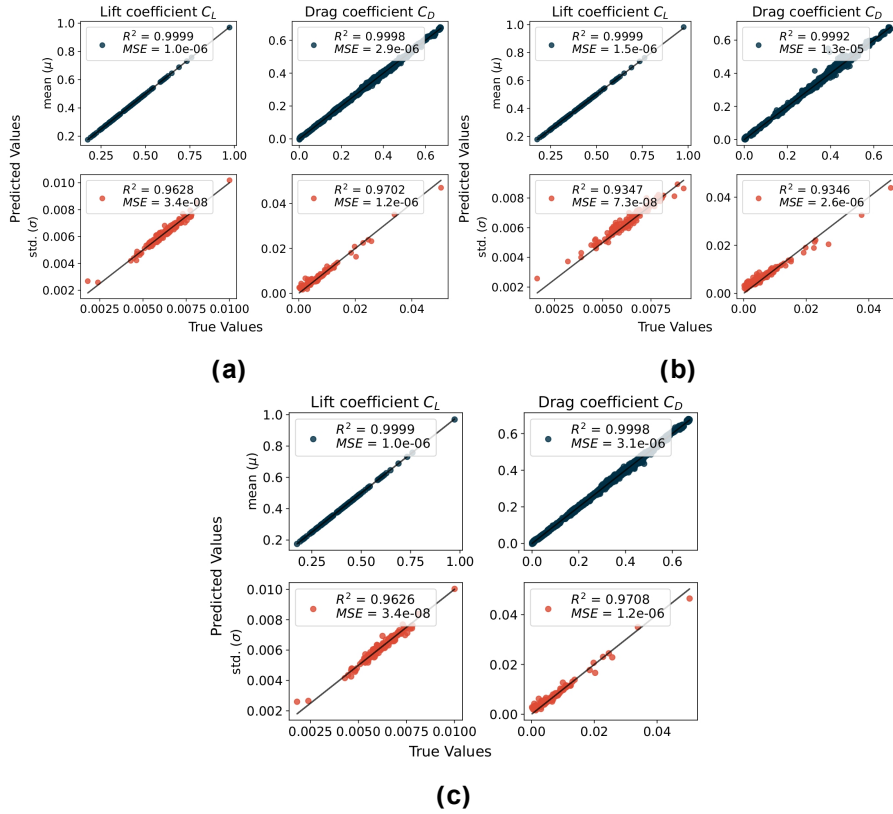
We trained the MDN model to predict the conditional distribution of lift and drag coefficients by explicitly considering α and Mach number as sources of



<Figure 4-11> Architecture of MDN for surrogate modeling

uncertainty. The architecture of the MDN model is shown in <Figure. 4-5>, with the latent vector (z), α , and M as inputs, and C_L and C_D as outputs. We evaluated MDN performance on test datasets using varying sample sizes of the random variables α and M for each airfoil in <Table 4-4>, employing the optimized architecture from <Table 4-5>. The model predictions were evaluated based on the conditional mean (μ) and standard deviation (σ), computed using Eq. (3.22) and Eq. (3.23), respectively, for each dataset size N , while maintaining a fixed model architecture. The evaluation followed the metrics described in Sec. 3.3.2, as our primary focus is on accurately predicting statistical properties relevant to uncertainty quantification.

Increasing the sample size generally improves prediction accuracy. The prediction results are presented in <Figure. 4-12>. As expected, larger sample sizes yield better predictions of σ , while the μ values are accurately predicted across all cases. Using 200 samples, the model achieved R^2 scores of 0.96 and 0.97 for the predicted standard deviations of lift coefficient C_L and drag coefficient C_D , respectively. Meanwhile, the predicted means achieved R^2 scores of 0.99, indicating a well-fitted probabilistic regression model suit-

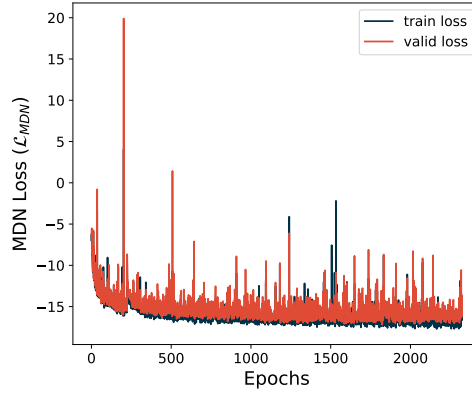


<Figure 4-12> MDN model predictions for the mean and standard deviation under random variables (α , M) for different sample sizes: (a) 50 samples, (b) 100 samples, and (c) 200 samples.

<Table 4-6> MDN model prediction with varying sample sizes

Sample	Metrics	C_L		C_D	
		μ	σ	μ	σ
50	R^2	0.999	0.827	0.998	0.394
	$MSE \times 10^{-6}$	4.22	0.219	21.2	23.3
100	R^2	0.999	0.927	0.999	0.933
	$MSE \times 10^{-6}$	1.56	0.0814	12.9	2.70
200	R^2	0.999	0.964	0.999	0.970
	$MSE \times 10^{-6}$	1.00	0.0389	1.19	0.224

able for downstream tasks in uncertainty quantification of generated airfoils and design variables.

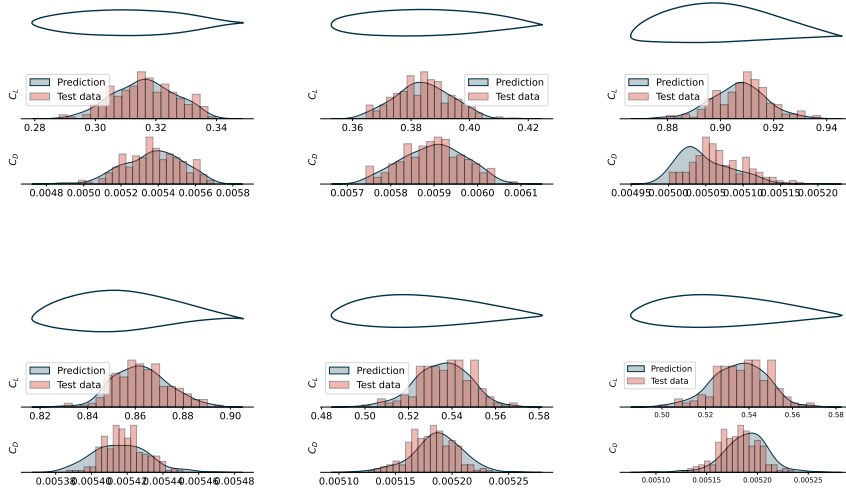


<Figure 4-13> Convergence of MDN training of optimal model

The convergence of the MDN model, as presented in <Table 4-5>, is illustrated in <Figure. 4-13> for both the training and validation datasets. The MDN loss converges after approximately 2300 iterations, achieving a negative log-likelihood (NLL) loss of -16.619 for the training data and -16.12 for the validation data.

4.3.3 Performance of MDN surrogate model

This section evaluates the probabilistic surrogate model using 200 random sample sizes on test datasets. Airfoils are generated via the VAE decoder $P_{\theta}(X|z)$, with uncertain inputs α and M taken from the test set and fed into the surrogate model. As shown in <Figure. 4-14>, the predicted PDFs for lift (C_L) and drag (C_D) closely match the simulation histograms across all airfoils, demonstrating accurate uncertainty prediction. These results confirm the model's effectiveness for uncertainty quantification and design under uncertainty in latent space, even with limited data.

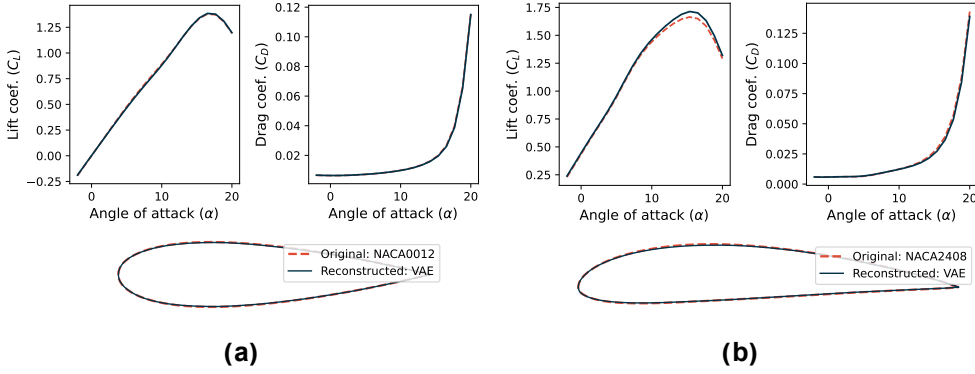


<Figure 4-14> Comparison of predicted and simulated conditional distribution PDFs for aerodynamic uncertainty

4.3.4 Validation on NACA airfoils

To evaluate the impact of geometric reconstruction errors on aerodynamic performance, we validated the variational autoencoder (VAE) using NACA airfoils. The NACA 0012 and NACA 2408 airfoils were input into the VAE and then reconstructed. As illustrated in <Figure 4-15>, the reconstructed airfoils deviate only slightly from the originals, with the most noticeable differences occurring near the maximum thickness regions on both the upper and lower surfaces. Nevertheless, aerodynamic predictions from NeuralFoil demonstrate strong agreement between the original and reconstructed airfoils. These results confirm that the VAE is capable of accurately reconstructing airfoils, as discussed in Section 4.2.4.1, with a typical error of approximately 1×10^{-6} , and that the aerodynamic analysis remains consistent.

The predictions of the surrogate model (MDN) are also validated for NACA airfoils by comparison with NeuralFoil results, as shown in <Figure 4-16>.

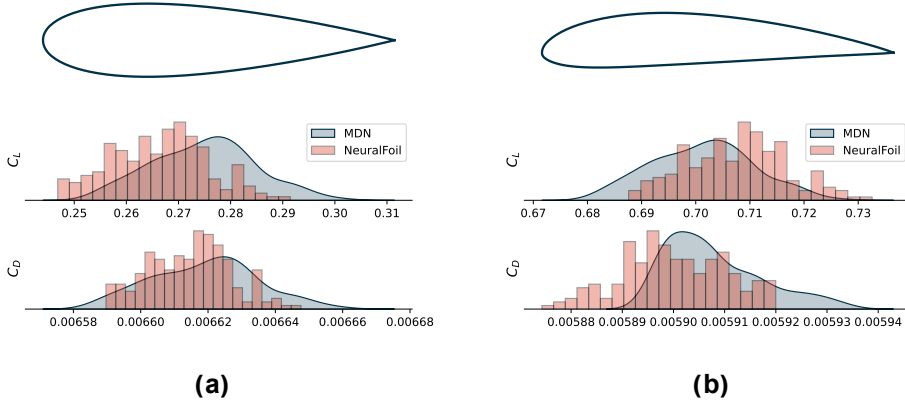


<Figure 4-15> Validation of original and reconstructed airfoil from VAE using NeuralFoil at $Re = 4.45 \times 10^6$, and $M = 0.1962$: (a) NACA 0012, (b) NACA 2408

The airfoils are reconstructed using our generative model (VAE), and latent variables for representative airfoils are obtained. The uncertain parameters—angle of attack α and Mach number M —are randomly sampled from normal distributions $\mathcal{N}(2.79, 0.1^2)$ and $\mathcal{N}(0.1962, 0.005^2)$, respectively, for 200 samples, since the surrogate model is trained on latent variables and uncertain parameters (see <Figure 4-11>). The results are compared with NeuralFoil outputs, demonstrating strong alignment in the predicted PDFs for lift and drag coefficients. The surrogate model achieves $R^2 = 0.999$ for mean prediction and $R^2 = 0.964$ for standard deviation prediction, indicating a well-fitted probabilistic regression model.

4.4 Optimization case studies

This section demonstrates how our approach overcomes the limitations of conventional airfoil design—including unrealistic shape generation, the curse of dimensionality, and high computational cost—by improving both solution quality (Section 4.4.3) and computational efficiency (Section 4.4.5).



<Figure 4-16> Validation of the MDN surrogate model uncertainty prediction for NACA airfoils at $Re = 4.45 \times 10^6$, $\alpha \sim \mathcal{N}(2.79, 0.1^2)$, and $M \sim \mathcal{N}(0.1962, 0.005^2)$. (a) NACA 0012; (b) NACA 2408.

4.4.1 Defining design space for fair comparison

Here, we compare our approach with the popular airfoil parameterization method CST, explained in Section 4.4.1.1.

4.4.1.1 Class/shape-function transformation

The CST method [12] is a parameterization technique based on class and shape functions. This method represents the airfoil shape using parameterized variables over a large design space. A mathematical representation of the airfoil is defined in Eq. (4.2).

$$y(x) = C(x) S(x) + x y'_{TE} \quad (4.2)$$

where $C(x)$ is the class function and $S(x)$ is the shape function, plus a term $x y'_{TE}$ that characterizes the thickness of the trailing edge (where we use $y'_{TE} = 0$). The variable x is defined over the interval $[0, 1]$, and the x, y coordinates of

the airfoil are normalized. The class function is defined in Eq. (4.3).

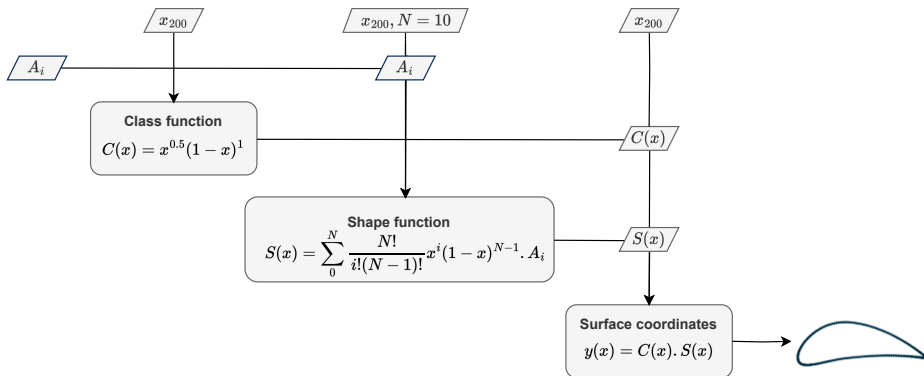
$$C(x) = x^{0.5}(1 - x)^1 \quad (4.3)$$

The shape function defines the geometric characteristics based on the class function and is typically constructed using Bernstein polynomials, as shown in Eq. (4.5).

$$S(x) = \sum_{i=0}^N A_i S_{i,N}(x) \quad (4.4)$$

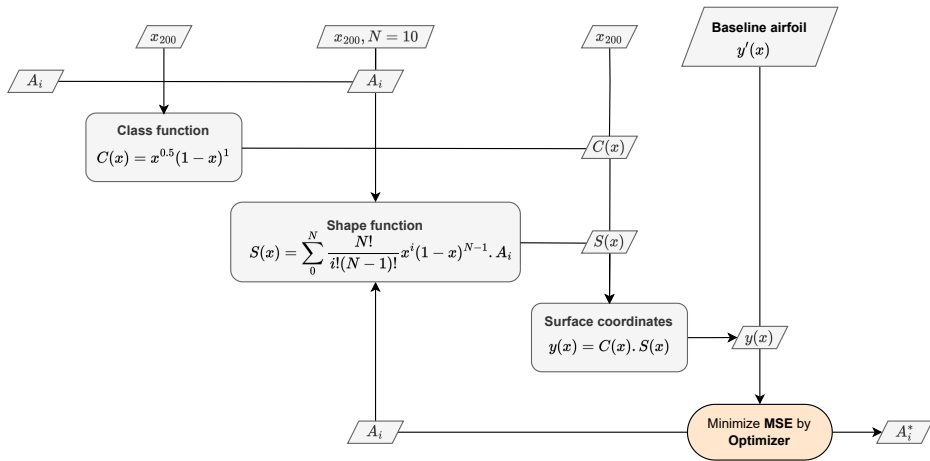
$$S_{i,N}(x) = \frac{N!}{i!(N-1)!} x^i (1-x)^{N-1} \quad (4.5)$$

where A_i are parameters for generating the airfoil coordinates x, y . Two functions are needed to represent the upper and lower surfaces of the airfoil, so the number of CST parameters is $2(N+1)$. In our comparison, we used the 10th-order CST parameterization method. A flow chart for the CST parameterization method is shown in <Figure. 4-17>.

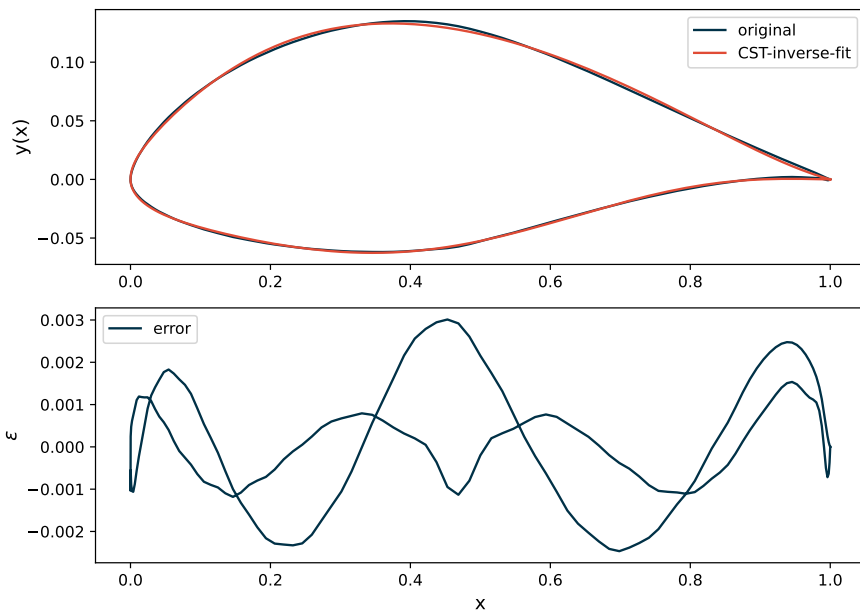


<Figure 4-17> Process of 10th order CST airfoil parametrization

Using the above CST method, inverse fitting (<Figure. 4-18>) is performed by minimizing the difference between the target airfoil surface coordinates and



<Figure 4-18> Process of inverse fitting



<Figure 4-19> Inverse fitting of CST parameterization

those generated by the CST method to obtain the CST polynomial coefficients. An example is shown in <Figure. 4-19>, demonstrating good alignment between the fitted and target airfoil surface coordinates.

4.4.1.2 Defining the design space

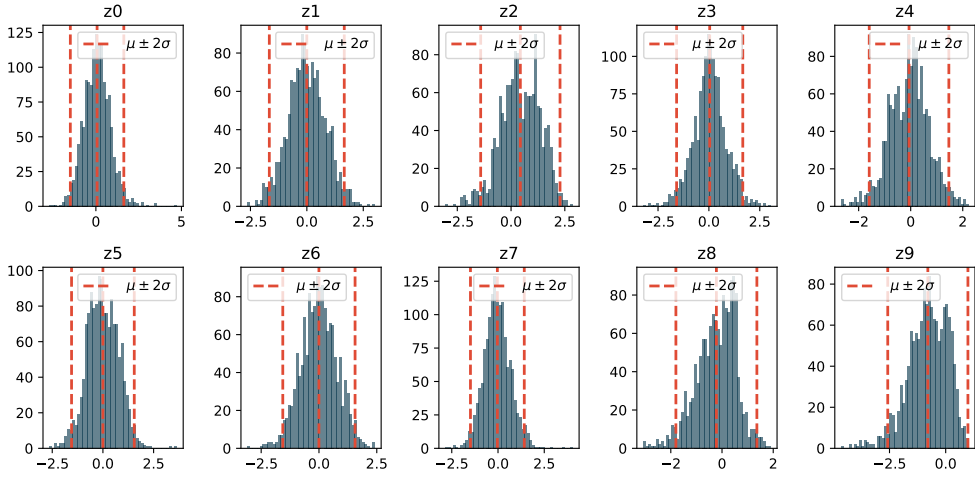
For a fair comparison with other methods, it is essential to maintain the same design space to ensure reliable results. Therefore, we use the VAE encoder to obtain the latent variables (z_i), while the CST method is fitted using a 10th-order polynomial through inverse fitting to obtain the fitting coefficients (A_i) for the entire UIUC airfoil dataset.

Histograms of the latent variables and CST coefficients for the entire dataset are shown in <Figure. 4-20>. As observed, the design variables approximately follow a normal distribution. Therefore, we define the lower and upper bounds as $\mu \pm 2\sigma$, which cover most of the design space for both methods used in the comparison.

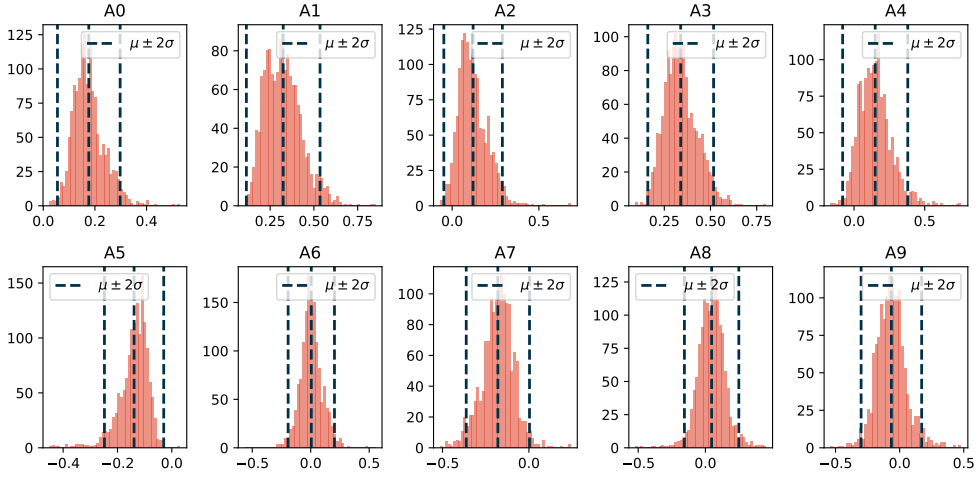
4.4.2 Design feasibility

We demonstrate the feasibility of our generative model (VAE) compared to the popular CST parameterization method [12], using the design space defined by $\mu \pm 2\sigma$ as described above. We generate 10,000 random airfoils and calculate the feasibility ratio using Eq. (4.6), which is defined as the number of non-intersecting airfoils divided by the total number of airfoils generated within the design space.

Examples of airfoils generated by the VAE and CST methods are shown in <Figure. 4-21>. The CST method produces some self-intersecting airfoils, whereas the VAE does not. Accordingly, the feasibility ratios in <Figure. 4-22> indicate 100% feasibility for the VAE (no self-intersections) and 71.9% for the CST method within the same design space. This result demonstrates that the VAE can generate new airfoil designs by sampling from the latent space without self-intersection, effectively eliminating unrealistic geometries and pro-



(a)

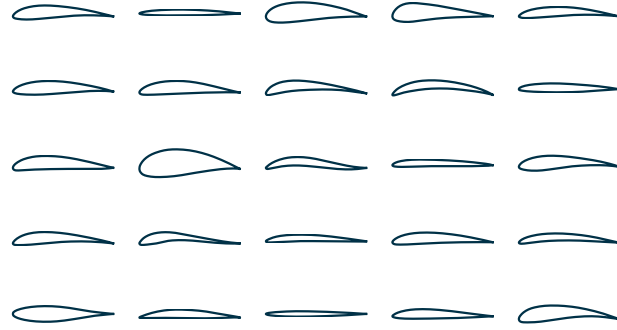


(b)

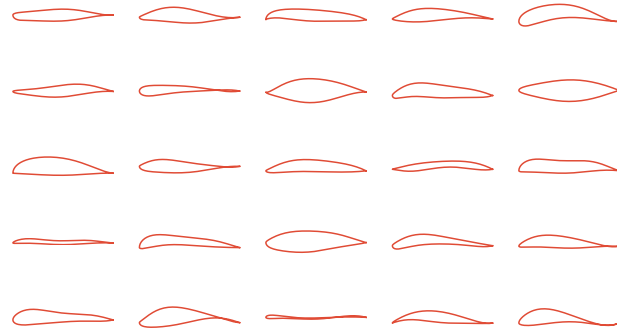
<Figure 4-20> Design space of UIUC airfoil datasets (a) VAE latent variables z_i , (b) CST coefficient A_i

viding high-quality solutions for the optimization process.

$$\text{Feasibility Ratio} = 1 - \frac{N_{\text{self-intersections}}}{N_{\text{total}}} \times 100\% \quad (4.6)$$



(a)

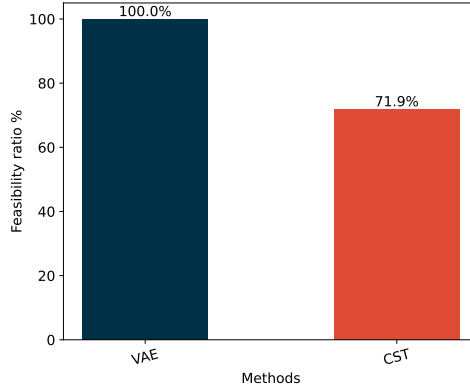


(b)

<Figure 4-21> Random airfoil generated within the design space (a) VAE (b) CST

4.4.3 Quality solutions

In the previous sections, we showed that the VAE learns a compact, lower-dimensional, disentangled latent space and can generate 100% feasible designs. In this section, we demonstrate the benefits of these improved properties for aerodynamic shape optimization. We compare the traditional parameterization method (CST) with the generative model (VAE) for maximizing the lift-to-drag ratio as an unconstrained optimization problem. We utilized Neu-



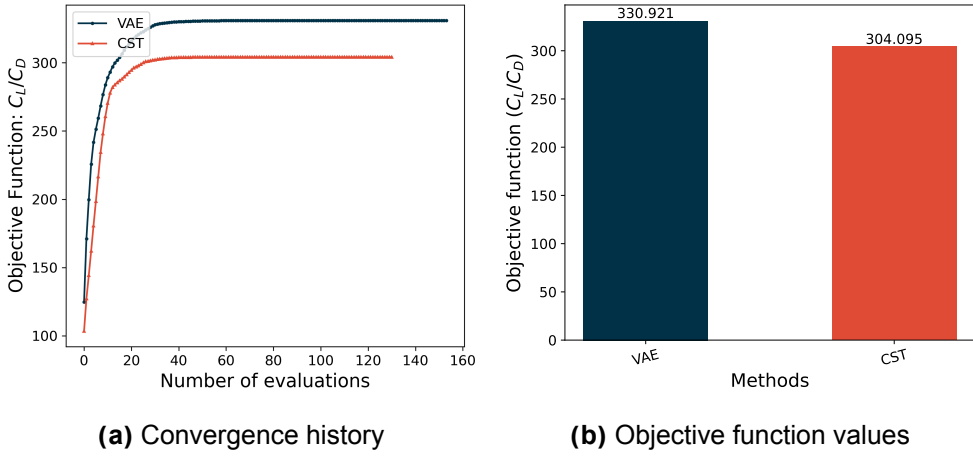
<Figure 4-22> Comparison of feasibility ratio

ralFoil[109] for aerodynamic analysis of airfoils (see Section 4.3.1.2), and employed a genetic algorithm (GA) to find the optimum airfoil design. The aerodynamic analysis is performed at a Reynolds number of 4.45×10^6 , angle of attack $\alpha = 2.79^\circ$, and Mach number 1.962. The optimization formulation is shown below in Eq. (4.7).

$$\begin{aligned} \max_{\mathbf{X}} \quad & J(\mathbf{X}) = C_L/C_D \\ \text{w.r.t.} \quad & \mathbf{X}^{\text{LB}} \leq \mathbf{X} \leq \mathbf{X}^{\text{UB}} \end{aligned} \quad (4.7)$$

We used latent variables (z) as design variables for VAE-based optimization, as described in Sec. 3.4.2, and CST parameters (A_i) as design variables for CST-based optimization, as described in Sec. 4.4.2.

The optimization convergence history is compared in <Figure. 4-23a>, where the maximum lift-to-drag ratios are 304.09 for CST and 330.92 for VAE, as summarized in <Figure. 4-23>. These results demonstrate that our VAE-based optimization achieves an 8.82% improvement over the conventional CST method. The optimized airfoil designs are shown in <Figure. 4-24>, indicating that both airfoils exhibit sharp leading and trailing edges. However, the VAE-based airfoil has greater camber compared to the CST-based design, resulting in higher



<Figure 4-23> Comparison of VAE-based optimization and CST-based optimization



<Figure 4-24> Optimum airfoils for VAE and CST method

aerodynamic efficiency and higher quality solutions.

4.4.4 Optimization formulations considering nonlinear aerodynamic effects

This section demonstrates how the GenAI optimization approach accounts for nonlinear aerodynamic effects, such as the maximum lift coefficient ($C_{L_{max}}$). We solve constrained optimization problems for different required lift coefficients ($C_{L_{req}}$), ranging from 0.7 to 1.0, at an angle of attack of 2.79° and a Mach number of 0.1962. A $C_{L_{max}}$ value of 1.65 is considered for all cases. The optimization formulation is presented in Eq. (4.8).

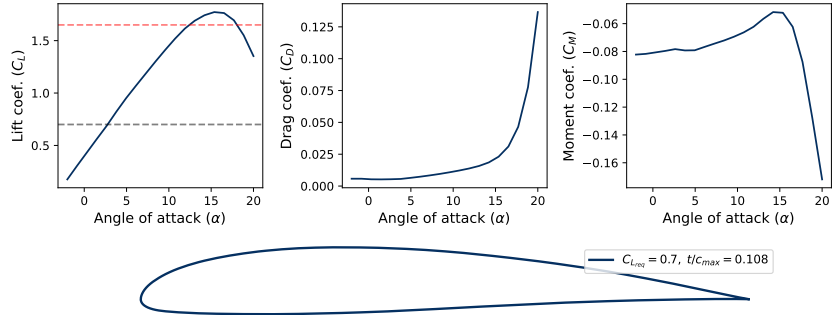
$$\begin{aligned}
& \min_{\mathbf{z}} \quad J(\mathbf{z}) = C_D \\
& \text{w.r.t.} \quad \mathbf{z}_{lb} \leq \mathbf{z} \leq \mathbf{z}_{ub} \\
& \text{s.t.} \quad C_L = C_{L_{req}}, \\
& \quad \quad C_{L_{max}} \geq 1.65, \\
& \quad \quad 0.8t_{max} \leq t \leq 3.0t_{max}
\end{aligned} \tag{4.8}$$

where, z represents the latent vectors, and t_{max} denotes the maximum thickness of the airfoil. The optimization results are shown in Figure 4-25.

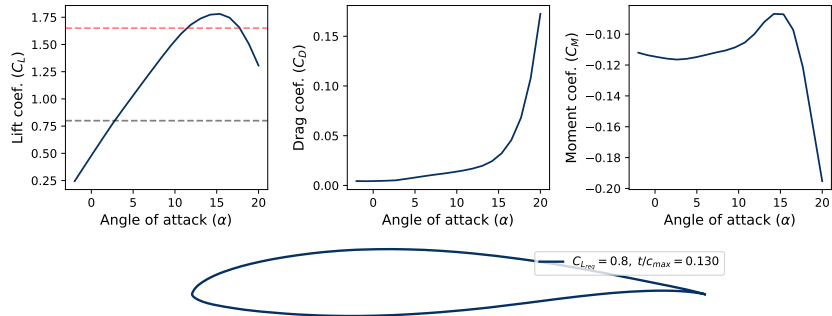
The optimization results demonstrate that, as shown in <Figure. 4-25a>, when the required lift coefficient is set to 0.7, the airfoil camber is the lowest among all optimization cases. The $C_{L_{max}}$ constraint is satisfied, as indicated by the red dotted line. The analysis achieved a $C_{L_{max}}$ value of 1.7 at an angle of attack of 15 degrees, with a drag coefficient (C_D) of 0.00533. In <Figure. 4-25c>, the airfoil camber increases significantly to meet the higher lift requirement, while still satisfying the maximum lift coefficient constraint. These results illustrate that GenAI optimization can be applied not only to a single analysis condition but also to account for nonlinear aerodynamic effects through systematic optimization formulations.

4.4.5 GenAI optimization for airfoil design under uncertainty

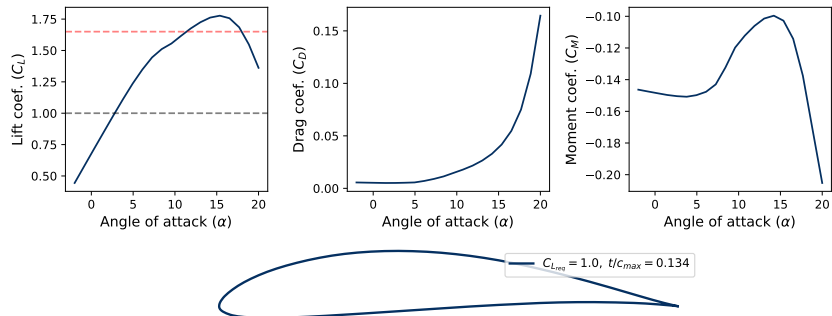
Here, we demonstrate that our approach can find optimal solutions more efficiently than conventional simulation-based methods by comparing the number of samples required for uncertainty-based optimization. We conducted a comparison between surrogate-based optimization using the MDN surrogate model (as described in Chapter 3, Section 3.4) and simulation-based optimization using NeuralFoil. The optimization formulations and comparative results



(a)



(b)



(c)

<Figure 4-25> Optimum airfoils and their corresponding aerodynamic analysis results for different C_{Lreq} (a) $C_{Lreq} = 0.7$, $C_D = 0.00533$, $C_M = -0.0776$ at $\alpha = 2.79^\circ$ (b) $C_{Lreq} = 0.8$, $C_D = 0.00520$, $C_M = -0.116$ at $\alpha = 2.79^\circ$ (c) $C_{Lreq} = 1.0$, $C_D = 0.00515$, $C_M = -0.150$ at $\alpha = 2.79^\circ$

are presented in Sections 4.4.5.1 and 4.4.5.2, respectively.

4.4.5.1 Optimization formulations

The design space from previous sections are utilized $\mu \pm 2z_\sigma$ for lower bounds and upper bounds, so that the latent distribution spans the airfoil design space. The E214 airfoil serves as the baseline for constraint formulation, with $C_L = 0.9362$, $C_D = 0.00543$, and maximum thickness $t_{\max} = 0.111$ at $\alpha = 2.79^\circ$, $M = 0.1962$, and $\text{Re} = 4.45 \times 10^6$.

A deterministic design optimization is formulated as follows:

$$\begin{aligned}
 \min_{\mathbf{z}} \quad & J(\mathbf{z}) = C_D \\
 \text{w.r.t.} \quad & \mathbf{z}_{\text{lb}} \leq \mathbf{z} \leq \mathbf{z}_{\text{ub}} \\
 \text{s.t.} \quad & C_L = C_{L_{\text{req}}}, \\
 & C_D \leq C_{D_{\text{base}}}, \\
 & 0.8t_{\max} \leq t \leq 3.0t_{\max}
 \end{aligned} \tag{4.9}$$

where, z represents the latent vectors, and $t_{\max}$ denotes the maximum thickness of the airfoil. We utilized the mean value from the surrogate model prediction in Eq. 3.22 with $N_{\text{mcs}} = 1$, as it does not consider the flight condition uncertainty.

A robust design optimization is performed using weight sum method for objective function calculation, with the formulation below:

$$\begin{aligned}
 \min_{\mathbf{z}} \quad & J(\mathbf{z}) = w \mu_{C_D} + (1 - w) \sigma_{C_D} \\
 \text{w.r.t.} \quad & \mathbf{z}_{\text{lb}} \leq \mathbf{z} \leq \mathbf{z}_{\text{ub}} \\
 \text{s.t.} \quad & \mu_{C_L} = C_{L_{\text{req}}}, \\
 & \mu_{C_D} \leq C_{D_{\text{base}}}, \\
 & 0.8t_{\max} \leq t \leq 3.0t_{\max}
 \end{aligned} \tag{4.10}$$

where w is the weighting factor used to balance between μ and σ in the robust design objective function, and we set $w = 0.1$. The same constraints are applied as in the deterministic formulation, except that random variables ξ are considered for the angle of attack and Mach number therefore, we use μ_{C_L} equality constraint calculation. Both parameters are sampled from normal distributions, $\alpha \sim \mathcal{N}(2.79, 0.1^2)$ and $M \sim \mathcal{N}(0.1962, 0.005^2)$, with Monte Carlo samples of $N_{\text{mcs}} = 5000$. The statistics of the mean and standard deviation are calculated using Eq. (3.22) and Eq. (3.23), respectively.

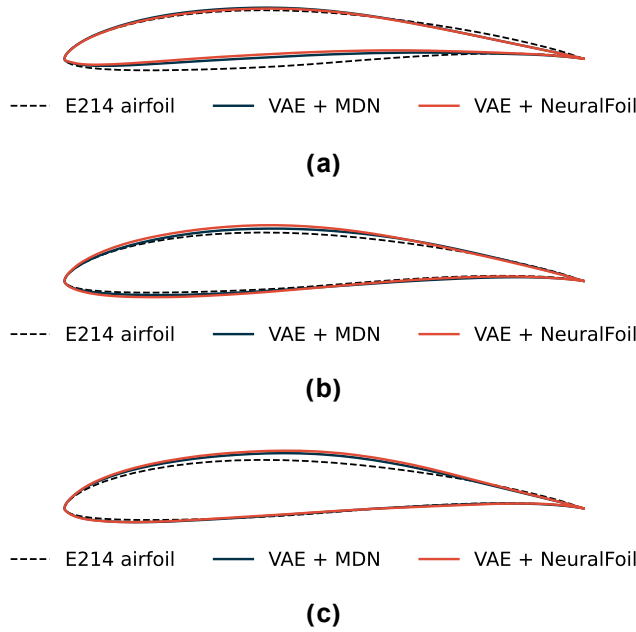
Next, a reliability-based design optimization is conducted with the formulation below;

$$\begin{aligned}
& \min_{\mathbf{z}} \quad J(\mathbf{z}) = \mu_{C_D} \\
& \text{w.r.t.} \quad \mathbf{z}_{\text{lb}} \leq \mathbf{z} \leq \mathbf{z}_{\text{ub}} \\
& \text{s.t.} \quad \mathbf{P}_{\mathbf{f}}[\mathbf{C}_{\text{L}_{\text{req}}} - \mathbf{C}_{\text{L}}(\mathbf{z}, \xi) \geq \mathbf{0}] \leq \mathbf{P}_{\mathbf{f}_{\text{req}}}, \\
& \quad \mu_{C_D} \leq C_{D_{\text{base}}}, \\
& \quad 0.8t_{\text{max}} \leq t \leq 3.0t_{\text{max}}
\end{aligned} \tag{4.11}$$

where $P_{f_{\text{req}}}$ is the required failure probability, which can be calculated from the cumulative distribution function (Φ) of the standard normal distribution using the reliability index β as $P_{f_{\text{req}}} = \Phi(-\beta)$. In our optimization, we set $\beta = 2.5$. We adopt the same strategy as in the robust design optimization to account for uncertainty, and the failure probability P_f is calculated using Eq. (3.25) with Monte Carlo samples of $N = 5000$.

4.4.5.2 Comparison of the optimization results

This section presents a comparison of the results from a surrogate-based optimization (VAE + MDN) and a simulation-based optimization (VAE + NeuralFoil)



<Figure 4-26> Optimization results comparing with VAE + MDN and VAE + NeuralFoil; (a) DO optimum (b) RDO optimum (c) RBDO optimum

using our framework with the formulation described in above section. The optimization results are compared in <Figure. 4-26> and <Table 4-7>.

The deterministic optimized airfoil geometries obtained from (VAE + MDN) and (VAE + NeuralFoil) models are presented in <Figure. 4-26a>, showing close agreement with thickness variations. Aerodynamic performance comparisons of baseline and optimized airfoils, summarized in <Table 4-7>, indicate minimal differences in, maximum thickness (0.00720), and drag coefficient (0.00015). These results demonstrate the surrogate optimization's accuracy and effectiveness, aligning closely with the simulation-based optimization.

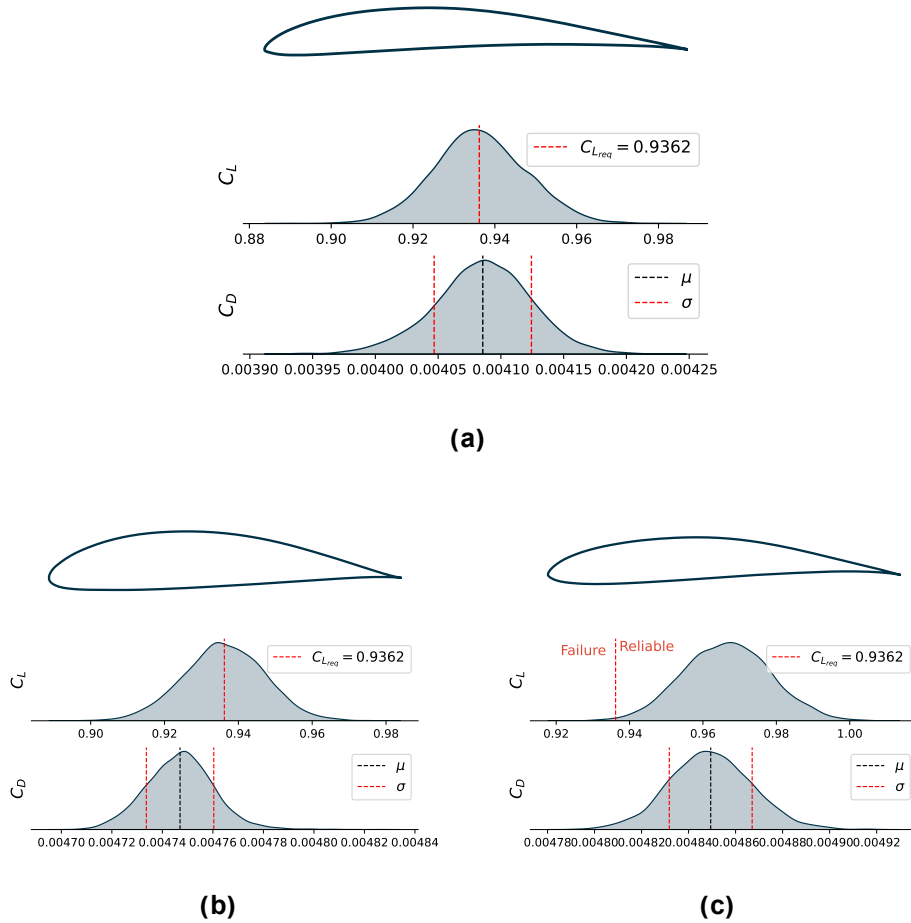
The robust optimum airfoil geometries obtained from surrogate-based and simulation-based robust designs are compared in <Figure. 4-26b>, showing good agreement, though the surrogate design exhibits slightly lower $t_{\max}$. The aerodynamic performance metrics (μ and σ) are summarized in <Table 4-7>.

<Table 4-7> Comparison of optimization results

Method	C_L		C_D		$t_{\max}$	$P_{f_{C_L}}$ (%)
	μ	$\sigma \times 10^{-4}$	μ	$\sigma \times 10^{-4}$		
E214 Airfoil	0.936	–	0.00543	–	0.111	50.0
DO Optimum						
VAE + MDN	0.936	11.2	0.00408	0.387	0.0898	50.0
VAE + NeuralFoil	0.936	10.4	0.00423	0.351	0.0970	50.0
Difference (ϵ)	0.000	0.800	0.00015	0.0460	0.00720	0.000
RDO Optimum						
VAE + MDN	0.935	11.2	0.00449	0.137	0.129	50.0
VAE + NeuralFoil	0.936	11.3	0.00514	0.0826	0.130	50.0
Difference (ϵ)	0.001	0.1	0.00065	0.0544	0.001	0.000
RBDO Optimum						
VAE + MDN	0.966	11.8	0.00484	0.177	0.121	0.360
VAE + NeuralFoil	0.967	11.4	0.00477	0.130	0.125	0.340
Difference (ϵ)	0.00100	0.400	0.00007	0.0470	0.00400	0.020

As expected, the predicted mean values (μ) show smaller errors compared to standard deviations (σ), consistent with the MDN model' s lower fitting accuracy for σ ($R^2 \approx 0.96$ – 0.97 ; Sec. 4.3.2), resulting in larger differences (ϵ) of 0.61 and 0.02479 for lift and drag, respectively. Nevertheless, the surrogate-optimized airfoil demonstrates enhanced robustness with lower sensitivity to uncertain parameter variations, whereas the simulation-based optimum airfoil yields higher drag ($C_D = 0.0052$) and greater maximum thickness ($t_{\max} = 0.174$).

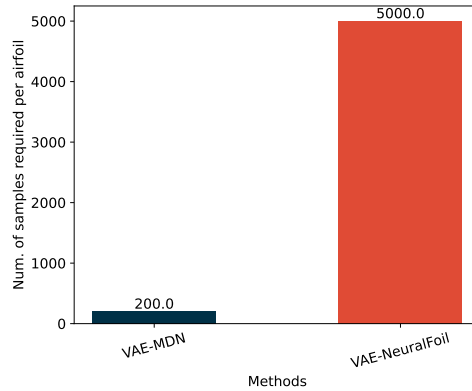
The RBDO-optimized airfoil geometries from surrogate-based and simulation-based designs are compared in <Figure. 4-26c>, showing closely matching shapes. Aerodynamic analyses indicate very small mean prediction errors for lift and drag coefficients (0.001 and 0.00007, respectively). However, standard deviation predictions exhibit slightly larger errors, with differences of 0.4 for lift and 0.047 for drag coefficients. Additionally, the failure probabilities are notably



<Figure 4-27> Probability density functions of lift and drag coefficient with 5000 samples; (a) DO optimum (b) RDO optimum (c) RBDO optimum

reduced to 0.36% (surrogate-based) and 0.34% (simulation-based), confirming the reliability improvements.

We illustrate the probability density distributions of the surrogate-based optimum airfoils compared to deterministic, robust, and reliable designs in <Figure. 4-27>. The deterministic optimum exhibits the lowest drag and thickness, whereas the robust design achieves lower drag with reduced variability under uncertain flight conditions, maintaining the required lift performance. In con-



<Figure 4-28> Sample required comparison for design under uncertainty

trast, the reliable optimum shifts the C_L distribution toward feasible regions, significantly reducing failure probability to 0.34% (reliability index of 2.5) at the target $C_L = 0.936$, compared to the 50% failure rates of other designs. This reliability enhancement comes at a slight penalty of increased drag and thickness relative to deterministic and robust designs.

This case study demonstrates that the proposed optimization framework can be effectively applied not only to deterministic but also to uncertainty-based design optimizations, even with limited data, as shown in <Figure. 4-28>, while maintaining accuracy compared to expensive simulation-based design optimization approaches.

4.5 Summary

We applied a GenAI optimization approach to reduce dimensionality, generate new designs, and predict aerodynamic uncertainty due to flight conditions through explicit modeling, thereby overcoming the limitations of conventional optimization approaches for airfoil design under uncertainties. The results show that our novel approach not only generates new designs without self-

intersecting airfoils, unlike traditional parameterization methods and can be utilized as a parameterization in a lower-dimensional latent space, but also predicts the uncertainty of aerodynamic performance, showing good alignment with simulation-based uncertainty quantification.

Furthermore, we performed design optimization comparing with CST methods within the same design space, achieving better-quality solutions. A surrogate-based and simulation-based optimization is also compared for deterministic, robust, and reliability-based design optimization through Monte Carlo simulation. The generative VAE model was employed as a parameterization tool for simulation-based design, providing a direct comparison with our proposed methodology. The results confirm that uncertainty-based design optimization can be effectively implemented within our framework, achieving a notably small deviation in standard deviation prediction while maintaining mean predictions closely aligned with simulation-based optimal results.

Our approach provides a valuable and efficient framework for accurately predicting uncertainty, even with limited data. Moreover, the proposed GenAI design optimization framework proves to be highly effective for robust and reliability-based design optimization in a reduced-dimensional design space.

5. Blended Wing Body Aircraft Design via GenAI Optimization Approach

In this chapter, we present our approach to a more challenging problem: aircraft design. A blended wing body aircraft is considered to evaluate the effectiveness of our method in terms of finding high-quality solutions, reducing computational cost, and adapting to large-scale problems.

First, we describe the problem definition in Section 5.1. Deep generative modeling for learning compact representations of complex aircraft geometry is discussed in Section 5.2, followed by surrogate modeling for uncertainty quantification from latent representations in Section 5.3. In Section 5.4, we demonstrate how our approach overcomes the limitations of conventional optimization methods and present the corresponding results.

5.1 Problem definition

The shape design of an aircraft significantly affects its overall performance, including aerodynamic efficiency. Optimal shape design relies heavily on optimization [22, 115], and a geometric design space with hundreds of design variables is usually required for fully efficient aerodynamic shape optimization[15]. This process leads to a larger design space, which can result in abnormal shapes that introduce unnecessary challenges to aircraft design [16], affecting both efficiency and computational cost during optimization. Therefore, learning a compact geometric design space with fewer variables is of great interest in

aerodynamic shape optimization.

In Chapter 4, it was demonstrated that optimization in a compact, lower-dimensional space offers higher efficiency than conventional parameterization methods such as CST. In previous research, approximately 198 design variables were reduced to 10. In aircraft design, the overall shape of an aircraft can be defined by hundreds of design variables [15], and methods such as Free-Form Deformation (FFD) [116] are often employed. However, these methods suffer from drawbacks such as geometric abnormalities and the need for a large number of FFD control points [117].

To overcome geometric abnormalities and the high-dimensional design space problem, recent deep generative models can be applied to learn compact, lower-dimensional representations of aircraft geometry directly from data. A few studies have explored the use of deep learning to represent wing geometries [74] and propeller blades [73], demonstrating the efficiency of design-space optimization and the potential to generate new designs. These works primarily employed Generative Adversarial Networks (GANs) to learn compact geometric representations; however, their focus was limited to demonstrating the effectiveness of GANs.

In this research, we employ Variational Autoencoders (VAEs) to explicitly learn compact, lower-dimensional representations of geometry for more realistic aircraft design problems. A blended wing body (BWB) aircraft is selected as the case study to demonstrate our methodology. The fundamental characteristics of the blended-wing-body (BWB) configuration, as an advanced technology demonstrator, include high aerodynamic efficiency, low operational empty weight, and potential noise reduction [118]. Several BWB configurations are shown in <Figure 5-1>. The design of a BWB aircraft can be fully controlled by shaping its sectional profiles, which are analogous to airfoil designs. Unlike

conventional tube-and-wing configurations, this makes the BWB an appealing subject for research, as its aerodynamic shape can be entirely defined through surface coordinates, allowing for a much larger design space. Several studies have used surface-coordinate-based representations to achieve full-geometry deformation of BWB configurations [15, 119, 120]. In contrast, other studies have adopted simplified parameterization models based only on planform geometry, such as wing-section chords, taper ratios, and spanwise distributions [50, 121–123]. These methods only simplify the optimization problem, which may not achieve maximum aircraft efficiency. Additionally, when dealing with uncertainty-based optimization, the computational cost is considerably high.

These challenges can be overcome using our GenAI optimization approach. This method learns a lower-dimensional representation of BWB geometries and performs design optimization using a surrogate model for robust and reliable design.

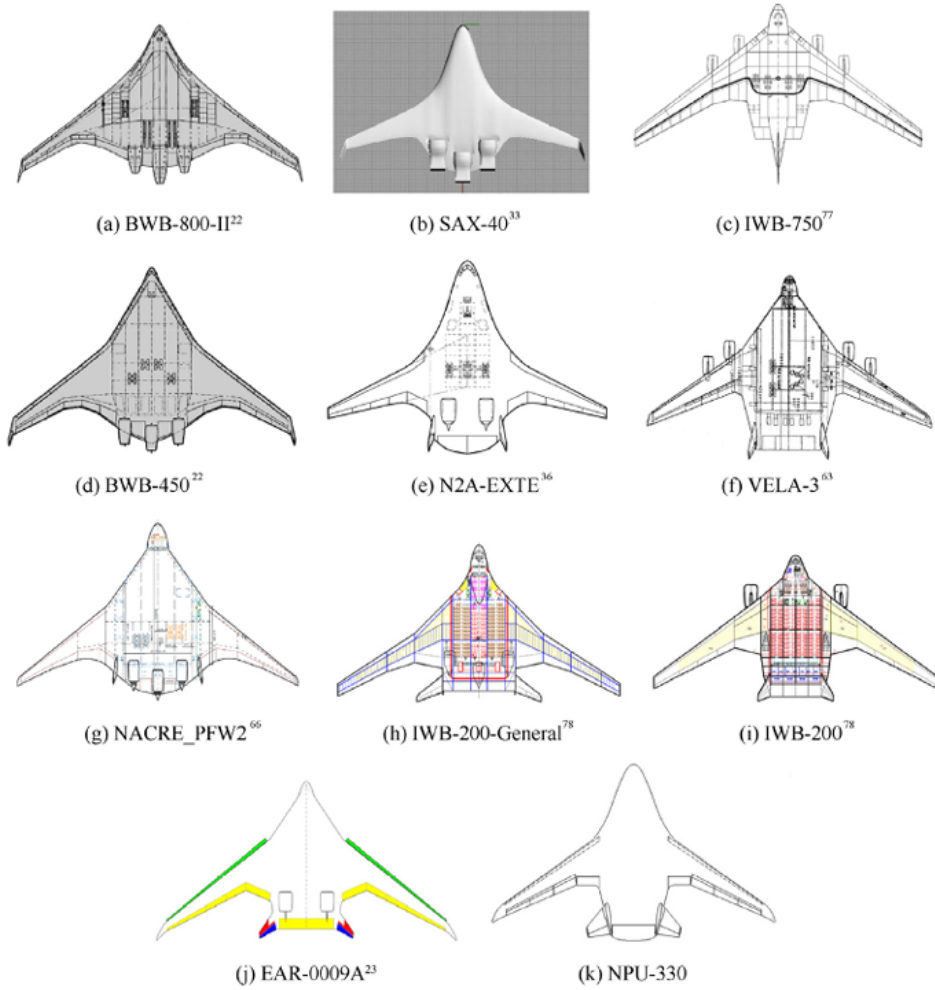
We follow the steps described in Chapter 3.

5.2 Deep generative modeling

As explained in Chapter 3, Sec. 3.2, we follow these steps for generative modeling of aircraft design.

5.2.1 Data pre-processing for blended wing body aircraft

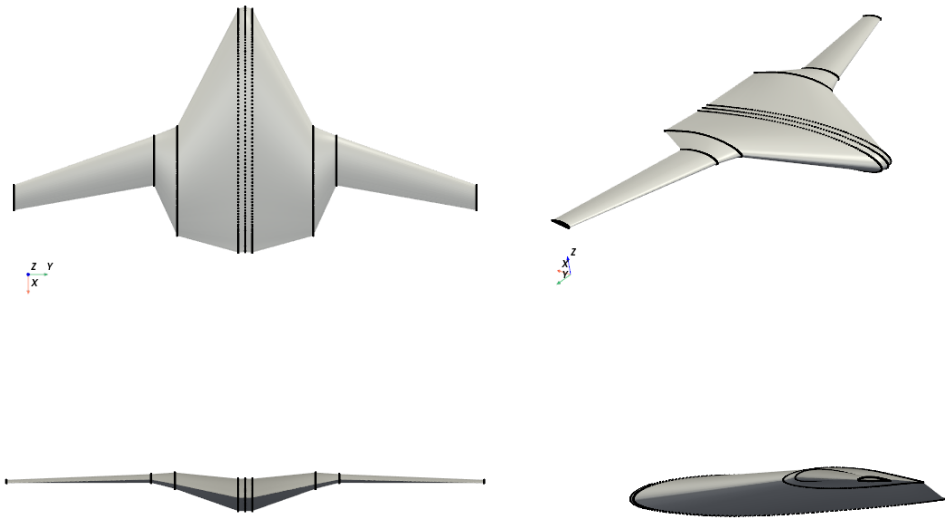
This section describes the data preparation process for training the VAE model to learn a lower-dimensional latent space. An example BWB configuration is shown in <Figure 5-2>. Because machine learning models—especially generative models—are highly sensitive to data quality, the training data must be



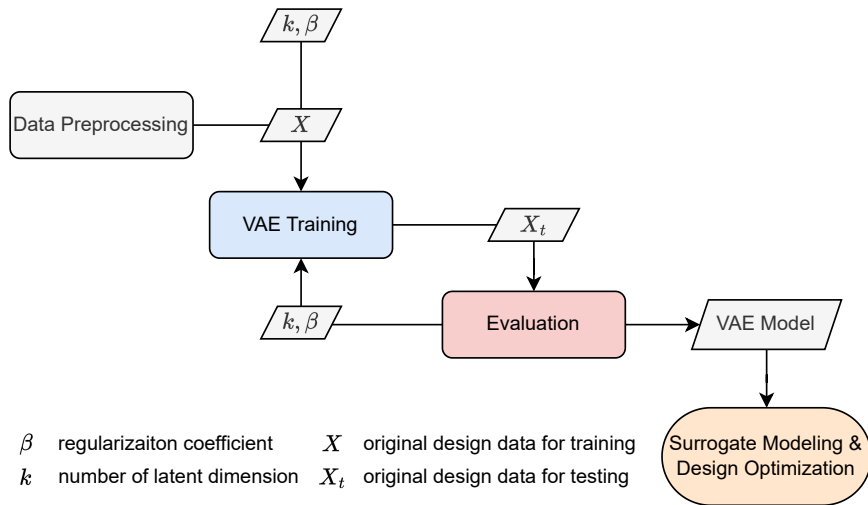
<Figure 5-1> Different BWB transport aircraft [118]

represented in a meaningful and consistent manner. Unlike the airfoil problem discussed in Chapter 4, which benefits from the availability of historical datasets for learning lower-dimensional representations, there are no publicly available datasets for BWB aircraft configurations. Consequently, a method that can cover a large design space for BWB aircraft is necessary.

Therefore, a constraint-based sampling strategy is developed to construct a comprehensive design space that represents various BWB configurations while avoiding the generation of abnormal or unrealistic shapes in the three-

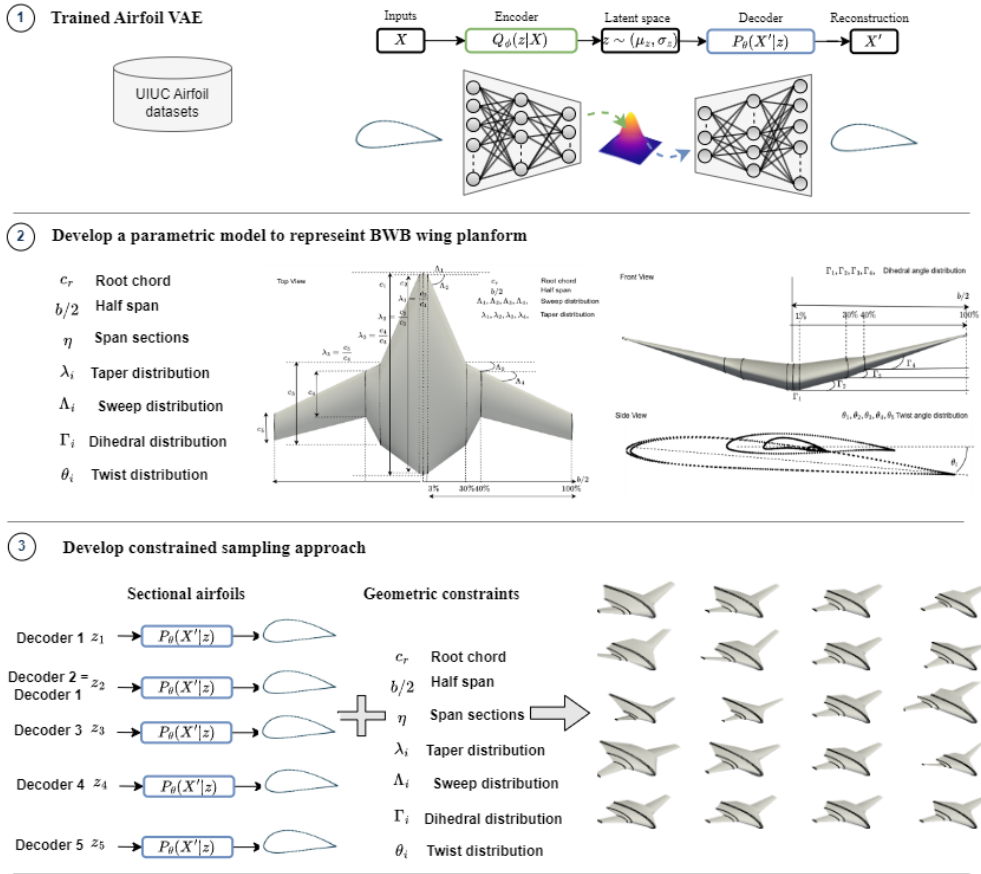


<Figure 5-2> Example of blended wing body aircraft



<Figure 5-3> Deep generative modeling

dimensional (3D) plane. Details of this strategy are presented in Section 5.2.1.2.



<Figure 5-4> Blended wing body parametrization and data preparation via constrained based geometric sampling

5.2.1.1 Constrained based geometric sampling

We develop a geometric constraint-based sampling strategy to generate different types of BWB configurations, including sectional surface coordinates to represent flexible and fully free-form deformation of aircraft shapes. The methodology for generating shape variations of the BWB configuration across a comprehensive design space is illustrated in <Figure 5-4>. The process consists of three main steps.

In the first step, an airfoil generator is trained to learn the latent representation of airfoils and capture the overall design space of existing airfoil shapes.

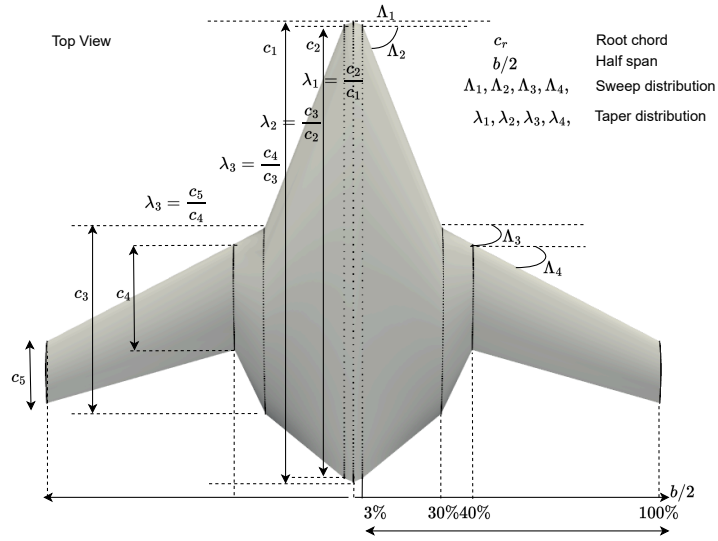
Here, we employ the airfoil VAE model developed in Chapter 4. The primary reason for using our previously trained VAE model is to ensure a complete and flexible representation of the sectional deformation of the aircraft. Unlike random airfoil generation approaches based directly on datasets such as those used in [73, 74], our trained model provides a compact latent representation capable of generating diverse and realistic airfoil shapes that extend beyond the original UIUC dataset.

Secondly, we developed a parametric model to represent the BWB aircraft configuration. A detailed representation of the parametric model is shown in <Figure 5-5>. The parametric model uses scaling factors for the root chord (c_r) and span (b), together with the spanwise distribution, to define the aircraft geometry. By expressing the spanwise distribution of each section as a percentage, we can represent the spanwise stations using the scaling factor of the wing span, as shown in Eq. (5.1).

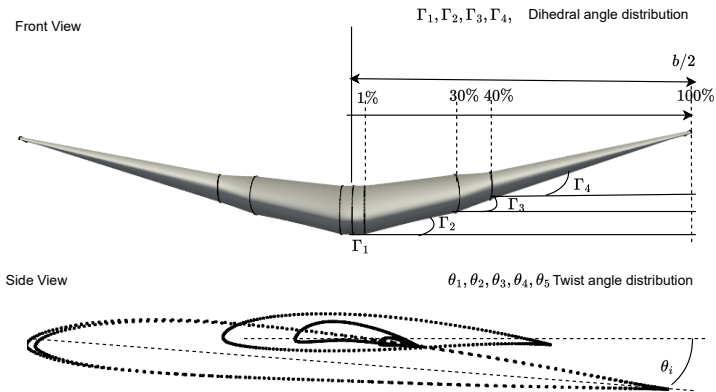
$$y_{j+1} = y_j + \Delta y_j, \quad \Delta y_j = \frac{b}{2}(\eta_j - \eta_{j-1}) \quad (5.1)$$

where y_j is the section location and η_j is the spanwise distribution in percentage. Taper (λ_j), sweep (Λ_j), dihedral (Γ_j), and twist (θ_j) distributions can be represented along the sections within the design space using design constraints, enabling a fully meaningful representation of aircraft geometry. For example, spanwise sections $\eta_j = [0\%, 3\%, 30\%, 40\%, 100\%]$ can be used to represent five sections of the aircraft.

Third, the constraint-based sampling method generates different airfoils along the spanwise distribution, incorporating the aircraft's parametric model parameters while observing geometric constraints. The step-by-step data flow is illustrated in <Figure 5-6>, and the detailed derivation of the geometry is provided in **Appendix 6.2**.



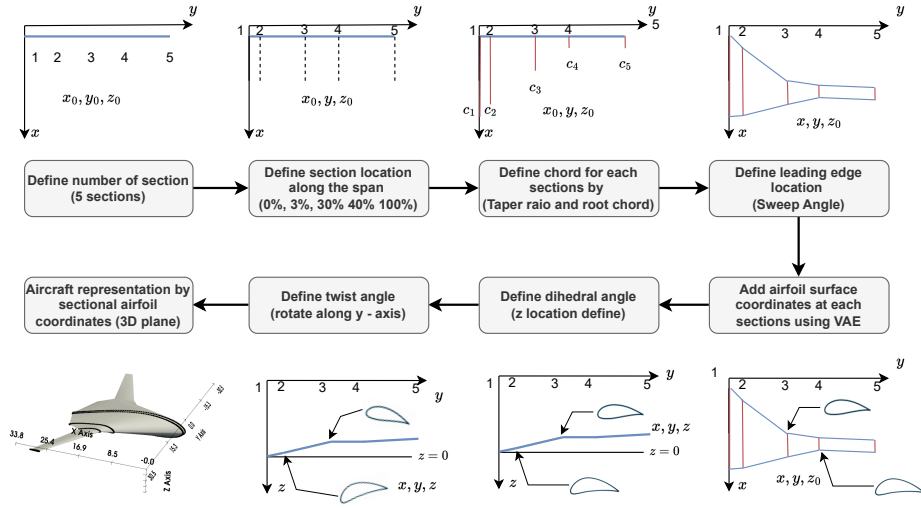
(a)



(b)

<Figure 5-5> Parametric representation of BWB aircraft model: (a) Top view, showing spanwise taper and sweep distributions; (b) Front view, showing spanwise dihedral distribution; (c) Side view, showing spanwise twist distribution

First, the aircraft is divided into multiple sections, and the sectional spanwise locations are defined as percentages along the span (b) using Eq. (5.1). Second, the chord lengths of each section (c_i) are determined from the taper



<Figure 5-6> Constrained based geometric sampling

ratio distribution (λ_i) based on the given root chord (c_r). Third, the leading-edge coordinates (x_{LE}, y_{LE}, z_{LE}) of each section are computed according to the sweep angle distribution (Λ_i). Fourth, at each section, airfoils containing 200 surface coordinates ($[\hat{x}, \hat{z}]_{200}$) are randomly generated using the VAE-airfoil model along the sectional chords. Fifth, the coordinates are rotated about the x -axis to account for the dihedral angle distribution (Γ_i). Sixth, the twist angle distribution (θ_i) is applied by rotating the coordinates about the y -axis. Finally, a three-dimensional representation ($[x, y, z]$) of the blended wing body aircraft is constructed by integrating all the steps described above. By incorporating parametric constraints together with VAE-airfoil latent variable constraints, this approach enables the systematic generation of diverse aircraft geometries while ensuring realistic configurations.

<Table 5-1> Baseline and design space for data generation

Parameters	Baseline [119, 124]	Lower bounds	Upper bounds
c_r (m)	45.10	- 30%	+ 30%
b (m)	64.92	- 30%	+ 30%
λ_1	0.9	0.9	1.0
λ_2	0.6	0.3	0.5
λ_3	0.5	0.3	0.6
λ_4	0.5	0.2	0.6
Λ_1 (deg)	-	18	18
Λ_2 (deg)	45	45	80
Λ_3 (deg)	50	-10	60
Λ_4 (deg)	20	-10	30
Γ_1 (deg)	-	0	0
Γ_2 (deg)	-	-5	20
Γ_3 (deg)	3	-5	20
Γ_4 (deg)	5	-5	20
θ_1 (deg)	-	0	0
θ_2 (deg)	-	0	0
θ_3 (deg)	0	-3	8
θ_4 (deg)	3	-3	8
θ_5 (deg)	2	-3	8

5.2.1.2 Data generation for generative model training

A constraint-based geometric sampling method is used to generate the data required for training. We considered a five-section wing, allowing flexible and free-form deformation of the wing shape geometry. [119] uses 40% of the wing span as the center body, with 40% to 100% designated as the wing geometry. For this reason, we utilize spanwise sections $\eta_i = [0\%, 3\%, 30\%, 40\%, 100\%]$, considering five sections to represent the aircraft. The parametric constraints are derived from the baseline model presented in [119, 124]. The corresponding baseline parameters and geometric constraints are summarized in <Table 5-1>.

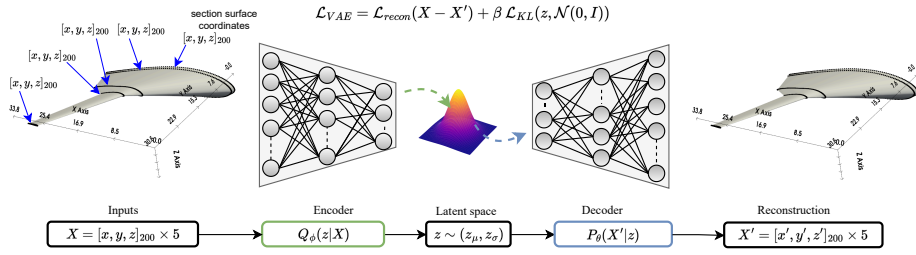


<Figure 5-7> Randomly generated bwb aircrafts by proposed sampling method

The airfoil coordinates for each section are generated by the VAE; however, we apply hard constraints for the first two sections (y_0, y_1) to ensure the airfoil thickness (t_{max}) is greater than 18% [119]. This results in a larger thickness at the center body of the BWB aircraft configuration and uses the same airfoil generator to ensure a smooth transition at the body center. By combining the VAE-generated airfoils at each section with the wing geometric constraints from <Table 5-1>, we are able to generate aircraft geometries represented by surface coordinates. An example of a randomly generated aircraft is shown in <Figure 5-7>.

For training the generative model, only half of the aircraft geometry is considered. Each aircraft is represented by five sectional profiles, with 200 coordinate points per section, resulting in a total of 3,000 surface coordinates per aircraft. These coordinates are directly used as input data for training the VAE. The training dataset is shown in Figure 5-8.

A total of 5,000 aircraft samples are randomly generated using Latin Hypercube Sampling (LHS) to train the VAE model, enabling it to learn a com-



<Figure 5-9> Architecture of VAE for blended wing body aircraft

<Table 5-2> BWB VAE input, latent representation, and output structure.

Encoder (Q_ϕ)		Latent space ($k = 16$) ($z \in \mathbb{R}^k$)	Decoder (P_θ)	
Input ($X \in \mathbb{R}^N$)	Output ($z_\mu \in \mathbb{R}^k, z_\sigma \in \mathbb{R}^k$)		Input ($z \in \mathbb{R}^k$)	Output ($X' \in \mathbb{R}^N$)
x_1, y_1, z_1				x'_1, y'_1, z'_1
x_2, y_2, z_2				x'_2, y'_2, z'_2
x_3, y_3, z_3	z_{μ_1}, z_{σ_1}	$z_1 \sim \mathcal{N}(z_{\mu_1}, z_{\sigma_1}^2)$	z_1	x'_3, y'_3, z'_3
x_4, y_4, z_4	z_{μ_2}, z_{σ_2}	$z_2 \sim \mathcal{N}(z_{\mu_2}, z_{\sigma_2}^2)$	z_2	x'_4, y'_4, z'_4
x_5, y_5, z_5	z_{μ_3}, z_{σ_3}	$z_3 \sim \mathcal{N}(z_{\mu_3}, z_{\sigma_3}^2)$	z_3	x'_5, y'_5, z'_5
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$x_{997}, y_{997}, z_{997}$	$z_{\mu_{15}}, z_{\sigma_{15}}$	$z_{15} \sim \mathcal{N}(z_{\mu_{15}}, z_{\sigma_{15}}^2)$	z_{15}	$x'_{997}, y'_{997}, z'_{997}$
$x_{998}, y_{998}, z_{998}$	$z_{\mu_{16}}, z_{\sigma_{16}}$	$z_{16} \sim \mathcal{N}(z_{\mu_{16}}, z_{\sigma_{16}}^2)$	z_{16}	$x'_{998}, y'_{998}, z'_{998}$
$x_{999}, y_{999}, z_{999}$				$x'_{999}, y'_{999}, z'_{999}$
$x_{1000}, y_{1000}, z_{1000}$				$x'_{1000}, y'_{1000}, z'_{1000}$

5.2.2 Training VAE model

A VAE mirror network architecture with four hidden layers is used for training. The VAE architecture for the blended wing body aircraft is shown in <Figure 5-9>. Each BWB aircraft wing configuration contains 200 coordinates per section, totaling 3,000 points. These coordinates are first normalized using Eq. 5.2, and the flattened coordinates (X) are mapped to the latent space $z \in \mathbb{R}^k$ via the encoder $Q_\phi(z|X)$. The decoder $P_\theta(X'|z)$ then reconstructs the surface coordinates X' from the latent vectors. The latent space is sampled from a Gaussian

distribution parameterized by the encoder's output mean z_μ and variance z_σ . During training, we minimize the loss function using Eq. (3.8), as described in Chapter 3, Section 3.2.1, by optimizing the trainable parameters ϕ and θ , corresponding to the encoder and decoder, respectively. The model is implemented in TensorFlow[100] using the Adam optimizer. A VAE architecture is shown in <Table. 5-3>.

The input, latent space, and output structure of the BWB VAE are summarized in <Table 5-2>. The encoder (Q_ϕ) receives the sectional airfoil surface coordinates represented in Cartesian form $[x, y, z]_{200}$ for each of the five sections shown in <Figure 5-8>, resulting in a total input dimensionality of 3,000 coordinates ($\mathbb{R}^{3000}$). The encoder maps this geometry into two k -dimensional latent vectors, z_{μ_k} and z_{σ_k} , where $k = 16$, as determined by the evaluation metrics in Section 5.2.3.2. The latent space provides a compact representation of the airfoil shape, modeled as a multivariate Gaussian distribution $z_k \sim \mathcal{N}(z_{\mu_k}, z_{\sigma_k}^2)$. The decoder (P_θ) reconstructs the airfoil surface coordinates $[x', y', z']_{200 \times 5}$ (also in $\mathbb{R}^{3000}$) from a sampled latent vector z_k . This reconstructed output represents the decoded BWB wing geometry produced by the trained VAE.

Appropriate hyperparameter selection is crucial for efficient reconstruction of aircraft geometry and for generating new designs. In a VAE architecture, aside from the network's layers and units, the most important hyperparameters are the number of latent dimensions (k) and the regularization coefficient (β). Choosing suitable values for k and β encourages a disentangled latent space that learns an interpretable representation of the design space, enabling accurate reconstruction. In such a setting, each latent variable becomes sensitive to a specific geometric factor while remaining relatively insensitive to others. First, the latent dimensions (k) are selected, and then the regularization coefficient (β) is chosen for the fixed latent dimension (k) to achieve a disentangled

latent representation, as explained in Section 5.2.3.

<Table 5-3> Optimal hyperparameter

Hyperparameter	Encoder (Q_ϕ)/Decoder(P_θ)
Hidden layer	4
Hidden units	1024, 512, 512, 215
Activation	relu, relu, relu, relu
Num. latent dims (k)	16
β	1×10^{-6}
Batch size	64
Learning rate	1×10^{-3}
Epochs	10000

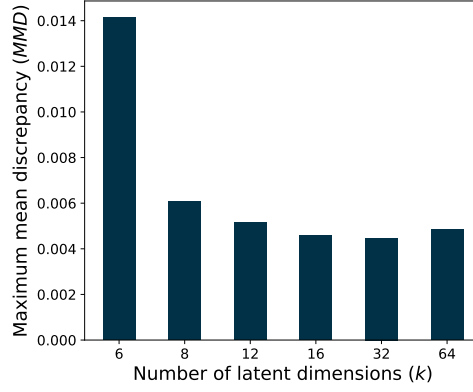
5.2.3 Evaluation of generative model

This section evaluates the latent representation learning model (VAE) according to the evaluation metrics mentioned in Chapter 3, Section 3.2.2, to achieve a disentangled latent representation. The evaluation is performed using the metrics introduced in the methodology section.

5.2.3.1 Effect of number of latent dimensions (k) on generative quality

To select an appropriate latent dimension, we first trained a VAE with varying latent dimensions (k), fixing β at 1×10^{-6} and using a mirrored network architecture with four hidden layers, as summarized in <Table 5-3>. The maximum mean discrepancy (MMD; Eq. (3.10)) was used as the selection metric, where a lower MMD indicates better generative quality. The variation of MMD across latent dimensions is shown in <Figure 5-10>. The results show that very small k values yield larger MMD, while beyond $k = 16$ the MMD stabilizes, even when k increases to 32 and 64.

A low latent dimension (e.g., $k = 6$) results in a VAE with insufficient ca-



<Figure 5-10> Effects of number of latent dimensions (k) on maximum mean discrepancy (MMD)

capacity to control the latent space, leading to poor generative quality and failure to capture key aircraft design features. In contrast, higher latent dimensions ($k \geq 16$) achieve better generative quality, as indicated by the lower MMD values. This suggests that a moderate latent dimension is sufficient to capture the essential features for generating realistic designs. Based on this trade-off, $k = 16$ was selected for subsequent analysis of the latent representation.

5.2.3.2 Effect of regularization coefficient (β) on reconstruction quality and disentanglement

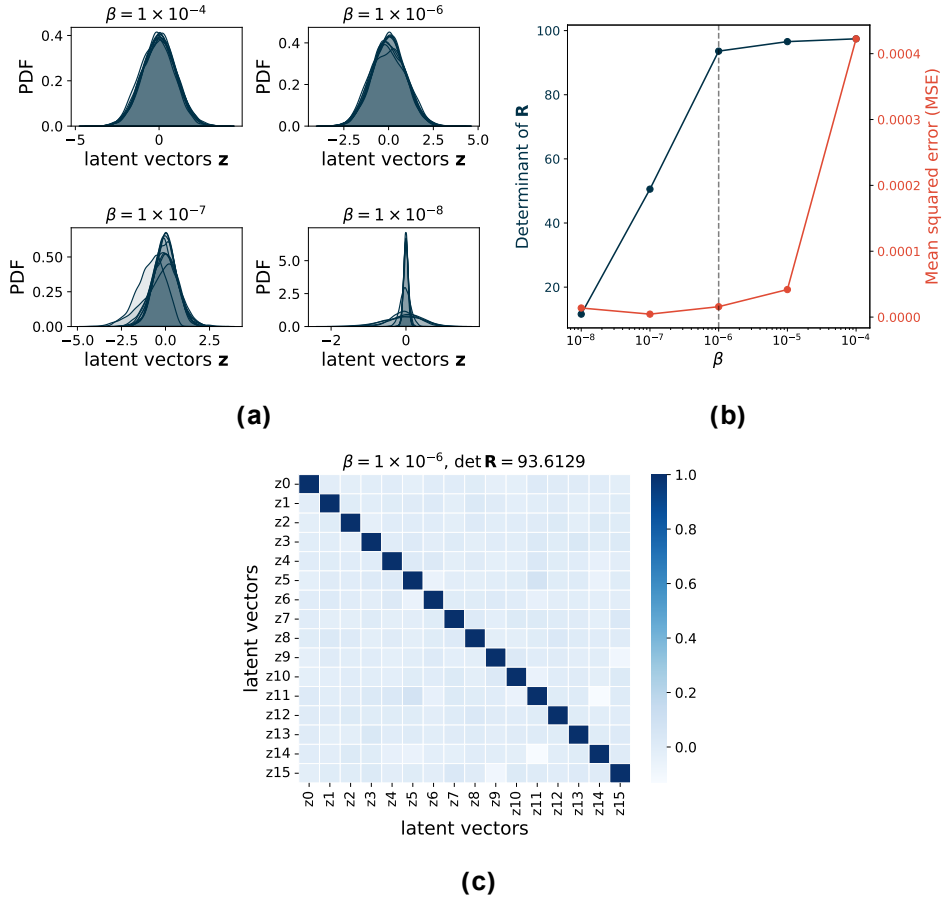
We describe the effects of β on the performance of the VAE according to the evaluation metrics. We use the determinant of correlation ($\det\mathbf{R}$) and reconstruction error (MSE) as evaluation metrics. Specifically, we consider $\beta = 1 \times 10^{-4}$, 1×10^{-5} , 1×10^{-6} , 1×10^{-7} , 1×10^{-8} . As shown in <Figure. 5-11a>, increasing β results in a more regularized latent space that aligns more closely with a standard normal distribution. When a very small value, such as $\beta = 1 \times 10^{-8}$, is used, the latent distribution becomes sharply peaked, indicating poor regularization. In contrast, a larger value like $\beta = 1 \times 10^{-4}$ produces

a distribution that more closely resembles the desired Gaussian prior, promoting better generalization. However, larger values of β lead to increased MSE, resulting in poorer reconstruction quality, as shown in <Figure. 5-11b>. Additionally, a higher value of $\det \mathbf{R}$ implies a more expressive and disentangled latent space, which tends to occur with lower β values. Considering the trade-off between MSE and $\det \mathbf{R}$, we selected $\beta = 1 \times 10^{-6}$, which achieves an MSE of 1.5×10^{-5} and a determinant value of $\det \mathbf{R} = 93.6129$, providing a good balance between reconstruction quality and disentanglement of the latent space. <Figure. 5-11c> presents the correlation matrix of the latent variables, where the strong diagonal dominance confirms that the latent dimensions are largely uncorrelated.

The convergence of the VAE, as presented in <Table 4-2>, is illustrated in <Figure. 5-12> for both the training and validation datasets. At the beginning of training, the total loss starts at approximately 0.016. After 300 epochs, it rapidly decreases to the order of 10^{-4} and subsequently stabilizes for both the training and validation datasets, as shown in <Figure. 5-12a> for the total loss, <Figure. 5-12b> for the reconstruction loss, and <Figure. 5-12c> for the KL divergence loss.

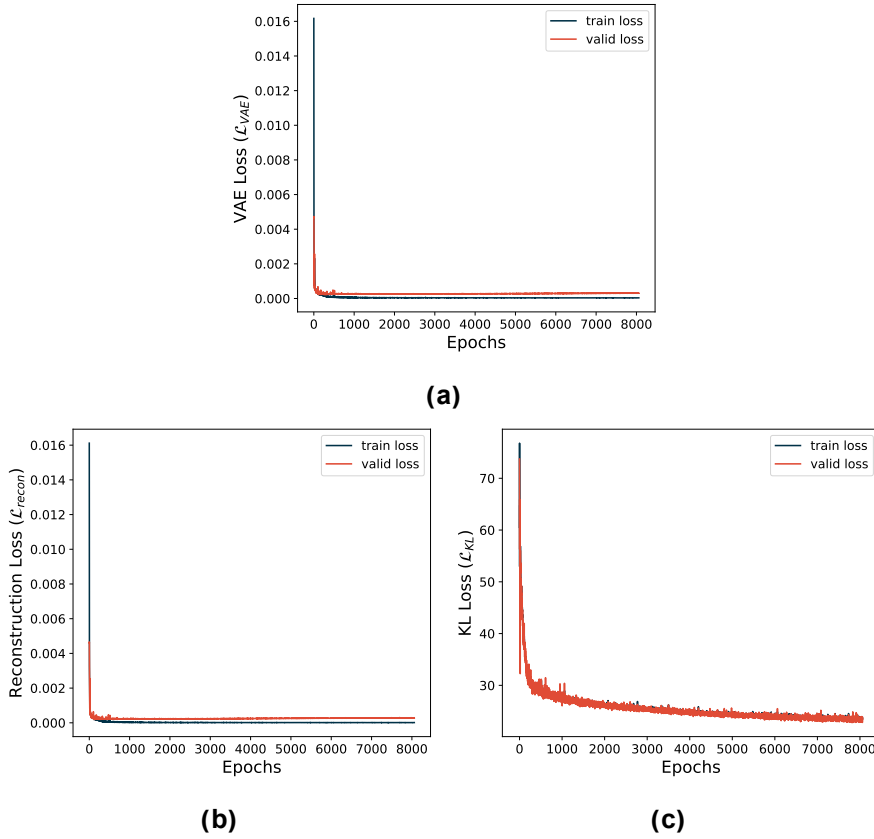
5.2.3.3 Verification of evaluation metrics

Here, we verify our method by comparing it with existing studies on 3D shape representation learning that utilize generative models. SDM-NET [125], which learns 3D representations of airplanes, cars, and tables for deformable mesh structures; MS-GAN [73], a point cloud-based GAN model for propeller blade generation; and AE-BUFE [75], which employs autoencoders as a parameterization model for aircraft shape design from point cloud data, are selected as comparison studies. These methods are the most similar in the open literature.



<Figure 5-11> Selection of hyperparameters by fine tuning optimized network architecture: (a) PDF of latent variables from different β ; (b) Effects of β on $\det \mathbf{R}$ and MSE (c) Correlation matrix for the latent variables at $\beta = 1 \times 10^{-6}$

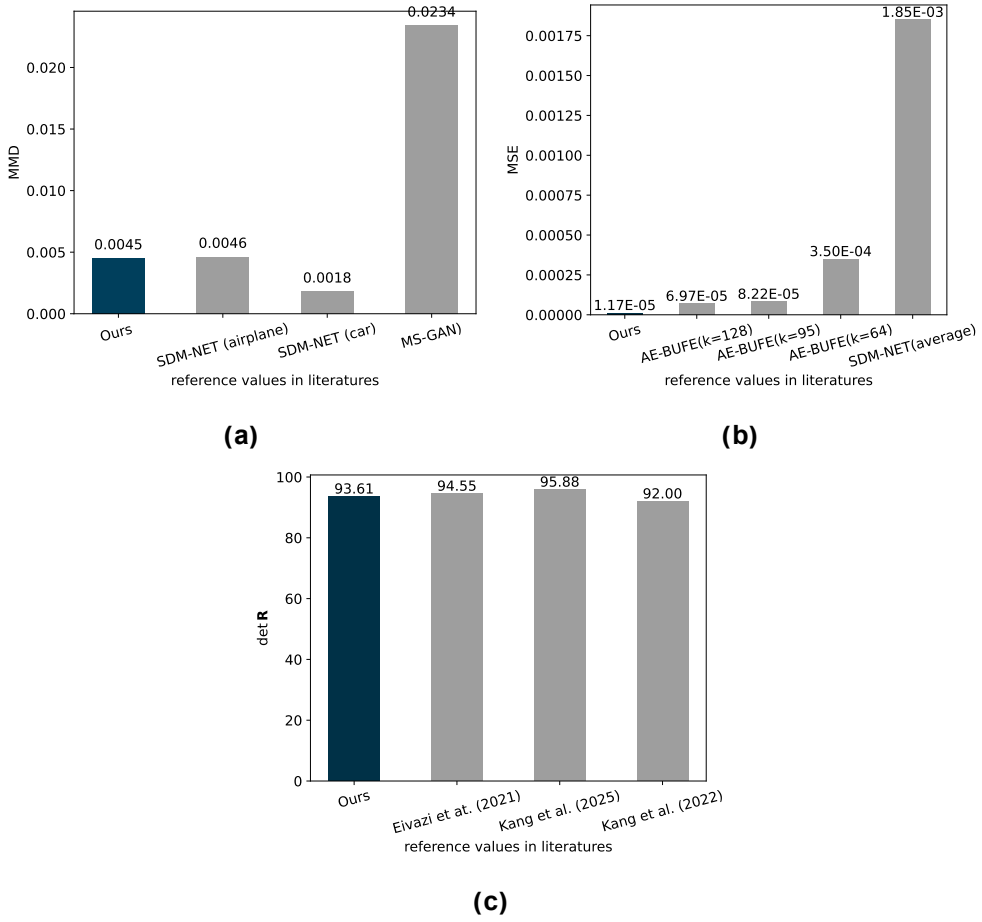
The MMD and MSE metrics are compared in <Figure 5-13a> and <Figure 5-13b>, respectively. Furthermore, Eivazi et al. [95], Kang et al. [71], and Kang et al. [102] are referenced for comparison of the latent correlation coefficient in <Figure 5-13c>. The comparison results indicate that the MMD value of the proposed model is comparable to those reported in the SDM-NET (airplane and car) studies, while the MS-GAN exhibits a significantly larger MMD value. This demonstrates that our VAE model achieves better generative quality than the SDM-NET (car) and MS-GAN models. In addition, the MSE results show



<Figure 5-12> Convergence of VAE training of optimal model: (a) Total VAE loss $\mathcal{L}_{VAE}$; (b) Reconstruction loss $\mathcal{L}_{recon}$ (c) KL divergence loss $\mathcal{L}_{KL}$

that our model provides better reconstruction accuracy compared to AE-BUFE, although AE-BUFE ($k = 128$) outperforms all other methods. The determinant of the correlation matrix ($\det \mathbf{R}$) also shows good agreement with existing literature, achieving a value of 93.610, which is comparable to that of the airfoil VAE model reported in [102].

These results confirm the reliability and robustness of the proposed generative model for subsequent applications and future investigations.



<Figure 5-13> Comparison of evaluation metrics with existing literature: (a) Maximum mean discrepancy (MMD); (b) Reconstruction error (MSE) (c) Determinant of correlation (det $\mathbf{R}$)

5.2.4 Performance of VAE model

In this section, we present the performance of the generative model in terms of reconstruction quality (Section 5.2.4.1) and disentanglement (Section 5.2.4.2).

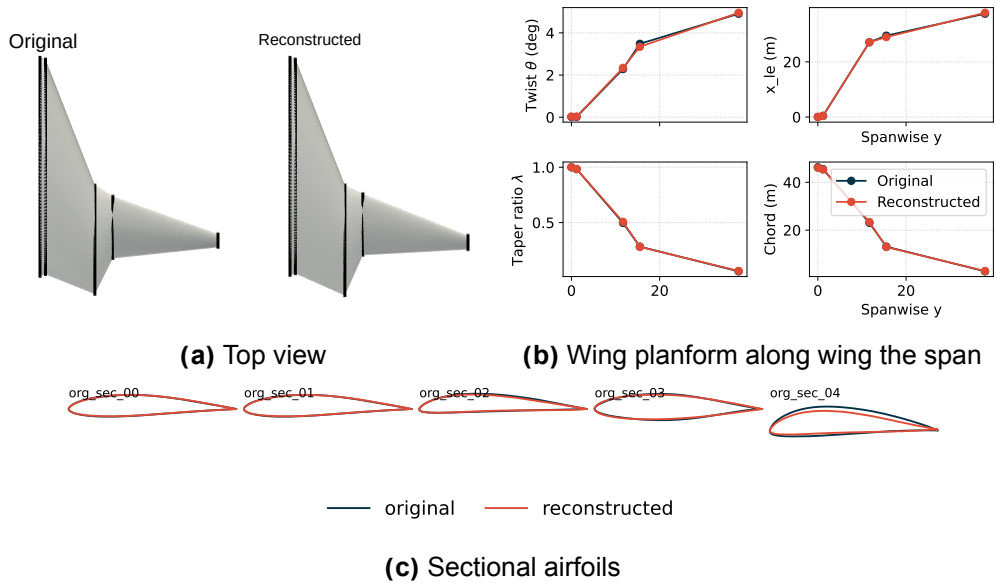
5.2.4.1 Reconstruction quality

This section shows how our VAE model learns to reconstruct the original design space. We randomly selected three aircraft from the test dataset, fed them to

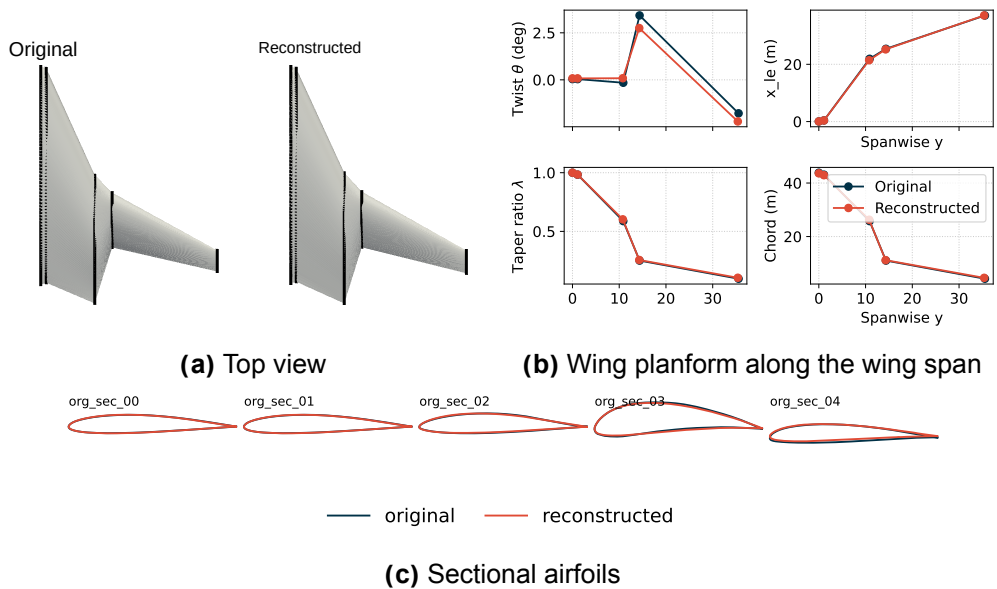
the VAE model, and reconstructed them in the original design space. We compare the whole geometry, sectional airfoils, and planform data along the span to provide an intuitive explanation for the reader, as shown in <Figures 5-14>, 5-15, and 5-16. The results indicate that although the reconstruction error in the datasets is close to 1×10^{-4} , the sectional planform exhibits some discrepancies—particularly for the twist angle (θ_i) distribution along the span and the reconstruction of airfoils at the wing tip. We also evaluated both the training and testing datasets with respect to reconstruction error (MSE). The reconstruction error results are shown in <Table 5-4>. The results indicate that our generative model can reconstruct the original design space well, with only small discrepancies. These could potentially be reduced by increasing the dataset size or training with a different type of machine learning model; however, we use this model for the remainder of our research.

<Table 5-4> Reconstruction quality of VAE

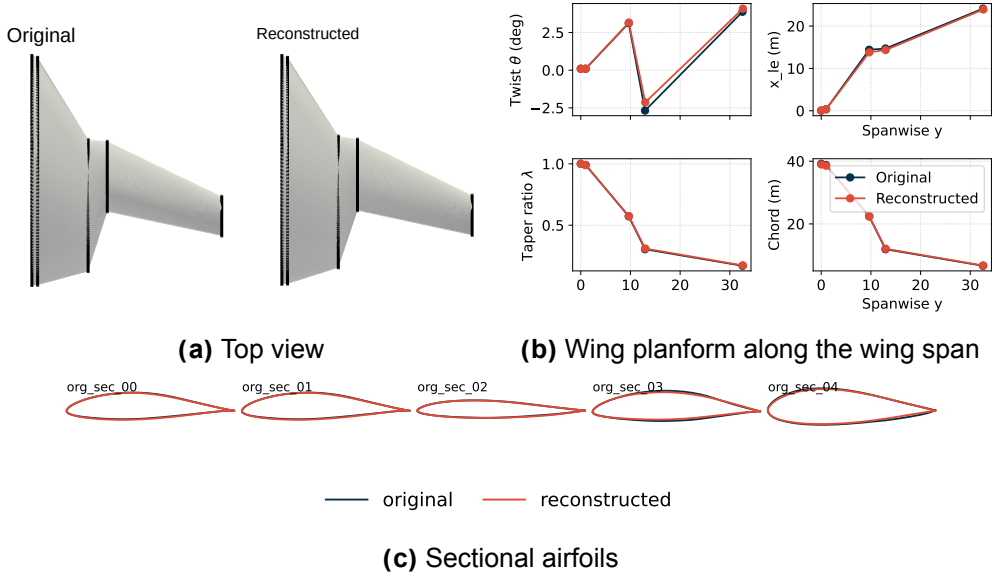
Metrics	Training dataset	Testing dataset
MSE $\times 10^{-4}$	0.117	3.105



<Figure 5-14> Randomly selected wing in the dataset showing comparison between original design (X) and reconstructed design (X')



<Figure 5-15> Randomly selected wing in the dataset showing comparison between original design (X) and reconstructed design (X')



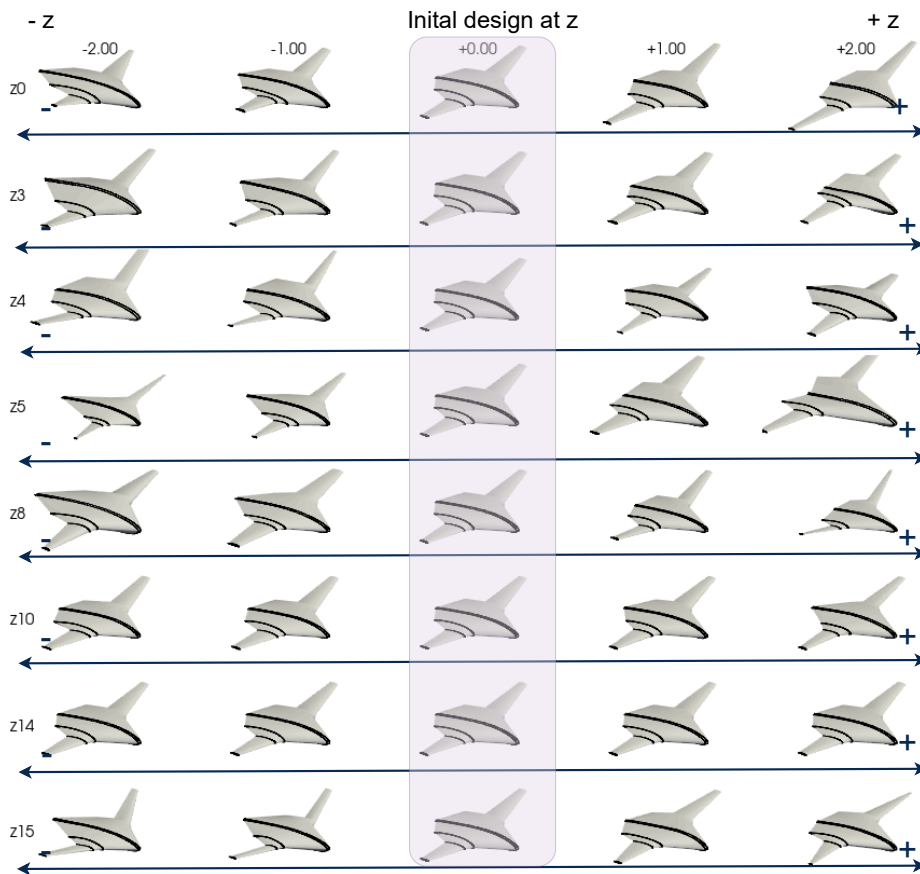
<Figure 5-16> Randomly selected wing in the dataset showing comparison between original design (X) and reconstructed design (X')

5.2.4.2 Disentanglement

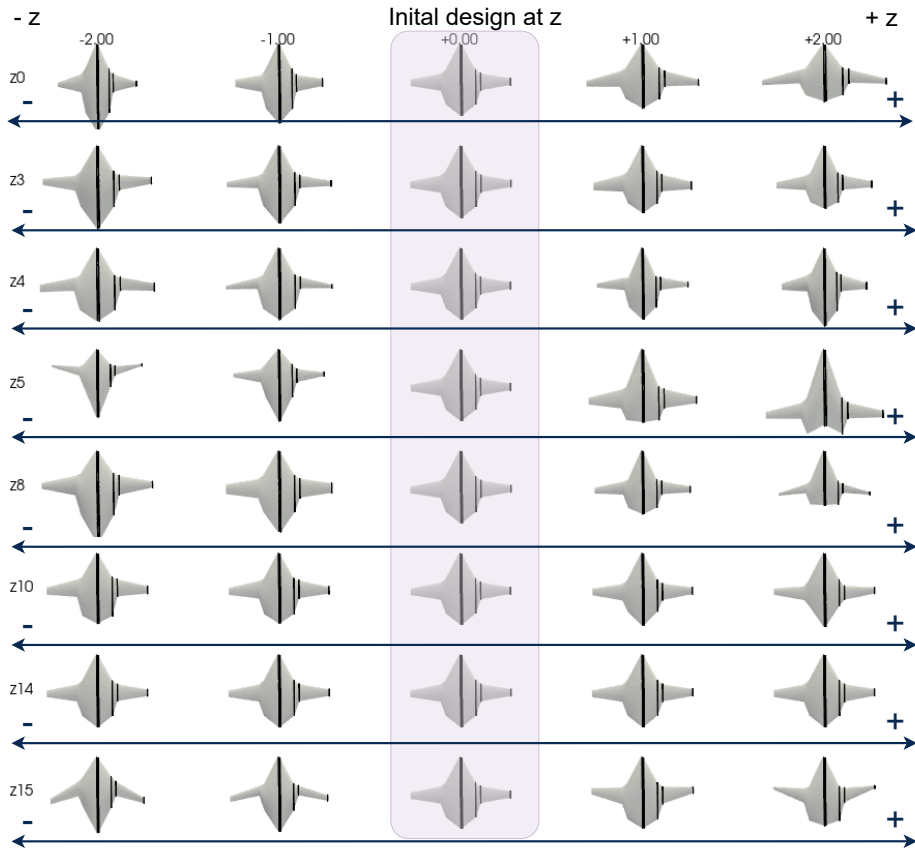
Here, we evaluate the disentanglement of the latent space, demonstrating that each latent vector is independent and interpretable with respect to the original data. Traversals through the latent space along different axes give rise to significantly different changes in aircraft geometry, which strongly affect aerodynamic performance. <Figure 5-17> shows an example of moving along eight different latent axes; each axis represents changes in the design geometry. The middle design is the initial design (X), which we input to the encoder ($Q_\phi(z|X)$) to obtain the latent design variables (z) for that aircraft. We then incrementally change each latent dimension while keeping the others fixed at the initial design and use the decoder ($P_\theta(X'|z)$) to generate the modified design in the original space (X). As shown in <Figures 5-17> and 5-18, the first latent dimension represents changes in root chord and span, the fifth latent dimension affects the position of the wing, and the final dimension represents the

sweep angle of the wing geometry, while the airfoil coordinates change along the axis.

This process demonstrates that our VAE model learns a disentangled, lower-dimensional representation that can be used to present design deformations in a way that admits human interpretation. Aircraft planform geometry and surface coordinates can be systematically altered, which is useful for downstream tasks such as identifying different geometric features and their aerodynamic performance, or for parametrization methods in design optimization using the lower-dimensional latent space.



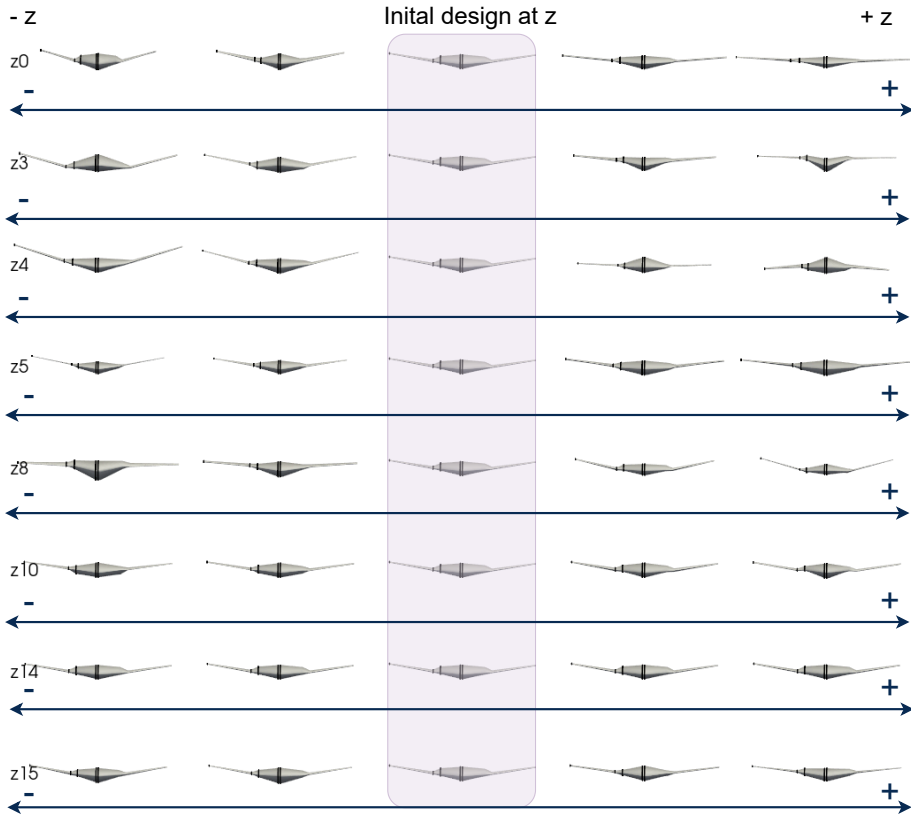
<Figure 5-17> Designs synthesized by varying the latent dimensions. ($k = 16$), Each row is a traversal of a single latent dimension (Isometric view)



<Figure 5-18> Designs synthesized by varying the latent dimensions. ($k = 16$), Each row is a traversal of a single latent dimension (Top view)

5.2.5 Sensitivity analysis

Sensitivity analysis is conducted to identify the effect of each latent variable on the aircraft's aerodynamic performance and geometric characteristics, including both planform and airfoil shape parameters. The percentage change is calculated using Eq. (5.4) by varying each latent variable within the range of $\mu \pm \sigma$ across 21 cases. Here, y_0 denotes the response evaluated at the mean latent variables, while y_{new} represents the response obtained after perturbing each

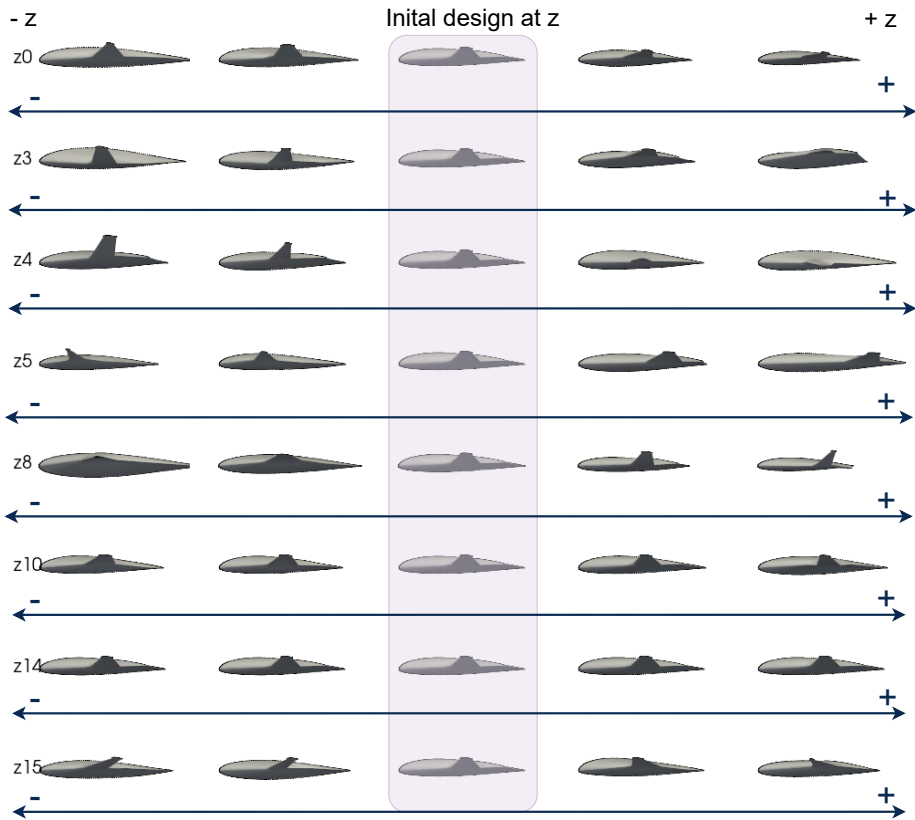


<Figure 5-19> Designs synthesized by varying the latent dimensions. ($k = 16$), Each row is a traversal of a single latent dimension (Front view)

latent variable. Because the latent variables (z) exhibit nonlinear relationships with the responses (y), the maximum and minimum percentage changes are extracted from the analysis results, and the corresponding tornado diagrams are presented for visualization.

$$\Delta\% = \frac{y_{new} - y_0}{|y_0|} \times 100 \quad (5.4)$$

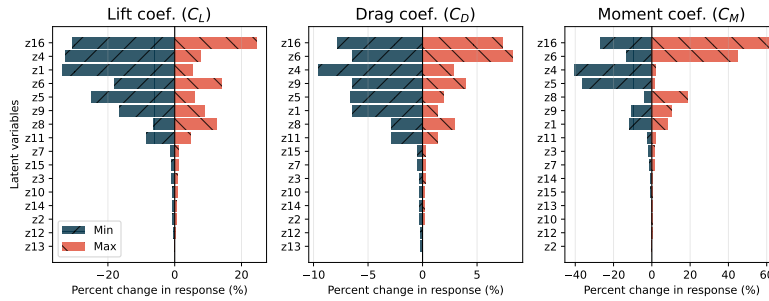
The aerodynamic response sensitivities of C_L , C_D , and C_M are illustrated in <Figure 5-21>. The lift coefficient is strongly influenced by z_{16} , z_4 , z_1 , z_6 , z_5 , z_9 , z_8 , and z_{11} , exhibiting maximum variations within $\pm 20\%$. These latent variables



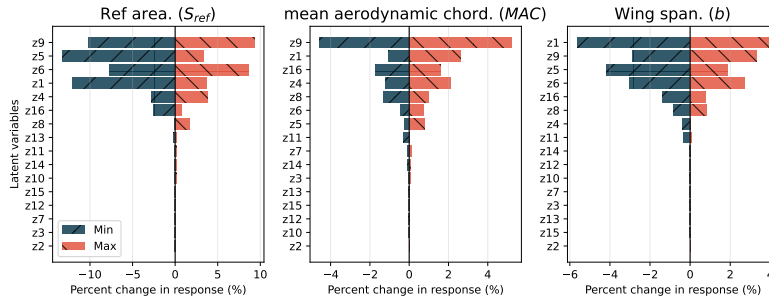
<Figure 5-20> Designs synthesized by varying the latent dimensions. ($k = 16$), Each row is a traversal of a single latent dimension (Side view)

correspond to shape and planform deformations that modify the camber, chord, and sectional area, all of which directly affect the lift distribution, as shown in <Figure 5-23>. The drag coefficient is highly sensitive to z_{16} , z_6 , z_4 , z_9 , z_5 , z_1 , and z_8 , with variations within $\pm 10\%$, as these variables induce changes in thickness distribution, camber, and planform sweep angles that affect friction and wave drag. Meanwhile, the moment coefficient demonstrates the strongest sensitivity among the aerodynamic metrics, with variations reaching $\pm 60\%$. This heightened sensitivity is primarily attributed to the effects of twist (θ) deformations and spanwise camber changes caused by z_{16} , z_6 , z_4 , and z_5 , which

directly modify the pitching moment distribution across the aircraft wing.



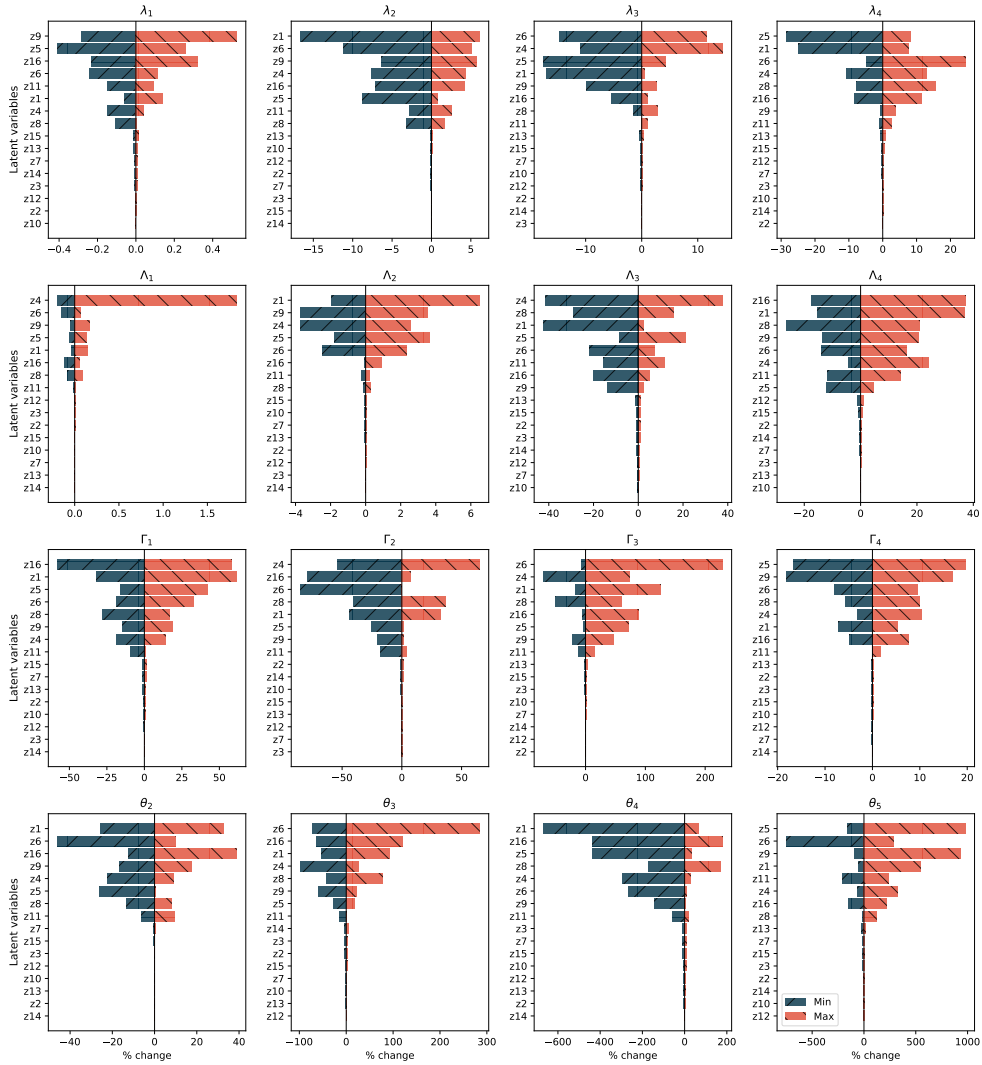
<Figure 5-21> Sensitivity of latent variables on aerodynamic parameters. Tornado diagrams illustrate the percentage change in each response variable when individual latent variables are perturbed within $\mu \pm \sigma$



<Figure 5-22> Sensitivity of latent variables on wing parameters. Tornado diagrams illustrate the percentage change in each response variable when individual latent variables are perturbed within $\mu \pm \sigma$

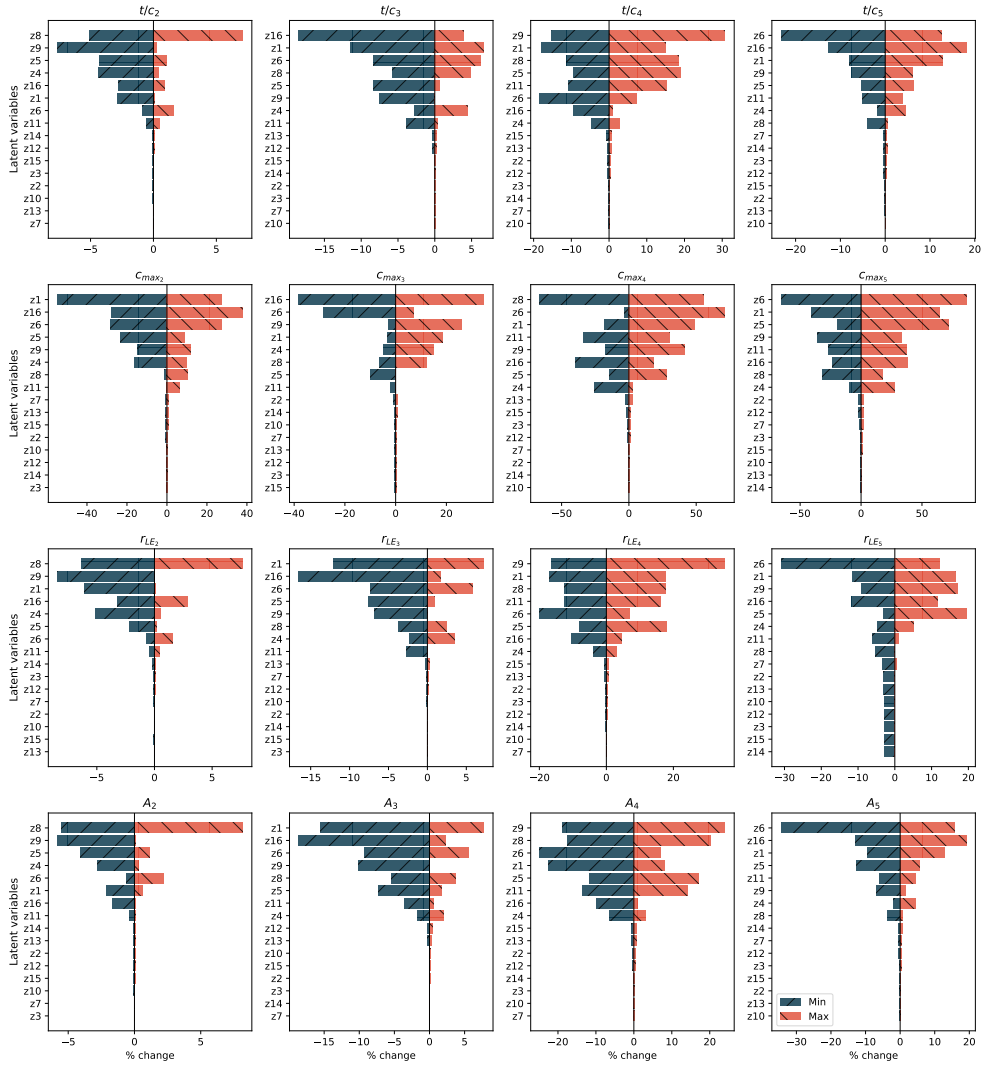
The aerodynamic sensitivity analysis results closely align with the underlying geometric sensitivities shown in <Figure 5-23> and <Figure 5-24>. These same dominant latent variables produce substantial variations in global plan-form parameters, as illustrated in <Figure 5-22>, including reference area ($\pm 10\%$), mean aerodynamic chord ($\pm 4\%$), and wingspan ($\pm 4\%$), reflecting their role in governing spanwise and chordwise distribution.

Notably, the leading-edge sweep angle $\Lambda_1 - \Lambda_4$ is primarily influenced by latent variables z_1, z_4, z_6, z_9 , which reshape the forward and aft displacement of sectional leading-edge positions. These modifications alter the effective wing



<Figure 5-23> Sensitivity of latent variables on wing planform parameters.

sweep, aerodynamic center, and overall spanwise loading, which are associated with observed changes in aerodynamic parameters. The sectional BWB planform parameters are also dominated by similar latent dimensions, showing significant variations—including taper ratio ($\lambda_1 - \lambda_4$), dihedral angles ($\Gamma_1 - \Gamma_4$), and twist angles ($\theta_2 - \theta_5$)—with the outboard sections showing the highest sen-



<Figure 5-24> Sensitivity of latent variables on sectional airfoil shape parameters.

sitivity. Airfoil sectional shape parameters also respond strongly to latent perturbations: thickness-to-chord ratio (t/c) varies by ± 20 – 30% , maximum camber C_{max} and its location vary by ± 20 – 40% , and section area and leading edge radius r_{LE} show moderate but meaningful variations up to $\pm 10\%$. These shape changes explain the pronounced aerodynamic sensitivity in C_L , C_D , and C_M ,

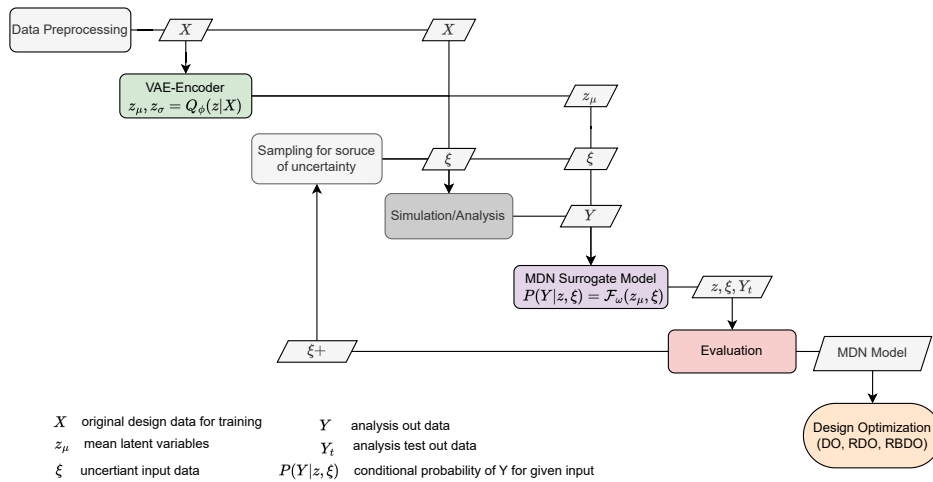
which are highly influenced by camber, thickness, and twist variations.

In contrast, while several latent variables dominate planform and shape parameters, some—such as z_2 , z_7 , z_{10} , z_{14} —produce the lowest sensitivity changes across aerodynamic, planform, and sectional airfoil shape parameters. This indicates that they represent subtle or redundant geometric variations with limited aerodynamic impact. Overall, the results confirm that the learned latent space is physically interpretable and dominated by a set of active latent variables that capture sweep, twist, taper, dihedral, and sectional airfoil-shape changes, validating its suitability for surrogate modeling and design optimization.

5.3 Probabilistic surrogate modeling from latent space

We present a combination of dimensionality reduction and probabilistic surrogate modeling methods to accurately predict uncertainties arising from flight conditions, even with limited data. This section describes the modeling of a probabilistic surrogate model from the lower-dimensional latent space, which is used to predict aerodynamic performance and its uncertainty. This approach reduces the computational cost of simulation models and accelerates the optimization process, as discussed in Chapter 3, Section 3.3. The process and data dependencies for surrogate modeling are shown in <Figure 5-25>.

In the surrogate modeling process, selecting an appropriate analysis tool is a crucial step, as it directly affects both the accuracy of the aerodynamic performance data and the associated computational cost. First, the selection of the analysis tool is described in Section 5.3.1.1, followed by the training and evaluation of the surrogate model in Section 5.3.2, and the uncertainty quantification using the trained model in Section 5.3.3.



<Figure 5-25> Process and data dependencies for surrogate modeling

5.3.1 Analysis tools and data generation

5.3.1.1 Aerodynamics analysis tool

<Table 5-5> Comparison of aerodynamic analysis approaches [126]

Primary Use	Accuracy (Average)	Methods
Conceptual design	$\pm 15\%$	DATCOM, Friction2K, etc.
Preliminary design	$\pm 10\%$	VSPAero, XFLR5, AVL, etc.
Detailed design	$\pm 5\%$	SU2, Fluent, ADflow, OpenFOAM, etc.

This section explains the selection of aerodynamic analysis tools and provides a brief description of the tools used for surrogate modeling preparation.

Computational aerodynamic tools can be classified into two categories: low-fidelity and high-fidelity methods. Low-fidelity methods, which use simplified assumptions about flow equations, can calculate aerodynamic data within seconds. Examples include semi-empirical and potential theory-based approaches. High-fidelity methods refer to CFD, which solve the complete Navier–Stokes equations for detailed geometries. A comparison of computational meth-

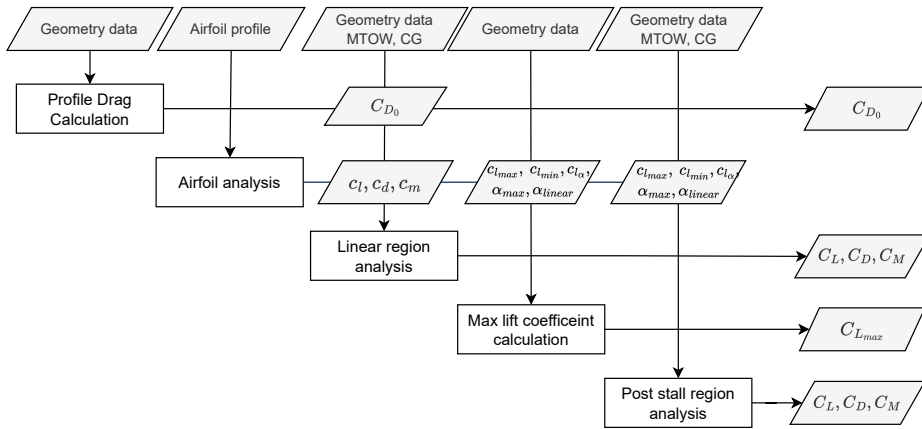
ods, design stages, and accuracy is shown in <Table 5-5>. As shown, in the preliminary design stage, potential flow methods are suitable within an accuracy of $\pm 10\%$. These methods include VSPAero[127], based on the panel method and is able to analyze the exact shape of an aircraft without the thin lifting-surface assumption; thus, the effect of thickness is not neglected. The panel method solves potential-flow equations (simplified Euler equations)[27]. Another method is the Vortex Lattice Method (VLM), which models lifting surfaces such as the wing and tail as an infinitely thin sheet of discrete vortices to compute lift and induced drag. Effects of thickness and viscosity are neglected, e.g., PyTornado[128]. A combined VLM and 3D analysis tool is also available for analyzing the exact aircraft shape with high accuracy, such as Athena Vortex Lattice (AVL) [129] and XFLR5[130]. AVL has been utilized for many research at the subsonic aircraft analysis and data-driven based design [122], [74], [131], [132] due to its accuracy and fast computation.

A comprehensive aircraft design analysis (CADE) program offers aerodynamics analysis for airfoils, linear-region analysis for low angle of attack, and post-stall region analysis. In CADE-aerodynamics linear-region analysis uses AVL for lift, drag and moment calculation and it's has been validated with CFD data [133]. The process of aerodynamics analysis is shown in <Figure. 5-26>. AVL only consider the lift induced drag (C_{D_i}), we utilized Friction2k[134], to calculate zero-lift drag (C_{D_0}) which include skin friction drag and form drag estimation using build-up method shown in Eq. (5.5).

$$C_{D_0} = \sum_i^N \frac{C_{F_i} F F_i S_{wet_i}}{S_{ref}} \quad (5.5)$$

where N is the number of components used to model the configuration. The form factor, FF , depends on whether the component is modeled as a planar surface or a body of revolution. The skin friction coefficient, CF , is determined

for either laminar or turbulent flow. S_{wet} is the wetted area of the component, and S_{ref} is the reference area of the aircraft.

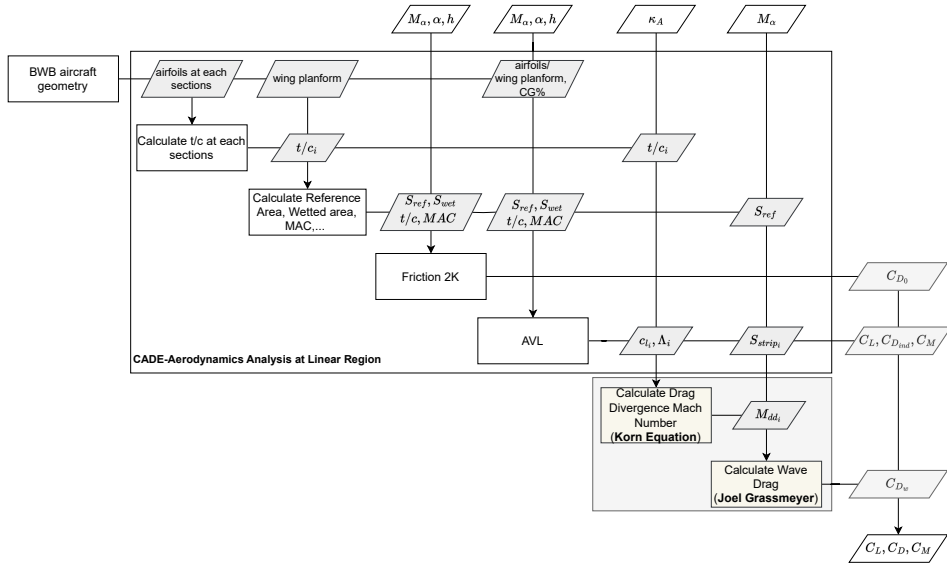


<Figure 5-26> CADE aerodynamics analysis module

Even though, CADE-aerodynamics analysis offer a wide range of angle of attack for flight regions at subsonic speed, wave drag estimation is not provided. Therefore, wave drag calculation module is developed using the method described in [135]. The estimation is calculated by Korn equation. It uses simple sweep theory to account the sweep effect of the wing in addition, wing geometry is divided into number of spanwise strips. Then, the wave drag model estimate as a function of each strips airfoil thickness to chord ratio, section lift coefficient, sweep angle and area of each strips. This method has been applied for the conceptual design of blended wing body aircraft as well as conventional aircraft which are at the transonic flight regions [136]. [137], [138].

The implementation of wave drag estimation is shown in Figure. 5-27. Firstly, aircraft geometry are given to the analysis and each sections thickness to chord ratio, and wing planform parameters are transform to get reference area, wetted and MAC then, Friction 2K perfoms for zero-lift drag calculation (C_{D_0}) and AVL perfoms for lift included drag (C_{D_i}), lift coefficient (C_L) and (C_M) for a given

flight condition at (M_α, α, h) . In addition, CG is assumed to be 25% of the MAC for all BWB aircraft.



<Figure 5-27> Aerodynamics analysis method for data generation

The AVL provides each strips (c_{li}) and sweep angle (Λ_i) are then, feed to Drag-divergence Mach number (M_{dd}) calculation using Korn equation show in Eq. (5.6).

$$M_{dd} = \frac{\kappa_A}{\cos \Lambda_i} - \frac{(t/c_i)}{\cos^2 \Lambda_i} - \frac{c_{li}}{10 \cos^3 \Lambda_i} \quad (5.6)$$

where κ_A is an airfoil technology factor. It is set as 0.89 for the Boeing 747 and 0.955 for the Boeing 777 [139]. In our analysis, $\kappa_A = 0.95$ is used for all cases.

Then, critical Mach number (M_{crit}) is calculated by Eq. (5.7).

$$M_{crit} = M_{dd} - \left(\frac{0.1}{80}\right)^{1/3} \quad (5.7)$$

Finally, wave drag coefficient is calculated by Joel Grassmeyer[139] Equation. (5.8).

$$C_{D_w} = \sum_i^{n_{strip}} 20(M_\alpha - M_{crit_i})^4 \frac{S_{strip_i}}{S_{ref}} \quad (5.8)$$

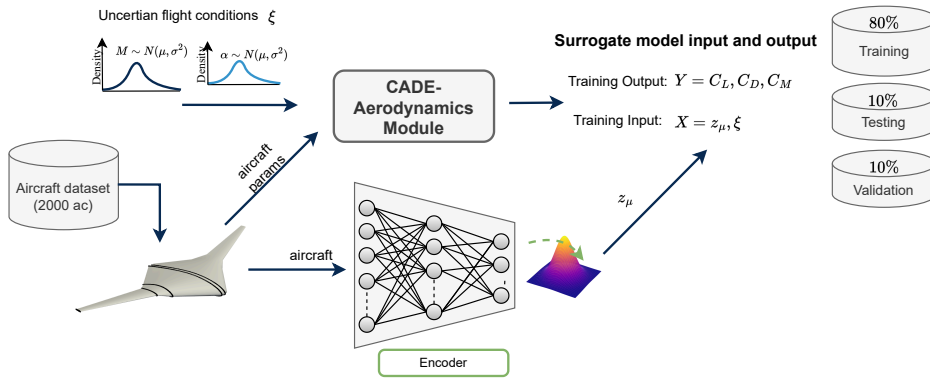
where S_{strip} is the area of each strip, and S_{ref} is the reference area. Then, total drag for aircraft can be calculated by summing induced drag, zero-lift drag and wave drag for aircraft aerodynamics analysis. This method is used for aerodynamics data generation and design optimization.

As our aerodynamic analysis method is strongly developed based on Friction2k, AVL, and wave drag correction equations, we use the following solver settings for data generation and design optimization. We follow the Athena Vortex Lattice Solver (AVL) 3.40 modeling guidelines and numerical recommendations from [129]. According to the manual, the BWB wings were discretized using 12 chordwise ($N_{chord}=12$) and 40 spanwise ($N_{span}=40$) horseshoe vortices, combined with cosine spacing in both directions ($Cspace=1.0$, $Sspace=1.0$), as this is recommended for achieving the best accuracy for a given number of vortices. This discretization strategy places control points near the leading and trailing edges where pressure gradients are strongest. Cosine spacing minimizes numerical instabilities and improves Trefftz-plane wake alignment, which is essential for induced-drag calculation within a vortex-lattice method.

While AVL is fundamentally an incompressible, linear aerodynamic solver, at transonic airspeeds (e.g., Mach 0.8), the Prandtl-Glauert compressibility correction is used along with consideration of the wing's sweep. However, the manual explicitly notes that at high subsonic speeds ($M > 0.7$), linearized methods may become invalid, potentially leading to inaccuracies in the lift-curve slope and moment prediction, especially for wings with low sweep. Therefore, we incorporate a separate wave-drag correction model using Eq. (5.8), ensuring that transonic drag rise and non-linear compressibility effects are ac-

counted for outside AVL's linear model. Although AVL is sensitive to compressibility-induced drag variations, compressibility effects on lift and moment still require additional consideration within our methodology.

5.3.1.2 Data generation for surrogate model



<Figure 5-28> Aerodynamics data generation using encoder and analysis tool

This section describes the process of preparing data for training the probabilistic surrogate model. <Figure 5-28> illustrates the dataset used. From the original VAE training dataset of 5,000 aircraft, we randomly selected 2,000 samples. Since running simulations for multiple cases is computationally expensive, and the surrogate model only requires the relationship between latent vectors and the quantities of interest, we limited the dataset to 2,000 aircraft for surrogate modeling.

The VAE encoder ($P_\theta(z|X)$) was used to obtain the latent variables. To characterize the stochastic properties of the latent space, we used the mean values of the latent variables (z_μ) as inputs to the surrogate model. Each design was also passed to the analysis model, where uncertainties were considered. The uncertain inputs were concatenated with the latent variables to form the surrogate model inputs, while the analysis results served as the surrogate

<Table 5-6> Aerodynamic analysis conditions and number of samples

h	α°	M	Total aircraft	Samples per aircraft
10668 m	$\mathcal{N}(2.0, 0.1^2)$	$\mathcal{N}(0.8, 0.05^2)$	2000	50, 100, 200

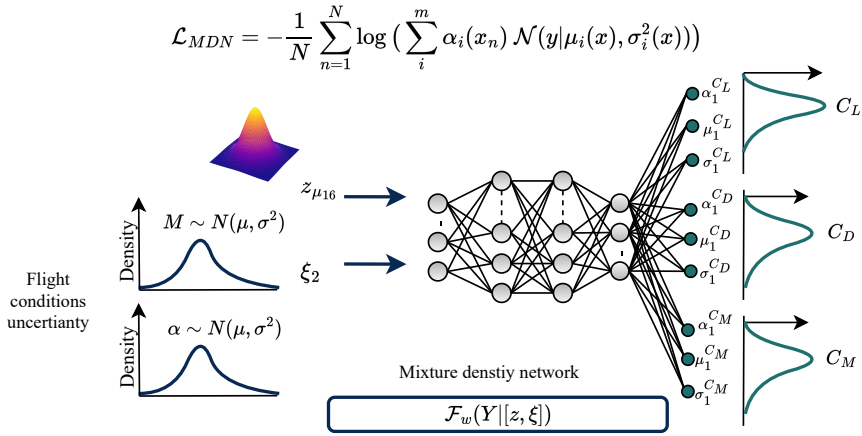
model outputs.

In aerodynamic performance analysis, uncertainties arise from various factors, such as fluctuations in Reynolds number, angle of attack, and Mach number. According to existing literature, angle of attack (α), Mach number (M), and geometric shape parameters are commonly considered as design or uncertain variables [15, 48, 50, 62, 140]. Following this approach, we treat α and M as uncertain parameters and use the MDN to predict the probability distributions of key aerodynamic coefficients (C_L , C_D , and C_M). The uncertain input bounds are taken from [50].

For each design, we sample uncertain parameters from a normal distribution, using different sample sizes to accurately predict the probability density function from the latent space by explicitly modeling uncertainty. The analysis conditions and sample sizes used to evaluate the prediction accuracy of the MDN network are summarized in <Table 5-6>. Note that the nominal flight conditions are derived from [15, 119].

5.3.2 Training and evaluation of surrogate model

We trained the MDN model in predicting the conditional distribution of lift and drag coefficient by explicitly considering α and Mach number as source of uncertainty. The architecture of MDN model is shown in <Figure. 5-29> having latent vector z , 16 dimensions and α and M as inputs while predicting C_L , C_D and C_M . We evaluate MDN performance on test datasets using varying sample sizes of the random variables α and M for each airfoil in <Table 5-6>, with



<Figure 5-29> MDN model training for uncertainty quantification

<Table 5-7> MDN model optimal hyperparameter

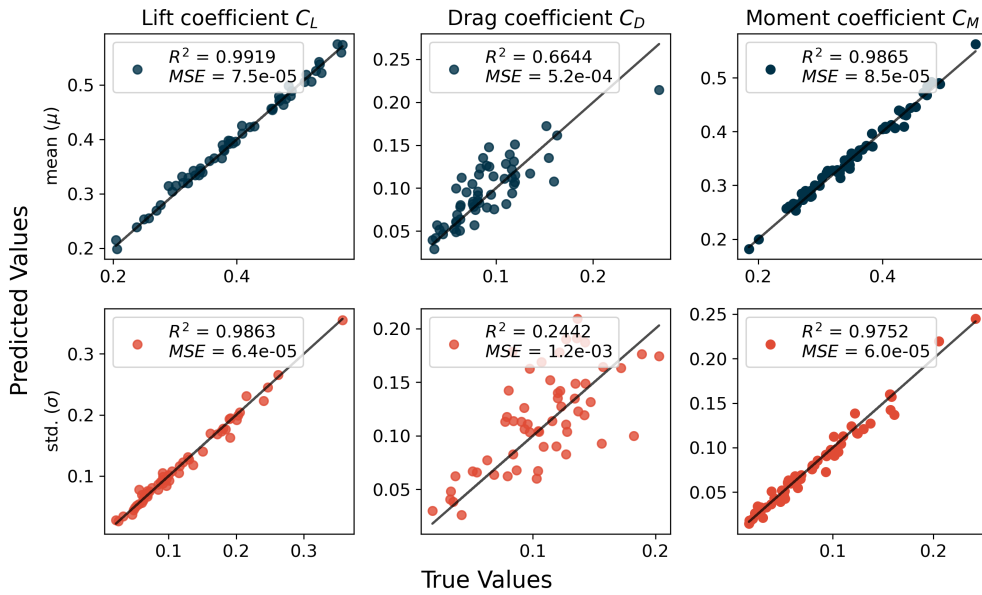
Hyperparameter	MDN($\mathcal{F}_w$)
Hidden layer	4
Hidden units	972, 468, 283, 128, 128
Activation	relu, tanh, relu, relu, relu
Num. of mixtures (m)	1
Batch size	512
Early stopping	200 epochs
Learning rate	1×10^{-4}
Epochs	5000

the optimized architecture from <Table 5-7>. Model predictions for the conditional mean (μ) and standard deviation (σ) were computed using Eq. (3.22) and Eq. (3.23), respectively, under a fixed model structure. Evaluation metrics, as defined in Section. 3.3.2, focused on the accuracy of predicted statistical properties essential for uncertainty quantification.

The prediction results are presented in <Table 5-8>. As expected, larger sample sizes yield better predictions of σ , while μ is accurately predicted across all cases. Using 100 samples, the model achieved R^2 scores of 0.998, 0.989,

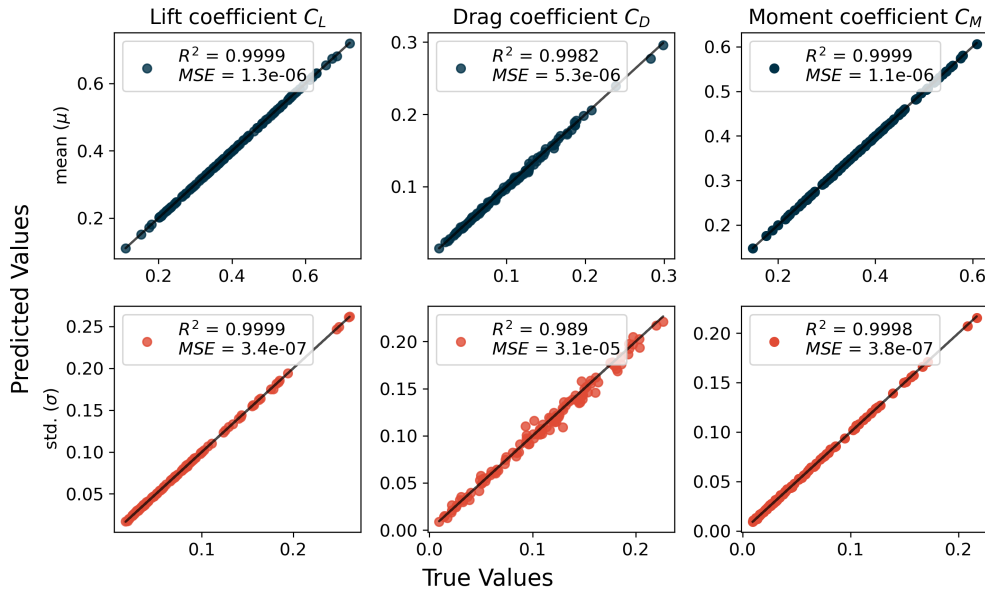
<Table 5-8> MDN model prediction with varying sample sizes

Sample	Metric	C_L		C_D		C_M	
		μ	σ	μ	σ	μ	σ
50	R^2	0.992	0.986	0.667	0.259	0.986	0.975
	$MSE \times 10^{-6}$	73.555	65.249	517.662	1196.245	84.918	59.743
100	R^2	0.999	0.999	0.998	0.999	0.999	0.998
	$MSE \times 10^{-6}$	1.300	0.342	5.229	31.755	5.229	0.38
200	R^2	.9998	0.999	0.998	0.994	0.999	0.990
	$MSE \times 10^{-6}$	1.905	0.236	1.987	13.113	1.369	0.220

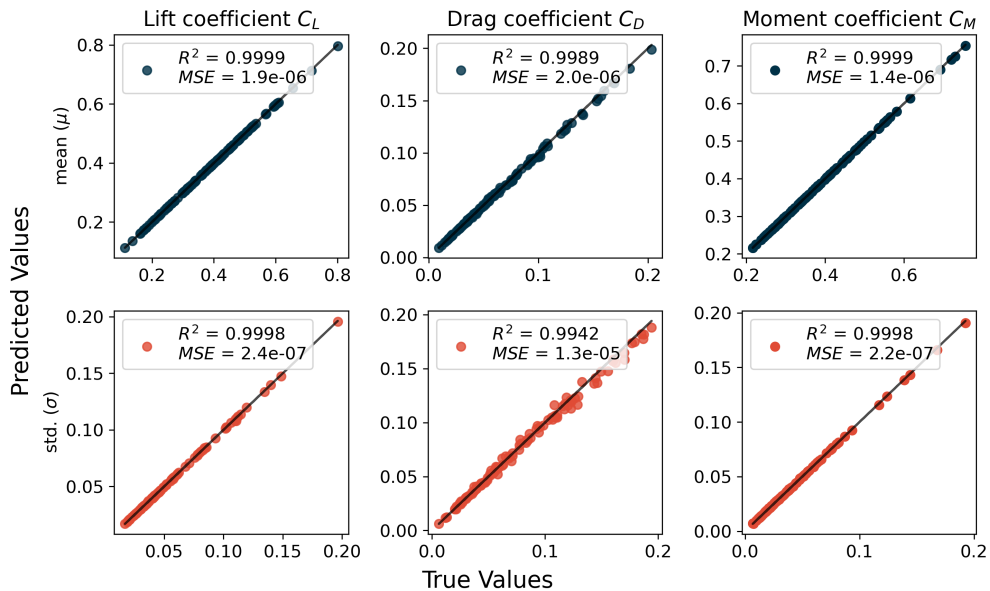


<Figure 5-30> MDN model predictions for the mean and standard deviation under random variables (α , M) for 50 samples

and 0.999 for the predicted standard deviations of the lift coefficient (C_L), drag coefficient (C_D), and moment coefficient (C_M), respectively, as shown in <Figure. 5-32>. The predicted means achieved R^2 scores of 0.99. With 200 uncertain samples per aircraft, the accuracy increased, with mean predictions achieving $R^2 = 0.999$ for all outputs. The C_L standard deviation reached $R^2 = 0.999$, and C_D reached $R^2 = 0.9974$, indicating a well-fitted regression

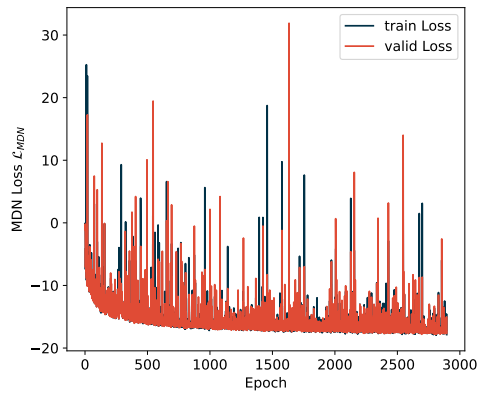


<Figure 5-31> MDN model predictions for the mean and standard deviation under random variables (α , M) for 100 samples



<Figure 5-32> MDN model predictions for the mean and standard deviation under random variables (α , M) for 200 samples

model suitable for downstream tasks in uncertainty quantification for generated aircraft and design variables. The convergence of the MDN model, as presented in <Table 5-7>, is illustrated in <Figure. 5-33> for both the training and validation datasets. The MDN loss converges after approximately 1203 iterations, achieving a negative log-likelihood (NLL) loss of -17.8492 for the training data and -17.660 for the validation data.

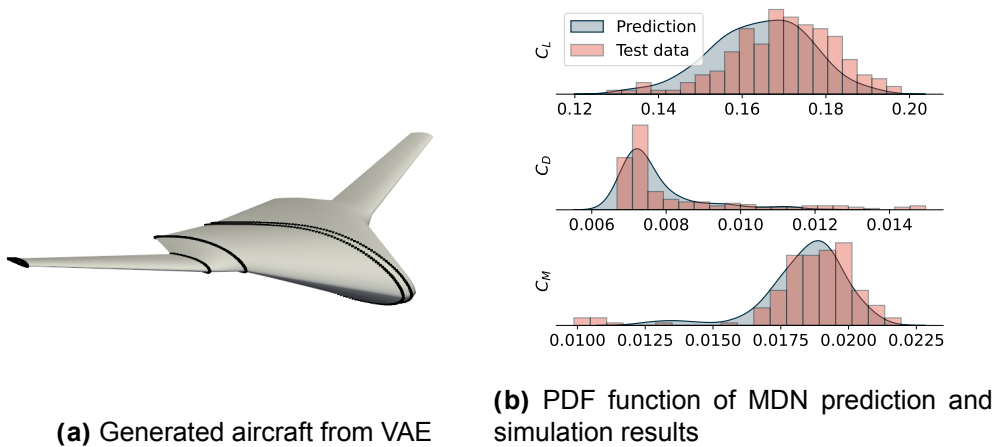


<Figure 5-33> Convergence of MDN training of optimal model for 200 samples

5.3.3 Performance of MDN surrogate model

This section presents the the uncertainty prediction from the lower dimensional latent space using generative model (VAE) and the probabilistic surrogate model (MDN). Aircraft design are generated via the VAE decoder $P_{\theta}(X|z)$, with uncertain inputs α and M taken from the test set and fed into the surrogate mode (MDN) are shown in <Figure. 5-34>, 5-35, 5-36, 5-37 As shown in the figure, the predicted PDFs for lift coefficient (C_L), the drag coefficient (C_D) and moment coefficient (C_M) align well with the histograms obtained from simulation results for all aircraft designs, indicating accurate uncertainty prediction that reflects the inherent variability in flight conditions.

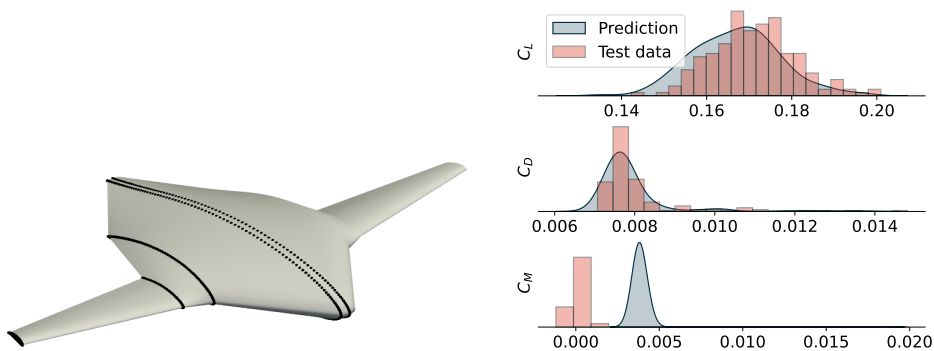
This process demonstrates that the system's stochastic behavior can be accurately predicted by a probabilistic surrogate model operating in a lower-dimensional latent representation, while also enabling continuous generation of new designs at lower computational cost. Together, these results highlight the effectiveness of our deep generative–probabilistic surrogate framework for design under uncertainty in latent space, even with limited data.



<Figure 5-34> Random aircraft generated from latent space comparison of predicted and simulated conditional distribution PDFs for aerodynamic uncertainty

5.4 Optimization case studies

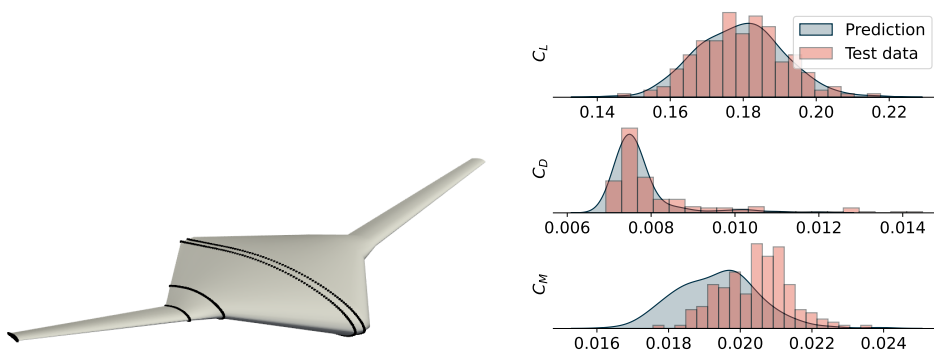
This section demonstrates how our approach overcomes the limitations of conventional aircraft design optimization, such as the curse of dimensionality and high computational cost, by improving both solution quality and computational efficiency. Improvements in solution quality are discussed in Section 5.4.2, where our approach is compared to parametric-based optimization. The reduction in computational cost is presented in Section 5.4.3, with comparisons to



(a) Generated aircraft from VAE

(b) PDF function of MDN prediction and simulation results

<Figure 5-35> Random aircraft generated from latent space comparison of predicted and simulated conditional distribution PDFs for aerodynamic uncertainty

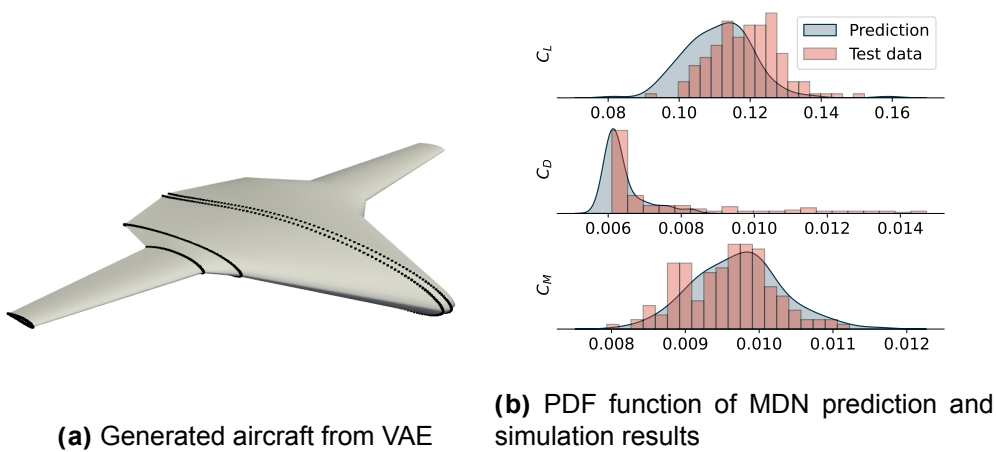


(a) Generated aircraft from VAE

(b) PDF function of MDN prediction and simulation results

<Figure 5-36> Random aircraft generated from latent space comparison of predicted and simulated conditional distribution PDFs for aerodynamic uncertainty

surrogate-based and simulation-based optimization. Furthermore, robust and reliability-based design optimizations are performed to show the effectiveness of the proposed approach.



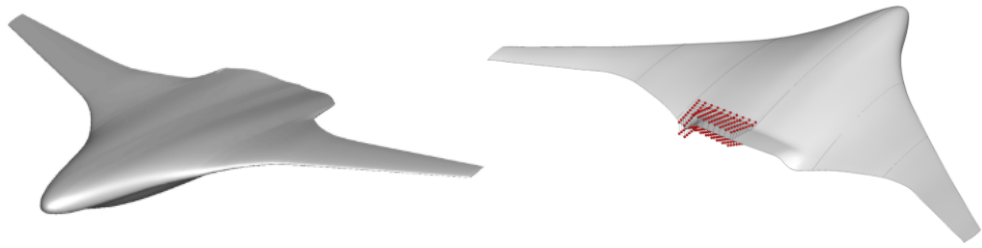
<Figure 5-37> Random aircraft generated from latent space comparison of predicted and simulated conditional distribution PDFs for aerodynamic uncertainty

5.4.1 Defining parametrization methods and design space for fair comparison

In this section, we define the baseline aircraft, parameterization methods, and corresponding design spaces to demonstrate how using a lower-dimensional latent representation can improve solution quality in aerodynamic shape optimization. A blended-wing-body (BWB) configuration [119] is adopted as the baseline design for optimization, targeting improved cruise efficiency at a Mach number of $M = 0.8$, an altitude of 10,668 m, and an angle of attack of $\alpha = 2^\circ$ [15, 119]. The reference BWB configuration and flight conditions used for this study are illustrated in Figure 5-38.

Three different parameterization methods are compared in this section:

- **Parametric method:** Considers only the wing planform design variables in a parametric model with a fixed airfoil across all sections, as shown in Figure 5-5 (14 design variables).



(a) NASA N2A hybrid wing-body [120, 124] (b) Boeing BWB aircraft [141],[119]

<Figure 5-38> Reference BWB aircraft for problem formulation

- **Parametric + VAE-airfoil:** Considers both planform design variables and section-wise airfoil coordinates generated by the VAE-based airfoil model (14 planform design variables + 40 airfoil latent design variables). A constraint-based geometric sampling method (Section 5.2.1.2) is employed.
- **VAE-BWB:** Considers a generative model that learns a lower-dimensional representation of the entire aircraft (16 latent design variables), as described in Section 5.2.2.

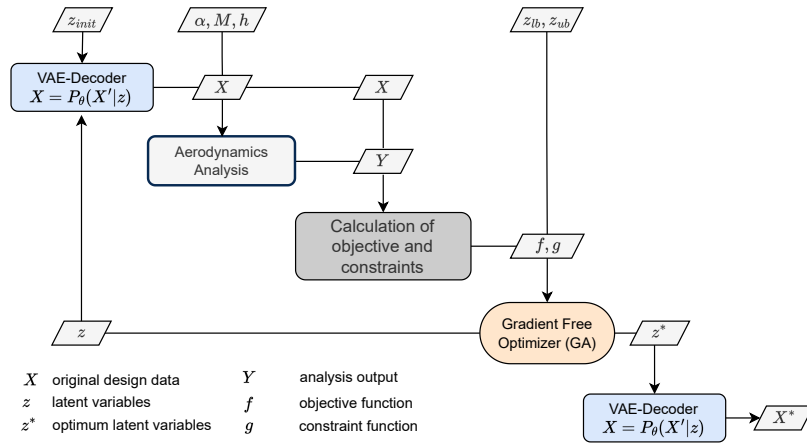
The design space bounds (upper and lower) are derived from <Table 5-1> for the **Parametric method** and **Parametric + VAE-airfoil**, where each sectional VAE-airfoil latent variable is defined within $\mu \pm 2\sigma$, as applied in the constraint-based geometric sampling method. Since the generative model is trained on data generated from this sampling method, which shares the same design space as the **Parametric + VAE-airfoil** case, the **VAE-BWB** method also considers latent variables within $\mu \pm 2\sigma$ for its 16 latent variables.

5.4.2 Quality solutions

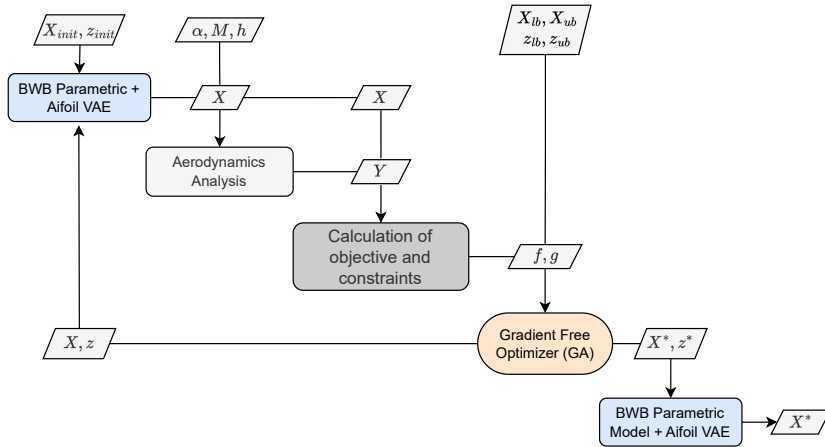
5.4.2.1 Optimization formulations

Here, we aim to maximize the lift-to-drag ratio (C_L/C_D) of the BWB aircraft for cruise flight conditions [119], corresponding to a Mach number of $M = 0.8$, an angle of attack of $\alpha = 2^\circ$, and an altitude of 10,668 m. Note that this optimization process is conducted solely to demonstrate improvements in optimization efficiency; therefore, other factors, such as aircraft stability and mass properties, are not considered in the optimization formulation. Additional constraints can also be incorporated if necessary. However, to ensure feasible designs and smooth transitions between each section of the wing configuration, thickness (t) constraints are applied. We use 70% of the baseline [119] thickness as the lower bound for the root section, with no upper bound imposed. The baseline thickness-to-chord ratio is 18% at the center plane; thus, with a chord length of 45 m, the absolute thickness (t) is $t_{base} = 8.1$ m. The tip thickness-to-chord ratio is 10%, so with a 5 m chord length, the tip thickness becomes $t_5 = 0.5$ m. The aerodynamic analysis was performed using the method described in Section 5.3.1.1. A genetic algorithm (GA), implemented via `pymoo` [97], was employed, with a population size of 50 and a maximum of 300 generations. The optimization was terminated early if the improvement in the objective function fell below 1×10^{-5} , which was set as the convergence criterion. The optimization data flow for VAE-BWB and parametrization is shown in <Figure 5-39>.

For the **Parametric method**, the optimization formulation is expressed as:



(a)



(b)

<Figure 5-39> GenAI optimization approach (a) VAE-BWB optimization (b) Parametrization based optimization

$$\begin{aligned}
 \max_{\mathbf{X}} \quad & J(\mathbf{X}) = \frac{C_L}{C_D} \\
 \text{w.r.t.} \quad & \mathbf{X}_{lb} \leq \mathbf{X} \leq \mathbf{X}_{ub} \\
 \text{s.t.} \quad & t_{i+1} \leq t_i, \quad i = 1, 2, \dots, n-2 \\
 & 0.7t_{base} \leq t_1
 \end{aligned} \tag{5.9}$$

where $\mathbf{X}$ represents the design variables listed in <Table 5-1>, including c_r , b , Λ_i , λ_i , Γ_i and θ , n represents the number of sections of wing.

For the **Parametric + VAE-airfoil** design optimization, the formulation is given as:

$$\begin{aligned}
 \max_{\mathbf{X}, \mathbf{z}} \quad & J(\mathbf{X}, \mathbf{z}) = \frac{C_L}{C_D} \\
 \text{w.r.t.} \quad & \mathbf{X}_{lb} \leq \mathbf{X} \leq \mathbf{X}_{ub} \\
 & \mathbf{z}_{lb,i} \leq \mathbf{z}_i \leq \mathbf{z}_{ub,i}, \quad i = 0, \dots, n-1 \\
 \text{s.t.} \quad & t_{i+1} \leq t_i, \quad i = 1, 2, \dots, n-2 \\
 & 0.7t_{base} \leq t_1
 \end{aligned} \tag{5.10}$$

where $\mathbf{X}$ denotes the planform design variables listed in <Table 5-1>, $\mathbf{z}_i$ represents the sectional airfoil latent variables, and t denotes the thickness for each wing section. n denotes the total number of wing sections.

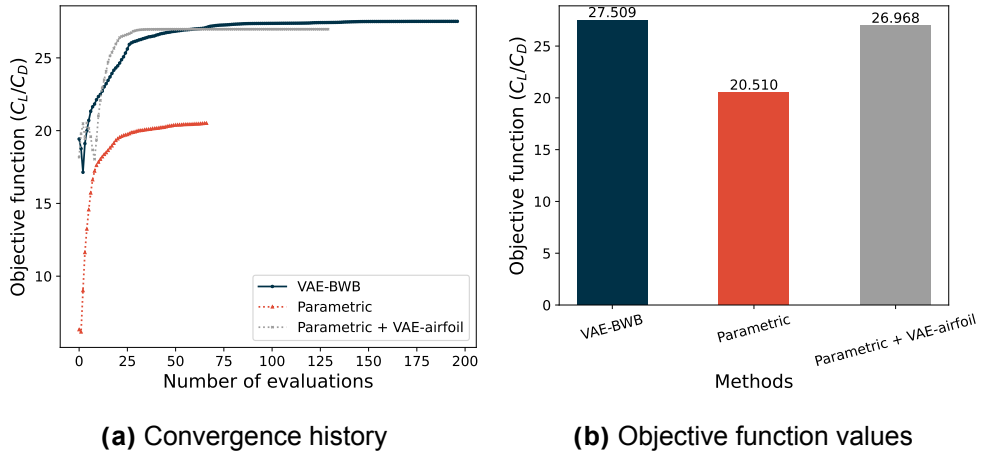
For the **VAE-BWB** design optimization, the formulation is defined as:

$$\begin{aligned}
 \max_{\mathbf{z}} \quad & J(\mathbf{z}) = \frac{C_L}{C_D} \\
 \text{w.r.t.} \quad & \mathbf{z}_{lb} \leq \mathbf{z} \leq \mathbf{z}_{ub} \\
 \text{s.t.} \quad & t_{i+1} \leq t_i, \quad i = 1, 2, \dots, n-1 \\
 & 0.7t_{base} \leq t_1
 \end{aligned} \tag{5.11}$$

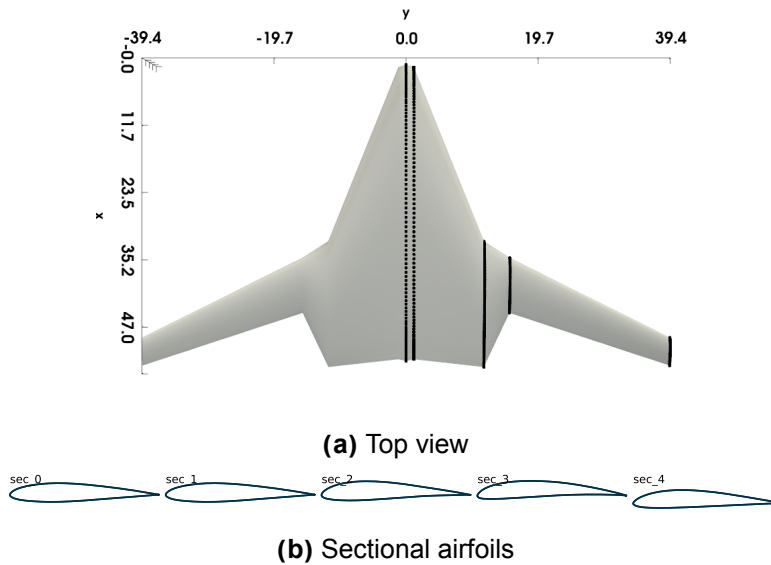
where $\mathbf{z}$ represents the latent design variables. The bounds on the latent variables are set to $z_\mu \pm 2z_\sigma$ to ensure that the latent distribution adequately spans the aircraft design space (see <Figure. 5-11a>) and t represents the thickness of each wing sections.

5.4.2.2 Optimization results

The convergence histories of the parametric model-based design and the VAE-based generative design are shown in <Figure. 5-40>. The planform using design variables and the reconstructed planform are compared in <Table 5-9>.



<Figure 5-40> Comparison of different parametrization method



<Figure 5-41> VAE-BWB Optimum Design

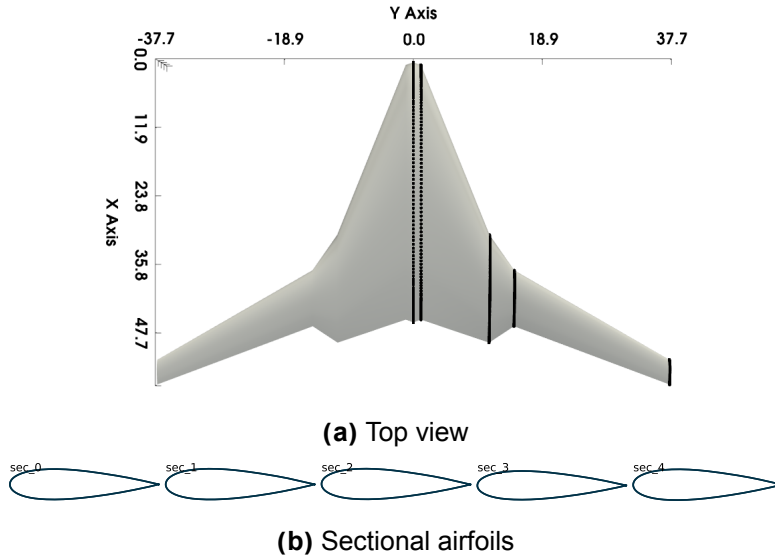
The **VAE-BWB** design improves the objective by **34.12%**, achieving $C_L/C_D =$

<Table 5-9> Comparison of optimization results

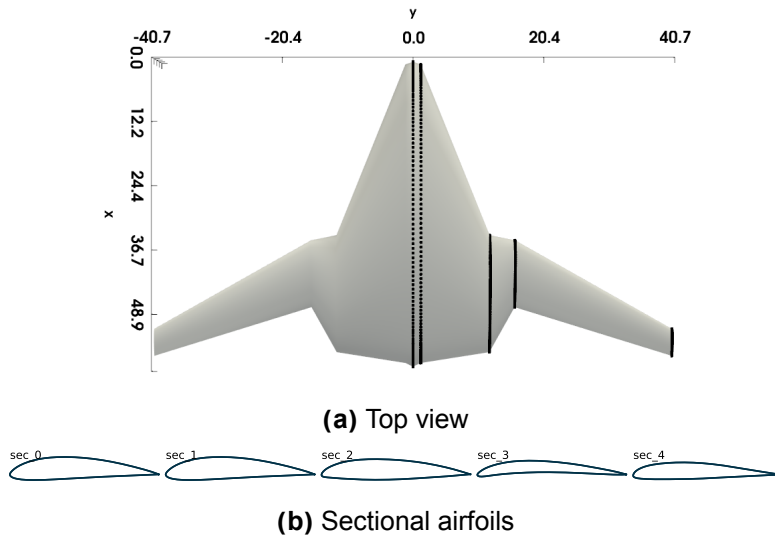
Parameters	Parametric	Parametric + VAE-airfoil	VAE-BWB
C_L/C_D	20.510	26.968	27.509
c_r (m)	42.395	48.789	45.936
b (m)	74.125	81.447	78.694
λ_1	0.979	0.979	0.983
λ_2	0.540	0.392	0.430
λ_3	0.402	0.570	0.436
λ_4	0.501	0.396	0.491
Λ_1 (deg)	18	18	18
Λ_2 (deg)	70.507	67.996	68.016
Λ_3 (deg)	56.147	11.173	32.655
Λ_4 (deg)	30.009	29.539	26.510
Γ_1 (deg)	0	0	0
Γ_2 (deg)	-2.117	-6.533	0.695
Γ_3 (deg)	22.149	10.454	11.984
Γ_4 (deg)	13.320	5.893	14.410
θ_1 (deg)	0	0	0
θ_2 (deg)	0	0	0
θ_3 (deg)	2.607	1.693	1.087
θ_4 (deg)	3.171	3.915	3.747
θ_5 (deg)	-1.469	-0.335	0.800

27.509, compared with $C_L/C_D = 20.510$ from the **Parametric** approach after 175 evaluations. There is also a **2.00%** improvement compared to the **Parametric+VAE-airfoil** optimum. Unlike the parametric model, which does not directly optimize surface coordinates, the VAE formulation enables full-shape deformation, exploring the design space with fewer variables and allowing for reduced-dimensional optimization with improved efficiency. The optimized results are shown in <Figure. 5-42>, <Figure. 5-43>, and <Figure. 5-41>, corresponding to the parametric optimum, the parametric+VAE-airfoil optimum, and the VAE-BWB optimum, respectively.

The parametric optimum exhibits a larger wing sweep angle, which reduces



<Figure 5-42> Parametric Optimum Design



<Figure 5-43> Parametric + VAE airfoil Optimum Design

drag while maintaining the sectional lift of the airfoil, thus delaying the critical Mach number and improving aerodynamic efficiency (C_L/C_D). However, it also introduces a dihedral angle of 21.508° , which increases lift but can lead to higher drag at high speeds. As the sweep angle increases, the onset of

transonic drag is further delayed, enhancing aerodynamic efficiency.

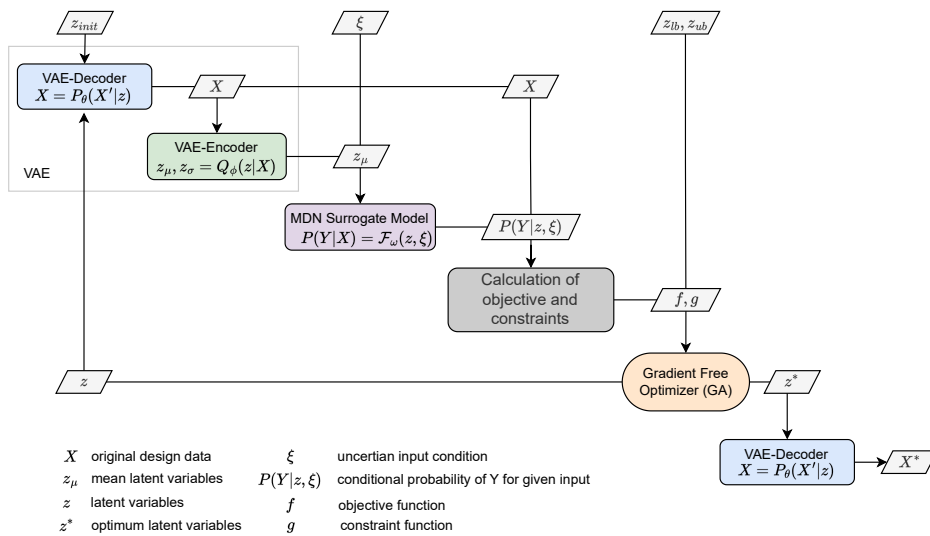
In the parametric+VAE-airfoil design, the airfoil shapes are deformed along the span due to the increased number of design variables, which makes the optimization more computationally expensive. However, this configuration achieved a C_L/C_D of 26.968. Additionally, no dihedral angle was introduced, while a swept-back wing was maintained, resulting in significantly reduced drag and increased lift. Although aerodynamic efficiency improved, the large number of variables imposed a considerable computational burden.

By contrast, the VAE-BWB configuration features a larger body area and a wingspan of 78.694 m, achieving a balanced chord and span distribution. The lower taper ratio, compared to the other configurations, decreases induced drag by creating a lift distribution closer to an ideal ellipse, thus improving aerodynamic efficiency. While exploring the design space, the VAE-BWB provides more realistic geometry and achieves a higher aerodynamic efficiency (C_L/C_D) compared with the parametric methods.

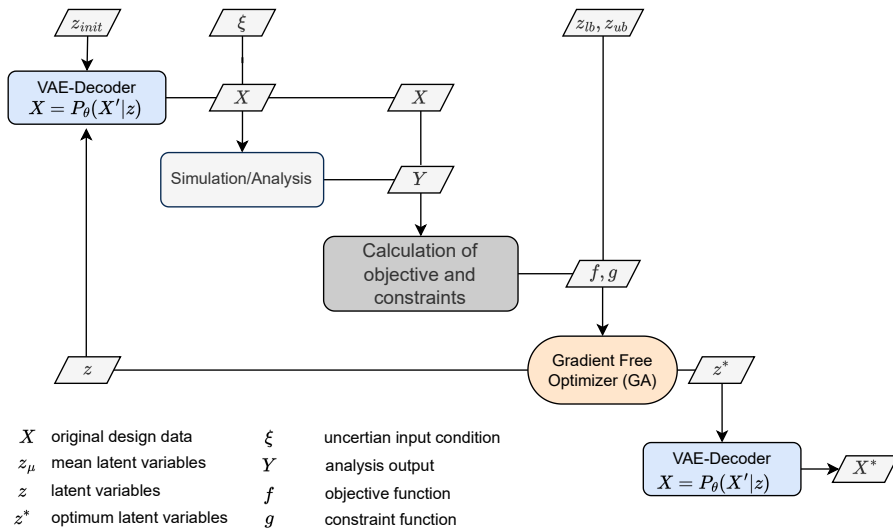
These findings indicate that the generative-AI optimization approach is substantially more efficient than high-dimensional optimization, where unrealistic designs often arise and exploration is constrained by feasible boundaries. This results in higher aerodynamic efficiency and higher-quality solutions with fewer design variables.

5.4.3 Aerodynamics shape optimization of blended wing body aircraft: Deterministic optimization

In order to evaluate the effectiveness and computational efficiency of the proposed GenAI optimization approach, we compare the surrogate-based optimization (VAE + MDN) <Figure. 5-44> with the simulation-based optimization



<Figure 5-44> GenAI optimization approach: Surrogate-based optimization (VAE + MDN)



<Figure 5-45> GenAI optimization approach: Simulation-based optimization (VAE + AVL)

(VAE + AVL) <Figure. 5-45>. Here, surrogate-based optimization uses our MDN surrogate model and simulation-based optimization uses aerodynamic

generation method.

5.4.3.1 Optimization formulation

In this section, we solve a constrained optimization problem using the same baseline BWB configuration. The lift and pitching-moment coefficients are imposed as constraints to ensure cruise flight conditions. For the BWB aircraft [119], the large wing planform area results in a lower required lift coefficient than conventional configurations. Thus, a lift coefficient of $C_L = 0.2$ is sufficient to maintain lift for the entire aircraft, and $C_M \leq 0$ is enforced for static stability. Additionally, thickness constraints are applied to ensure smooth transitions and to avoid overly thin airfoils along the wing span. The absolute thickness (t) at the root is $t_{base} = 8.1$ m, and the tip thickness is $t_5 = 0.5$ m, as considered in [119].

A deterministic design optimization is formulated as follows:

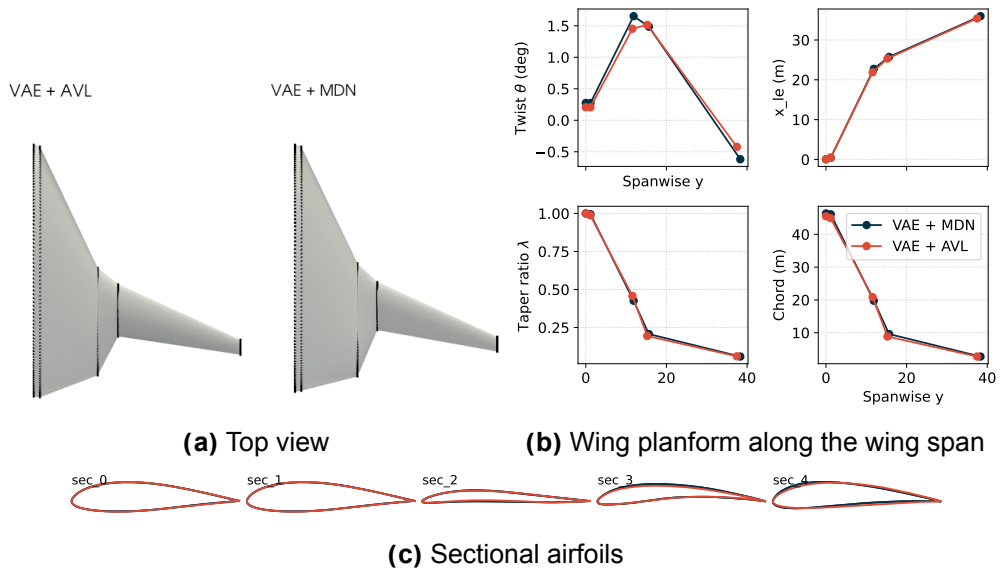
$$\begin{aligned}
 & \min_{\mathbf{z}} \quad J(\mathbf{z}) = C_D \\
 & \text{w.r.t.} \quad \mathbf{z}_{lb} \leq \mathbf{z} \leq \mathbf{z}_{ub} \\
 & \text{s.t.} \quad C_L = C_{L_{req}}, \\
 & \quad \quad C_M \leq 0 \\
 & \quad \quad t_{i+1} \leq t_i, \quad i = 1, 2, \dots, n-1 \\
 & \quad \quad 0.7t_{base} \leq t_1
 \end{aligned}$$

where, z represents the latent variables. We utilized the mean value from the surrogate model prediction in Eq. 3.22 with $N_{mcs} = 1$, as it does not consider the flight condition uncertainty.

5.4.3.2 Optimization results comparison

We assess the accuracy and efficiency of our framework by comparing simulation-based optimization (VAE + AVL) with surrogate-based optimization (VAE + MDN) in a deterministic setting. Because Monte Carlo sampling for design under uncertainty is computationally expensive, we omit full uncertainty-based optimization comparisons.

The optimization results for the surrogate-based and simulation-based approaches are shown in <Table 5-10>, and the corresponding configurations are illustrated in <Figure 5-46>. The results indicate that surrogate-based and simulation-based optimizations exhibit small discrepancies in root chord (c_r) and span (b), both on the order of approximately 2.0%. The aerodynamic performance shows a maximum error of about 14.5% in the moment coefficient, while lift and drag calculations have error percentages of 3.5%. In addition, the twist angle distribution differs by approximately 15%, demonstrating the accuracy level of the surrogate model.

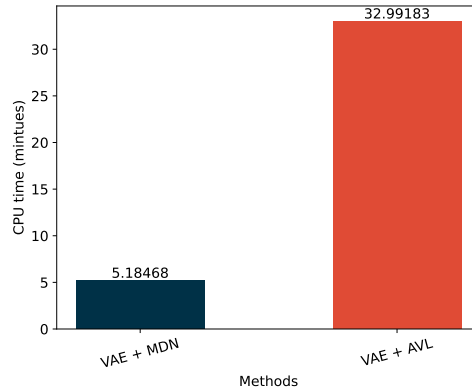


<Figure 5-46> Comparison of (VAE + MDN) and (VAE + AVL) optimum results

<Table 5-10> Comparison of optimization results with surrogate model and simulation model

Parameters	VAE + AVL	VAE + MDN	% diff
C_D	0.00826	0.00855	+3.507%
C_L	0.20134	0.20470	+1.668%
C_M	-0.00505	-0.00580	+14.851%
c_r (m)	46.365	45.498	-1.871%
b (m)	76.643	74.950	-2.207%
λ_1 (deg)	0.98	0.98	0.000%
λ_2 (deg)	0.428	0.463	+8.178%
λ_3 (deg)	0.485	0.424	-12.577%
λ_4 (deg)	0.285	0.310	+8.772%
Λ_1 (deg)	18	18	0.000%
Λ_2 (deg)	64.434	64.118	-0.490%
Λ_3 (deg)	38.392	40.143	+4.569%
Λ_4 (deg)	24.282	24.367	+0.351%
Γ_1 (deg)	0	0	0.000%
Γ_2 (deg)	-8.330	-7.873	+5.483%
Γ_3 (deg)	22.973	23.001	+0.122%
Γ_4 (deg)	16.594	15.964	-3.796%
θ_1 (deg)	0	0	0.000%
θ_2 (deg)	0	0	0.000%
θ_3 (deg)	1.653	1.453	-12.100%
θ_4 (deg)	1.485	1.514	+1.954%
θ_5 (deg)	-0.618	-0.521	+15.700%

<Figure 5-47> illustrates that the proposed generative-AI optimization approach can find optimal solutions much faster than the computationally expensive simulation-based approach. This case study demonstrates that the proposed optimization framework can be applied effectively while maintaining accuracy comparable to simulation-based design optimization.



<Figure 5-47> CPU compute time comparison for surrogate-based (VAE + MDN) and simulation-based optimization (VAE + AVL)

5.4.4 Aerodynamics shape optimization of blended wing body aircraft: Design under uncertainty

In the previous section, we showed that our surrogate model can perform deterministic optimization with less computational effort while maintaining accuracy. In this case study, we focus on design under uncertainty by considering variations in flight conditions. We have trained a probabilistic surrogate model that can accurately predict aerodynamic uncertainty, as demonstrated in Section 5.3.3. In this section, we conduct both robust design and reliability-based design optimization.

A robust design optimization is performed using weight sum method for objective function calculation, with the formulation below:

$$\begin{aligned}
& \min_{\mathbf{z}} \quad J(\mathbf{z}, \xi) = w \mu_{C_D} + (1 - w) \sigma_{C_D} \\
& \text{w.r.t.} \quad \mathbf{z}_{\text{lb}} \leq \mathbf{z} \leq \mathbf{z}_{\text{ub}} \\
& \text{s.t.} \quad \mu_{C_L} = C_{L_{\text{req}}}, \\
& \quad \mu_{C_M} \leq 0 \\
& \quad t_{i+1} \leq t_i, \quad i = 1, 2, \dots, n - 1 \\
& \quad 0.7t_{\text{base}} \leq t_1
\end{aligned} \tag{5.12}$$

where w is the weighting factor used to balance between μ and σ in the robust design objective function, and we set $w = 0.1$. The same constraints are applied as in the deterministic formulation, except that random variables ξ are considered for the angle of attack and Mach number. Therefore, we use μ_{C_L} and μ_{C_M} for equality constraints. Both parameters are sampled from normal distributions, $\alpha \sim \mathcal{N}(2.0, 0.1^2)$ and $M \sim \mathcal{N}(0.8, 0.05^2)$, with Monte Carlo samples of $N_{\text{mcs}} = 5000$. The mean and standard deviation statistics are calculated using Eq. (3.22) and Eq. (3.23), respectively.

Next, a reliability-based design optimization is conducted using the formulation below:

$$\begin{aligned}
& \min_{\mathbf{z}, \xi} \quad J(\mathbf{z}, \xi) = \mu_{C_D} \\
& \text{w.r.t.} \quad \mathbf{z}_{\text{lb}} \leq \mathbf{z} \leq \mathbf{z}_{\text{ub}} \\
& \text{s.t.} \quad \mathbf{P}_{\mathbf{f}}[\mathbf{C}_{\mathbf{L}_{\text{req}}} - \mathbf{C}_{\mathbf{L}}(\mathbf{z}, \xi) \geq \mathbf{0}] \leq \mathbf{P}_{\mathbf{f}_{\text{req}}}, \\
& \quad \mathbf{P}_{\mathbf{f}}[\mathbf{C}_{\mathbf{M}}(\mathbf{z}, \xi) - \mathbf{0} \geq \mathbf{0}] \leq \mathbf{P}_{\mathbf{f}_{\text{req}}}, \\
& \quad t_{i+1} \leq t_i, \quad i = 1, 2, \dots, n - 2 \\
& \quad 0.7t_{\text{base}} \leq t_1
\end{aligned} \tag{5.13}$$

where $P_{\mathbf{f}_{\text{req}}}$ is the required failure probability, which can be calculated from the cumulative distribution function (Φ) of the standard normal distribution using

the reliability index β as $P_{f\text{req}} = \Phi(-\beta)$. In our optimization, we set $\beta = 2.5$. We use the same strategy as in the robust design optimization to account for uncertainty, and the failure probability P_f is calculated using Eq. (3.25) with Monte Carlo samples of $N = 5000$.

5.4.4.1 Optimization results comparison

The uncertainty-based design optimization results, including both robust and reliability-based optima, are summarized in <Table 5-11>. <Figure 5-48>, <Figure 5-49>, and <Figure 5-50> present the final configurations, illustrating the planform geometry, sectional airfoils, and three-dimensional wing shapes. Additionally, the probability density functions (PDFs) of the aerodynamic uncertainty predictions are shown for the deterministic, robust, and reliability-based designs.

The deterministic optimum features a smaller wingspan and root chord, with a notched center-body section and a dihedral angle of approximately 7° across the wing segments. This configuration achieves moderate drag while satisfying lift and moment constraints, as shown in <Figure. 5-48a>.

The robust design optimum exhibits straighter leading edges, reduced dihedral angles, and slightly smaller aerodynamic surfaces. It produces the lowest drag among the three optimum designs and demonstrates minimal sensitivity to variations in operating conditions, achieving a drag standard deviation of just 0.000319.

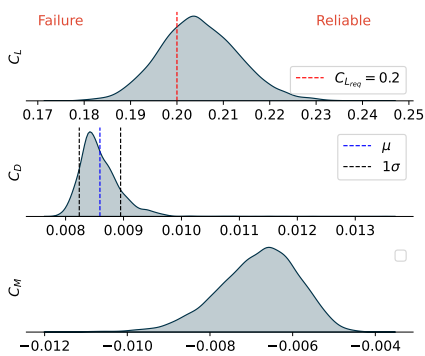
The reliability-based optimum (RBDO) maintains a planform similar to the DO and RDO designs, but adopts increased airfoil camber and thickness toward the outer wing. This leads to improved lift performance while keeping drag at a moderate level. The lift distribution moves into a safer region of the feasible design space, reducing the failure probability to 0.34% ($\beta = 2.5$) while

<Table 5-11> Comparison of optimization results for deterministic, robust and reliable designs

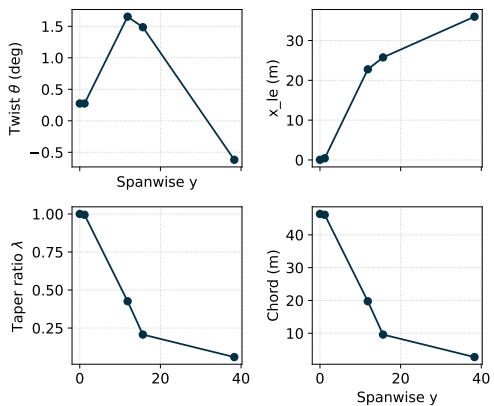
Parameters	DO	RDO	RBDO
C_D	0.00855	0.0077	0.00873
σ_{C_D}	0.000456	0.000319	0.000352
C_L	0.204	0.1994	0.228
C_M	-0.00580	-0.00387	-0.00481
t_1 (m)	5.931	5.908	6.440
t_2 (m)	5.899	5.819	6.355
t_3 (m)	2.316	1.915	1.934
t_4 (m)	1.161	1.332	1.029
t_5 (m)	0.477	0.570	0.434
c_r (m)	46.365	47.695	46.360
b (m)	76.643	74.554	74.707
λ_1 (deg)	0.980	0.986	0.980
λ_2 (deg)	0.428	0.436	0.446
λ_3 (deg)	0.485	0.500	0.483
λ_4 (deg)	0.285	0.413	0.426
Λ_1 (deg)	18	18	18
Λ_2 (deg)	64.434	66.436	63.528
Λ_3 (deg)	38.392	41.862	50.048
Λ_4 (deg)	24.282	26.566	22.204
Γ_1 (deg)	0	0	0
Γ_2 (deg)	-8.330	-5.743	-5.458
Γ_3 (deg)	22.973	23.031	15.657
Γ_4 (deg)	16.594	16.818	16.478
θ_1 (deg)	0	0	0
θ_2 (deg)	0	0	0
θ_3 (deg)	1.653	1.297	0.575
θ_4 (deg)	1.485	2.344	2.044
θ_5 (deg)	-0.618	-0.064	-2.00

the DO and RDO designs exhibit failure rates of approximately 50%.

The aerodynamic analyses for the optimized configurations are presented in <Figure 5-51>. Although the overall planform shape remains similar across the deterministic, RDO, and RBDO designs, the sectional shape parameters



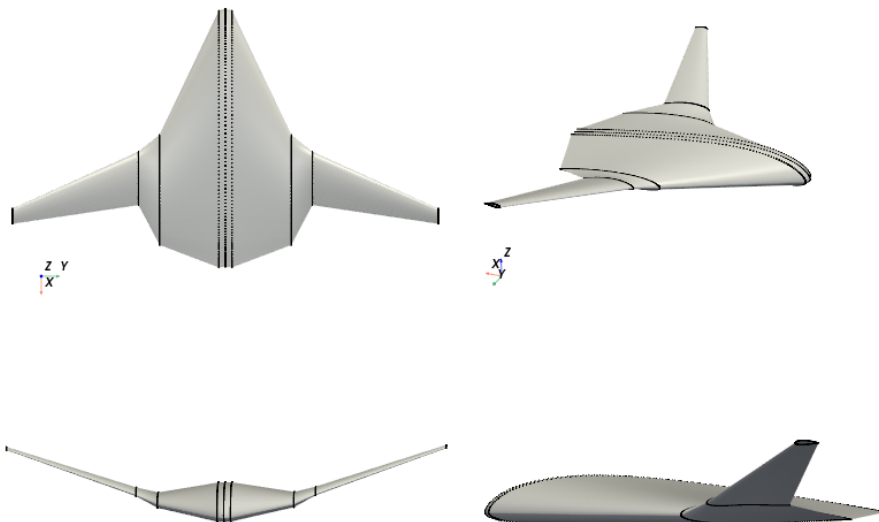
(a) PDF functions of aerodynamics uncertainty



(b) Wing planform along wing the span



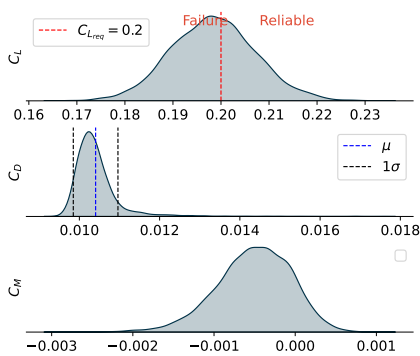
(c) Sectional airfoils



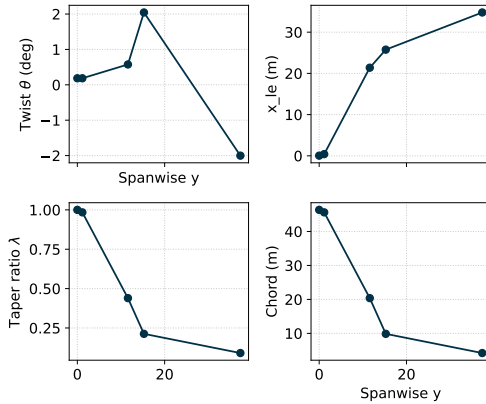
(d) Deterministic Optimum configuration

<Figure 5-48> Deterministic optimization (DO) results

—particularly camber distribution and thickness—strongly influence aerodynamic performance at high subsonic speeds (e.g., $M = 0.8$). Because the deterministic optimum does not incorporate Mach number uncertainty, <Fig-



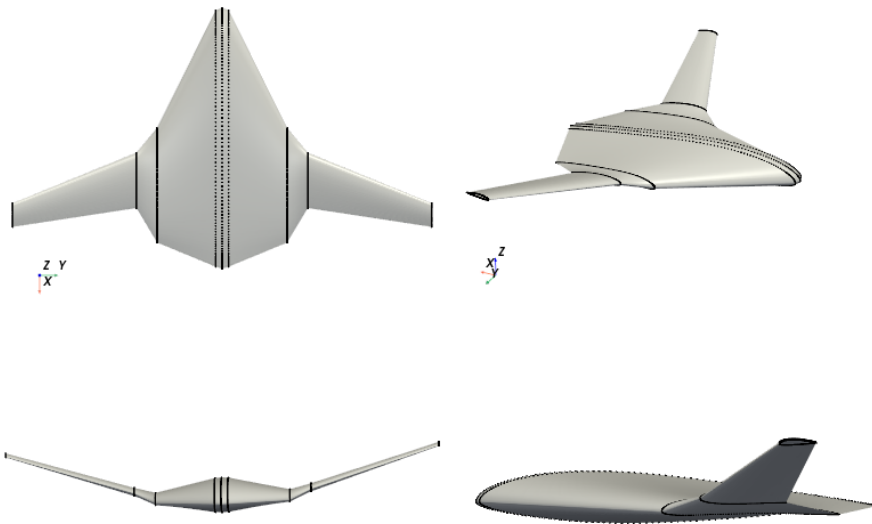
(a) PDF functions of aerodynamics uncertainty



(b) Wing planform along wing the span



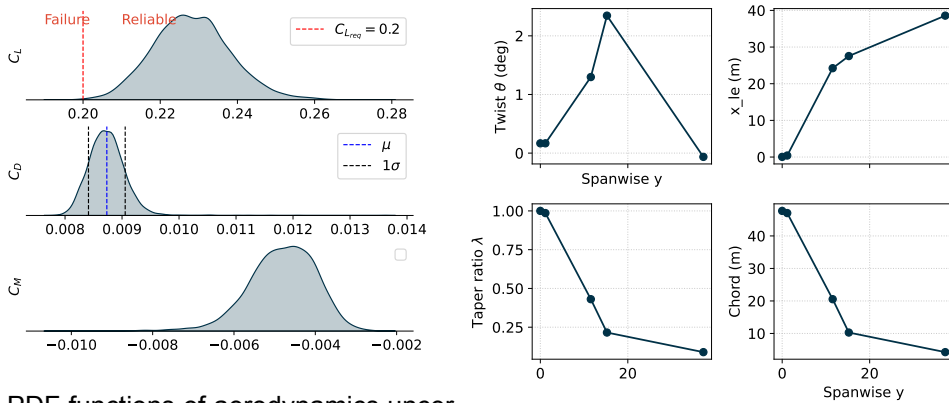
(c) Sectional airfoils



(d) Robust Optimum configuration

<Figure 5-49> Robust design optimization (RDO) results

ure 5-51b> shows an early rise in wave drag, with drag divergence occurring near $M \approx 0.85$. In contrast, the robust design, which considers Mach number uncertainty with a standard deviation of ± 0.05 , successfully delays the drag divergence Mach number. This delay results from sectional shape adjustments

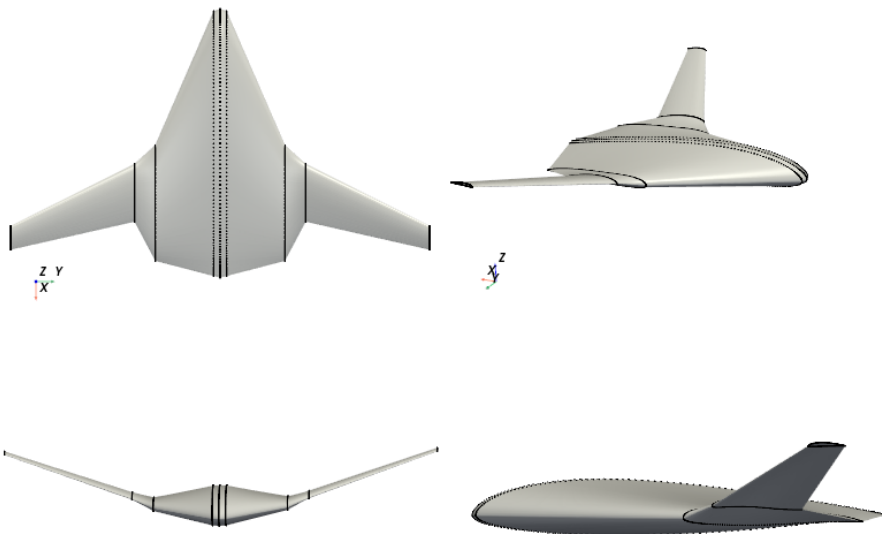


(a) PDF functions of aerodynamics uncertainty

(b) Wing planform along wing the span



(c) Sectional airfoils

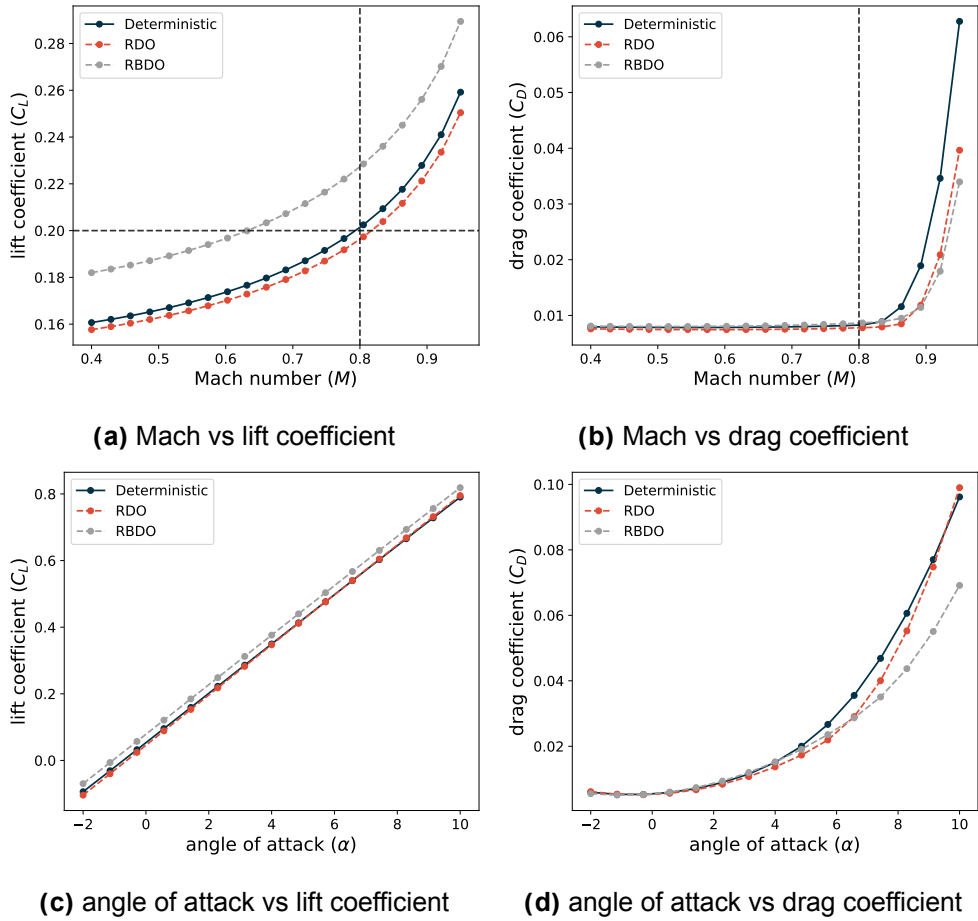


(d) Reliable optimum configuration

<Figure 5-50> Reliability-based design optimization (RBDO) results

that reduce sensitivity to variations in flight conditions.

Both the deterministic and robust optima exhibit similar lift coefficients, C_L . However, the RBDO optimum achieves a higher lift coefficient ($C_L = 0.228$) due to the reliability constraint, which encourages the optimizer to adopt slightly



<Figure 5-51> Aerodynamics analysis of optimum configurations

more cambered airfoils in the wing sections. Meanwhile, the body-section airfoil shapes remain similar to those in the robust optimum, allowing the RBDO design to maintain a delayed wave-drag onset comparable to that of the robust solution.

In addition, since we solely rely on AVL for lift and induced-drag calculations, we also evaluate the Prandtl–Glauert (PG) compressibility correction[129] for the optimum designs. Based on AVL’s documented criteria, the optimum wing configurations fall within the “PG-suspect” regime of the correction model at $M = 0.8$. Although the deterministic, robust, and reliability-based optima ex-

hibit variations in sweep, chord distribution, and dihedral, their effective compressibility behavior are suspected the subsonic PG relation, $\beta = \sqrt{1 - M^2 \cos^2 \Lambda}$, PG correction factor ($\frac{1}{\beta}$). According to AVL's guidelines, the C_L predictions for these optimum designs are classified as "PG-suspect" at $M = 0.8$, indicating that compressibility effects may introduce additional uncertainty. The computed PG-correction factors for all optimum designs are summarized in <Table 5-12>. These results indicate that the lift predictions at $M = 0.8$ should be interpreted with caution due to compressibility-related uncertainty.

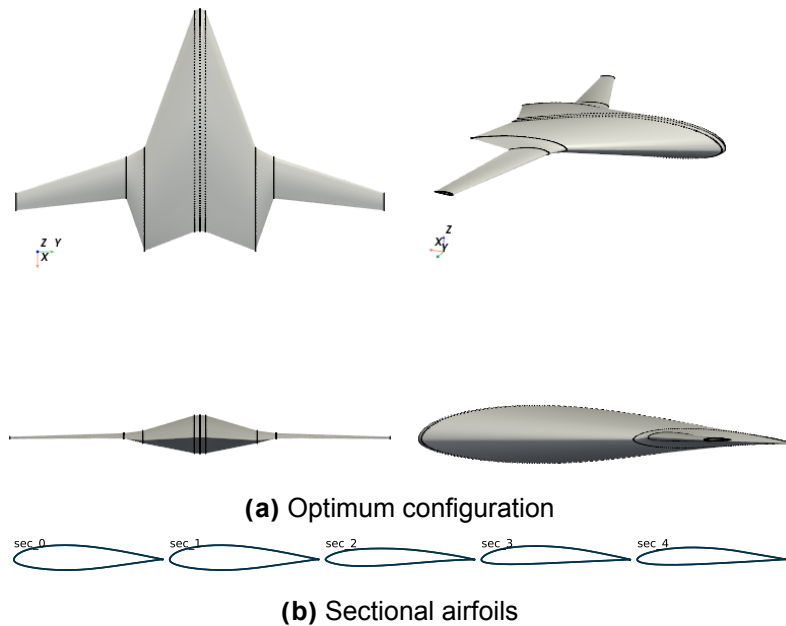
<Table 5-12> Prandtl–Glauert compressibility evaluation for optimum designs using leading-edge sweep angles at $M = 0.8$ and $M = 0.6$.

Design	Section	$M_n = M \cos \Lambda_i$	$1/\beta_n$	AVL Validity
DO	Λ_1	0.760	1.540	PG suspect
	Λ_2	0.348	1.063	PG valid
	Λ_3	0.736	1.510	PG suspect
	Λ_4	0.730	1.500	PG suspect
RDO	Λ_1	0.760	1.540	PG suspect
	Λ_2	0.321	1.049	PG valid
	Λ_3	0.736	1.510	PG suspect
	Λ_4	0.714	1.476	PG suspect
RBDO	Λ_1	0.760	1.540	PG suspect
	Λ_2	0.353	1.066	PG valid
	Λ_3	0.770	1.577	PG suspect
	Λ_4	0.739	1.513	PG suspect
Design at $M = 0.6$	Λ_1	0.571	1.218	PG valid
	Λ_2	0.205	1.021	PG valid
	Λ_3	0.530	1.179	PG valid
	Λ_4	0.570	1.217	PG valid

AVL Prandtl–Glauert (PG) Validity Conditions:

PG Valid: $1/\beta \leq 1.25$; PG Suspect: $1/\beta \approx 1.3\text{--}1.5$; PG Unreliable: $1/\beta > 1.6$.

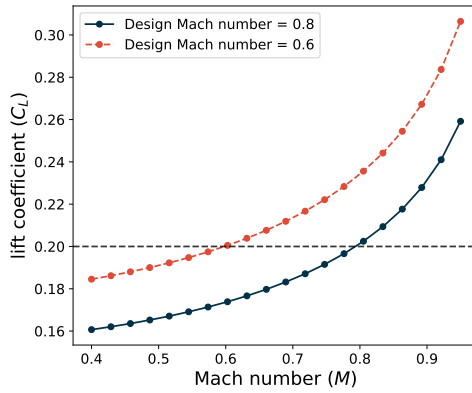
We also assess an additional optimization at reserve-mission flight condition at **Mach = 0.6** and an altitude of **6080 m**, following the criteria in [142]. Using the same optimization formulation as in the Mach 0.8 case, the result-



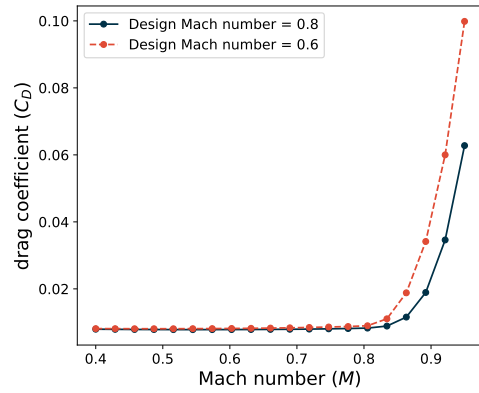
<Figure 5-52> Optimum design at Mach 0.6, $\alpha = 2^\circ$, altitude = 6080 m

ing optimal configuration is presented in <Figure. 5-52>. The computed PG-correction factors indicate that the Mach 0.6 design operates entirely within AVL' s "PG-valid" regime, suggesting that the aerodynamic predictions at this condition are more reliable than those for the Mach 0.8 design. As shown in <Figure. 5-53>, the Mach 0.6 optimum employs increased airfoil camber, reduced leading-edge sweep, and a slightly increased wingspan to satisfy the lift requirement at the lower Mach number. These geometric adaptations enable the configuration to achieve greater lift across the Mach range, delay drag rise, and maintain smoother aerodynamic behavior at elevated angles of attack. Overall, the Mach 0.6 design demonstrates improved robustness under subsonic compressibility effects, providing a more reliable and aerodynamically efficient solution for reserve-mission operation.

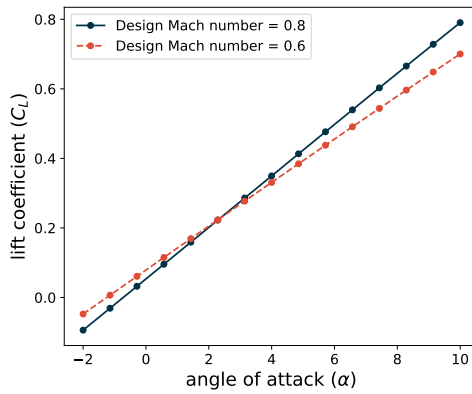
This case study demonstrates that the proposed optimization framework is applicable not only to small-scale problems such as airfoil design, but also to



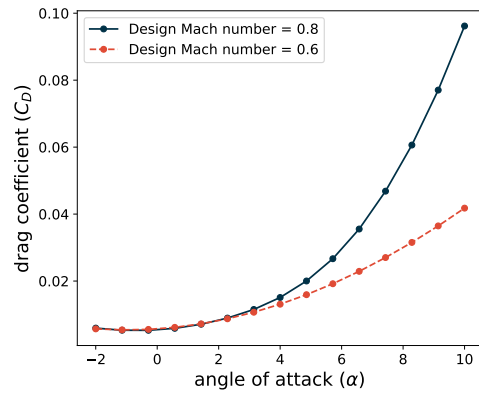
(a) Mach vs lift coefficient



(b) Mach vs drag coefficient



(c) angle of attack vs lift coefficient



(d) angle of attack vs drag coefficient

<Figure 5-53> Aerodynamics analysis of optimum configurations

large-scale aircraft configurations. It supports both deterministic and uncertainty-based optimization and remains effective even with limited training data, achieving performance comparable to high-fidelity simulation-driven approaches.

5.5 Summary

In this chapter, the developed method was applied to a more challenging problem—the design of a blended-wing-body (BWB) aircraft—to demonstrate the adaptability of our approach. A total of 3,000 design variables describing the

aircraft geometry were reduced to a compact 16-dimensional latent space that effectively captures the essential features of the aircraft shape parameters. Optimization performed in this lower-dimensional latent space significantly enhanced design exploration within the same design boundaries compared to conventional parameterized models. Specifically, the proposed approach achieved a 34.12% improvement over the parametric-only model (13 variables) and a 2.00% improvement over the combined parametric-plus-airfoil model (54 variables).

Next, a probabilistic surrogate model was developed. Instead of modeling from the original design space, we utilized latent variables and uncertain parameters as inputs and used aerodynamic performance as outputs. The model achieved an accuracy of $R^2 = 0.999$ in mean prediction, with the worst case being a standard deviation prediction of $R^2 = 0.9973$ for the drag coefficient. In addition, we compared the uncertainty predictions from the latent space with Monte Carlo simulation results, showing good alignment in the PDF functions for aerodynamic predictions.

Furthermore, we proposed an uncertainty-based design optimization framework that leverages trained models to perform deterministic, robust, and reliability-based design optimization. The latent representation model was utilized as a parameterization scheme and compared with simulation-based optimization, demonstrating comparable accuracy in deterministic optimization. The uncertainty-based optimization case studies further verified that employing a probabilistic surrogate model effectively reduces computational cost, even with limited data availability. The PG compressibility correction effects were also evaluated for the optimum design to assess the robustness of the aerodynamic analysis method. Additionally, we conducted a design study at Mach 0.6 conditions, in which lift calculation compressibility is more reliable than at high-speed sub-

sonic Mach 0.8.

Overall, the proposed approach provides a valuable and efficient framework for design space exploration and accurate uncertainty prediction under data constraints. Moreover, the uncertainty-based optimization framework proves to be highly effective for robust and reliability-based optimization within a reduced-dimensional design space.

6. Conclusions

In this final chapter, we summarize the contributions of this work, discuss recommendations, its limitations, and present potential avenues for future research.

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6.1 Dissertation summary

In this dissertation, we proposed a new way of solving uncertainty-based optimization problems by enhancing the problem-solving framework in terms of solution quality, speed, and adaptability, through the integration of generative AI and probabilistic machine learning models. The outcomes of this dissertation are summarized as follows.

Firstly, we demonstrate how the high-dimensional design spaces of airfoil and aircraft design can be reduced to compact, lower-dimensional representations without losing effectiveness, by employing deep generative models—specifically, Variational Autoencoders (VAE). This approach transforms large-scale optimization problems into low-dimensional ones, thereby improving solution quality and computational efficiency.

Second, through deep generative modeling, we learn a systematic representation of airfoil and aircraft geometries. As a result, design feasibility is improved compared to conventional parameterization methods. Consequently, abnormal shape generation—which often burdens the optimization process—is effectively eliminated.

Third, the probability density of aerodynamic performance is modeled using

a mixture density network (MDN) from the lower-dimensional representation of design parameters, thereby improving model training and eliminating the need for repeated computations. Here, variations in Mach number and angle of attack are considered sources of uncertainty, and we have demonstrated this approach for both low-speed and high-speed subsonic flight conditions.

Fourth, a constraint-based geometric sampling strategy is proposed for generating diverse aircraft shapes without producing infeasible designs during deep generative model training. This method can be effectively applied not only to aircraft but also to wing-like shapes such as propeller blades, and wind turbine blades.

Finally, an efficient optimization approach is developed that can adapt to different scales of optimization problems—from airfoil design to complex aircraft design—through generative modeling and uncertainty prediction using probabilistic surrogate modeling.

These results demonstrate that incorporating generative AI and probabilistic surrogate models overcomes the limitations of conventional optimization approaches, such as limited design space, infeasible design generation, high dimensionality, and high computational cost. The proposed framework improves problem-solving efficiency by facilitating the discovery of high-quality solutions, accelerating convergence, and ensuring adaptability to both small- and large-scale problems, representing a significant step forward in aerospace optimization research.

6.2 Recommendations and limitations of study

This section outlines the key recommendations and limitations of the proposed generative AI based optimization framework. Since the generative model com-

presses high dimensional geometric data into a low dimensional latent space, the optimization problem can be formulated in a manner similar to a conventional design optimization task. However, instead of directly manipulating physical design variables, it is recommended to perform optimization using latent variables while enforcing the necessary physical and geometric constraints to ensure feasible designs.

Although the proposed framework represents a step forward in aerospace design optimization, several limitations remain. As with most machine-learning-based approaches, data quality plays a crucial role, especially for deep generative models. High-dimensional geometric data must be consistently and systematically represented; otherwise, the generative model may produce unrealistic or infeasible shapes, reducing optimization performance and reliability.

Another limitation arises from the fidelity of the aerodynamic data used during surrogate model training. The probabilistic surrogate models in this work were trained primarily using low fidelity analysis tools, such as NeuralFoil for airfoil analysis, AVL for three dimensional aerodynamic prediction, and empirical equations for high subsonic blended wing body aircraft. To improve the accuracy and dependability of the predicted aerodynamic performance distributions, future studies should incorporate higher fidelity aerodynamic analysis methods.

Finally, the latent space learned by the generative model is inherently non-convex and may contain disconnected or irregular regions. As a result, global optimization algorithms, such as evolutionary algorithms or other gradient free metaheuristic methods, remain necessary to explore the latent design space effectively and avoid suboptimal local minima.

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Appendix

Appendix 1: Nomenclature

A	Airfoil area
A_w	Wing planform area
A_i	Bernstein polynomial coefficients (CST)
B	Wing span
C	Chord length
C_{D_0}	Zero-lift drag coefficient
$C_{D_{ind}}$	Induced drag coefficient
C_{D_w}	Wave drag coefficient
C_L	Lift coefficient
$C_{L_{max}}$	Maximum lift coefficient
C_M	Moment coefficient
$C(x)$	Class function of the CST method
C_{ij}	Covariance matrix element
c	Sectional chord length
c_r	Root chord
c_{ref}	Reference chord length
D_{KL}	Kullback–Leibler divergence
FF	Form factor
h	Altitude
h_n^l	Neuron at the n -th position of layer l

k	Number of latent dimensions
M	Mach number
M_{crit}	Critical Mach number
M_{dd}	Drag divergence Mach number
m	Number of mixture components (MDN)
N_{mcs}	Number of Monte Carlo samples
P_f	Failure probability
$P(Y X)$	Conditional probability density of Y given X
Q_ϕ	VAE encoder network
P_θ	VAE decoder network
R^2	Coefficient of determination
Re	Reynolds number
S_{wet}	Wetted area
S_{ref}	Reference area
$S(x)$	Shape function of the CST method
t_{max}	Maximum thickness
t/c	Thickness-to-chord ratio
α	Angle of attack
β	Reliability index
β_{target}	Target reliability index
β_{reg}	Regularization coefficient in VAE loss
$\mathbf{b}$	Bias vector of a neural network
Γ_i	Dihedral angle distribution
γ	Dihedral angle
Λ_i	Sweep angle distribution
λ	Taper ratio distribution (or chord-scaling parameter)
$\mathcal{L}_{KL}$	KL divergence loss component

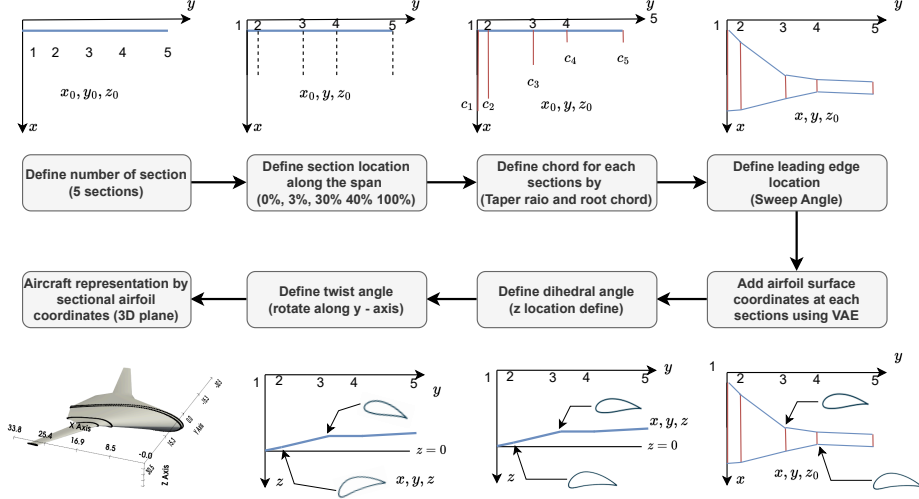
$\mathcal{L}_{MDN}$	MDN training loss
$\mathcal{L}_{recon}$	Reconstruction loss of VAE
$\mathcal{L}_{VAE}$	Total VAE loss function
μ	Mean value (latent / aerodynamic parameters)
η_j	Spanwise distribution parameter
σ	Standard deviation
$\sigma(z)$	Activation function
θ_i	Twist angle distribution
b	Bias vector
W	Weight matrix of a neural network
X	Input data (original design)
X'	Reconstructed design
X_t	Test input data
Y	Target output data
Y_t	Test output data
z	Latent variable vector
z	Latent variable
z_μ	Mean of latent variable distribution
z_σ	Standard deviation of latent variable distribution
ξ	Random variables
detR	Determinant of correlation matrix
x_{LE}	Leading-edge x -coordinate
y	Spanwise coordinate

Appendix 2: Abbreviations

AE	Autoencoder
ASO	Aerodynamic Shape Optimization
AVL	Athena Vortex Lattice
BNN	Bayesian Neural Network
BWB	Blended-Wing Body
CADE	Comprehensive Aircraft Design Environment
CDF	Cumulative Distribution Function
CST	Class–Shape Transformation
DNN	Deep Neural Network
DOE	Design of Experiments
FFD	Free-Form Deformation
GenAI	Generative AI
GAN	Generative Adversarial Network
GA	Genetic Algorithm
GPR	Gaussian Process Regression
LHS	Latin Hypercube Sampling
LSO	Latent Space Optimization
MCS	Monte Carlo Simulation
MDO	Multidisciplinary Design Optimization
MDN	Mixture Density Network
MMD	Maximum Mean Discrepancy
MSE	Mean Squared Error
NLL	Negative Log-Likelihood
PDF	Probability Density Function
PINN	Physics-Informed Neural Network

PCA	Principal Component Analysis
QoI	Quantity of Interest
RBDO	Reliability-Based Design Optimization
RDO	Robust Design Optimization
Re	Reynolds Number
SVD	Singular Value Decomposition
UQ	Uncertainty Quantification
UBO	Uncertainty-Based Optimization
VAE	Variational Autoencoder
VLM	Vortex Lattice Method

Appendix 3: Parametrization for BWB aircraft



<Figure 6-1> Constrained based geometric sampling

This section presents the details derivation of the parametric modeling of constrained based geometric sampling strategy. Step-by-step data flow is illustrated in Figure. 6-1.

Firstly, the aircraft wing is divided into multiple sections, and the sectional spanwise locations are defined as percentages along the span (b) using Eq. (6.1).

$$y_{i+1} = y_i + \Delta y_i, \quad \Delta y_i = \frac{b}{2}(\eta_i - \eta_{i-1}) \quad (6.1)$$

where, y_i is the section location, η_j is the spanwise distribution in percentage. Here, span wise sections $\eta_j = [0\%, 3\%, 30\%, 40\%, 100\%]$ considering five sections to represent aircraft. Second, chord lengths of each section (c_j) are determined by taper ratio distribution (λ_j) based on the root chords (c_r) using Eq. (6.2).

$$c_{j+1} = c_j \lambda_j, \quad \Delta c_j = c_{j+1} - c_j \quad (6.2)$$

where c_0 is the root chord and $\lambda_j = c_j/c_{j-1}$ represents the local taper ratio between adjacent sections. Then, leading edge locations (x_{LE}, y_{LE}, z_{LE}) of each sections are defined by sweep angle distribution (Λ_j) . The offset of each sections can is calculated by using below Eq. (6.4), Eq. (6.5) and Eq. (6.6).

$$\Delta x_j = y_j \tan \Lambda_j \quad (6.3)$$

$$x_{j+1} = x_j + \Delta x_j, \quad \Delta x_j = s_j \tan \Lambda_j \quad (6.4)$$

$$y_{j+1} = y_j + \Delta y_j, \quad \Delta y_j = \frac{b}{2}(f_j - f_{j-1}) \quad (6.5)$$

$$z_{j+1} = z_j, \quad z_j = 0 \quad (6.6)$$

The leading edge of each sections along the y-axis becomes,

$$x_{LE,j} = x_j, \quad y_{LE,j} = y_j, \quad z_{LE,j} = z_j, \quad x_{TE,j} = x_j + c_j. \quad (6.7)$$

In the next step, airfoils are added at the leading edge location using trained VAE decoder and surface coordinates are added. Here, airfoil surface coordinates are present as $([\hat{x}, \hat{z}(\hat{x})]_{200})$, where, $\hat{x}$ is the normalized x coordinates of local airfoil and $\hat{z}$ is the surface coordinates of local airfoil along $\hat{x}$ axis. Here, each section airfoils becomes, u, w using Eq. (6.8).

$$u = c_j \hat{x}, \quad w = c_j \hat{z}(\hat{x}), \quad (6.8)$$

Now, we present, each section airfoil coordinates and leading edge location by Matrix form as below.

$$\mathbf{r}_{\text{sec}}(\hat{x}) = \begin{bmatrix} u \\ 0 \\ w \end{bmatrix}, \quad \mathbf{p}_{0,j} = \begin{bmatrix} x_{\text{LE},j} \\ y_{\text{LE},j} \\ z_{\text{LE},j} \end{bmatrix}. \quad (6.9)$$

where, $\mathbf{r}_{\text{sec}}$ is each section airfoil along the y-axis. and $\mathbf{p}_{0,j}$ is the location of the leading edge of each section.

In the next step, dihedral distortion Γ_j is added along each sections. As we have added the airfoil coordinates, here, we use rotational matrix along x-axis using Eq. (6.10) with dihedral angle.

$$\mathbf{R}_x(\Gamma_j) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \Gamma_j & -\sin \Gamma_j \\ 0 & \sin \Gamma_j & \cos \Gamma_j \end{bmatrix}, \quad \mathbf{r}^{(d)} = \mathbf{R}_x(\Gamma_j) \mathbf{r}_{\text{sec}} = \begin{bmatrix} u \\ -w \sin \Gamma_j \\ w \cos \Gamma_j \end{bmatrix}. \quad (6.10)$$

Combining the location of the leading edge $\mathbf{p}_{0,j}$ and the output of rotation for each section $\mathbf{r}^{(d)}$. All the coordinates can be represented as Eq. (6.11).

$$x_j^{(d)}(\hat{x}) = x_{\text{LE},j} + u, \quad y_j^{(d)}(\hat{x}) = y_{\text{LE},j} - w \sin \Gamma_j, \quad z_j^{(d)}(\hat{x}) = z_{\text{LE},j} + w \cos \Gamma_j. \quad (6.11)$$

where, x^d represents x locations and d denotes the dihedral angle rotation.

Next, twist angle distribution θ_j are added at each section j . This process is done by rotating along the y-axis using Eq. (6.12)

$$\mathbf{R}_y(\theta_j) = \begin{bmatrix} \cos \theta_j & 0 & \sin \theta_j \\ 0 & 1 & 0 \\ -\sin \theta_j & 0 & \cos \theta_j \end{bmatrix}, \quad \mathbf{r}^{(t)} = \mathbf{R}_y(\theta_j) \mathbf{r}^{(d)}. \quad (6.12)$$

Finally, a global surface coordinates in three dimensional plane can be presented as Eq. (6.13).

$$\begin{aligned}x_j(\hat{x}) &= x_{LE,j} + u \cos \theta_j + (w \cos \Gamma_i) \sin \theta_j, \\y_j(\hat{x}) &= y_{LE,j} - w \sin \Gamma_j, \\z_j(\hat{x}) &= z_{LE,j} - u \sin \theta_j + (w \cos \Gamma_j) \cos \theta_j,\end{aligned}\tag{6.13}$$

where, u and w are sectional airfoil surface coordinates as presented in Eq. 6.8. Through this process, and given the constraints on wing planform parameters c_r , b , λ , Λ , and θ , diverse wing geometries can be generated in 3D surface coordinates, which can then be used for training the generative model.

Here, we presents geometric parameters calculation for subsequent process such as wing area, wetted area and MAC calculation. Wing area (S) is calculated by integrating along the span wise distribution using Eq. (6.14).

$$S = 2 \int_0^{b/2} c(y) dy,\tag{6.14}$$

and the aspect ratio is given by

$$AR = \frac{b^2}{S}.\tag{6.15}$$

Using wing area (S) and spanwise chord c , mean aerodynamic chord MAC is calculated by below equation;

$$MAC = \bar{c} = \frac{2}{S} \int_0^{b/2} c(y)^2 dy, \quad S = 2 \int_0^{b/2} c(y) dy.\tag{6.16}$$

Here, spanwise average $(t/c)_{max}$ is calculated using below equation;

$$\left(\frac{t}{c}\right)_{max} = \frac{2}{S_{ref}} \int_0^{b/2} \left(\frac{t}{c}\right)_{max}(y) c(y) dy\tag{6.17}$$

The wetted area of aircraft for friction drag calculation is calculated by using below equation;

$$S_{wet} = 2 S_{ref} (1 + 0.25 (t/c)_{max}) \quad (6.18)$$

Abstract (in Korean)

대규모 항공기 불확실성 설계를 위한 GenAI 최적설계 기법

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설계 최적화는 항공기 설계와 같은 대규모 공학 문제에서 필수적인 요소이다. 실제 설계 과정에는 본질적으로 불확실성이 존재하며, 불확실한 조건 하에서의 성능 확률을 평가하기 위해서는 반복적인 해석 또는 시뮬레이션이 요구된다. 이러한 요구로 인해 불확실성 기반 공력 형상 최적화는 매우 높은 계산 비용을 수반한다. 최적 해는 설계 공간의 표현 방식에 크게 의존하며, 설계 변수의 차원은 최적화 과정의 효율성에 중요한 영향을 미친다. 항공기 설계에서는 설계 공간이 일반적으로 고차원적이고 복잡하여 고품질의 해를 도출하기 어렵고, 계산 비용이 크게 증가하는 문제가 발생한다. 이러한 한계는 다양한 규모와 복잡도를 갖는 최적화 문제에 대해 효율적으로 고품질 해를 도출할 수 있는 불확실성 기반 최적화 프레임워크의 필요성을 강조한다.

본 학위논문은 해의 품질, 계산 속도, 그리고 다양한 최적화 문제에 대한 적응성을 중심으로 불확실성 기반 최적화의 효율성을 향상시키는 것을 목표로 한다. 이를 위해 생성형 인공지능(Generative AI, GenAI) 기반 최적화 프레임워크를 제안한다. 변분 오토인코더(Variational Autoencoder, VAE)를 활용하여 복잡하고 고차원적인 기하학적 설계 공간을 물리적으로 의미 있는 특징을 보존하면서도 원래의 설계 공간을 효과적으로 포괄하는 저차원의 잠재 공간(latent space)으로 축소한다. 이후 혼합 밀도 신경망(Mixture Density Network, MDN)을 학습시켜 잠재 공간 내에서 비행 조건의 변화로 인해 발생하는 공력 성능의

불확실성을 예측함으로써 확률 분포를 효율적으로 추정한다. 이러한 모델들은 불확실성 기반 최적화 과정에 통합되어, 계산 부담을 크게 줄이면서도 축소된 설계 공간에서 강건 최적화 및 신뢰성 기반 최적화를 수행할 수 있도록 한다.

제안된 프레임워크는 에어포일 설계와 블렌디드 윙 바디(Blended-Wing-Body, BWB) 형상의 복잡한 항공기 설계를 대상으로 한 두 가지 사례 연구를 통해 검증되었다. 그 결과, 본 방법은 실현 가능하고 다양한 설계안을 생성할 수 있으며, 공력 성능 불확실성을 정확하게 예측하고, 기존 접근법 대비 최적화 효율성과 해의 품질을 크게 향상시킬 수 있음을 확인하였다. 또한 다수의 설계 변수를 포함하는 최적화 문제에도 적용 가능함을 보여주어, 항공기 설계 분야에서의 적응성과 실용성을 입증하였다.

이러한 결과는 제안된 방법이 단순한 문제부터 복잡한 대규모 문제에 이르기까지 불확실성 기반 최적화를 효과적으로 수행할 수 있음을 보여주며, 계산 효율성과 해의 품질 측면에서 상당한 향상을 달성할 수 있음을 시사한다. 종합적으로, 본 연구에서 제안한 방법론은 항공우주 최적화 연구 분야에서 중요한 진전을 이루는 성과라 할 수 있다.

주제어 : 생성형 인공지능, 확률적 기계학습, 대규모 설계 최적화, 불확실성 하 설계, 에어포일 설계, 블렌디드 윙 바디 항공기 설계